XIZI PUMPS

Submersible Sewage Pump Series

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Hangzhou Xizi Pumps Co., Ltd. has been dedicated to the research&development and manufacturing of submersible sewage pumps and related systems for over thirty years. Since its establishment in 1994, the company has consistently adhered to the business philosophy of "Customer First, Quality Foremost." We focus on perfecting every product and are committed to providing superior pump products and services for every partner.

The company has collaborated with relevant universities to build a professional R&D and technical team, obtaining over 40 national utility/invention patents and computer software copyrights. In 2022, Xizi Pump was among the first to receive the **"National Grade One Energy Efficiency"** certification for submersible sewage pumps. Thousands of Xizi pump products are successfully operating worldwide, delivering excellent Xizi solutions in fields such as wastewater treatment, sewage transportation, flood control and drainage, water conservancy projects, irrigation, and water supply and drainage.

Since its inception, Xizi Pump Industry has emphasized the learning and application of modern scientific management methods, having successively obtained ISO9001, ISO14001, and ISO45001 management system certifications. It has also been recognized for multiple years as a **"National High-Tech Enterprise," "Zhejiang Provincial Specialized, Refined, Unique, and New Enterprise," "Zhejiang Manufacturing Certification,"** and **"Hangzhou Enterprise Technology Center."**

Honor & Qualification

Factory Overview

The factory is located in the Tangqi Industrial Park, Linping District, Hangzhou, with a construction area of 30,000 square meters. It houses comprehensive production, manufacturing, R&D, and testing centers. Key functional areas include metal processing, motor manufacturing, pump assembly, pump station and sewage lift workshops, and a "National Grade One Precision" pump testing center.
The factory can customize personalized products and services for clients with diverse needs, providing optimal pump and system solutions.

Professional
Professional dedication, with thirty years of R&D and manufacturing experience in sewage pumps and systems.

High-quality
Possesses a strict quality management system and advanced industry detection equipment, including a water performance laboratory. 100% of products are tested and qualified before leaving the factory.

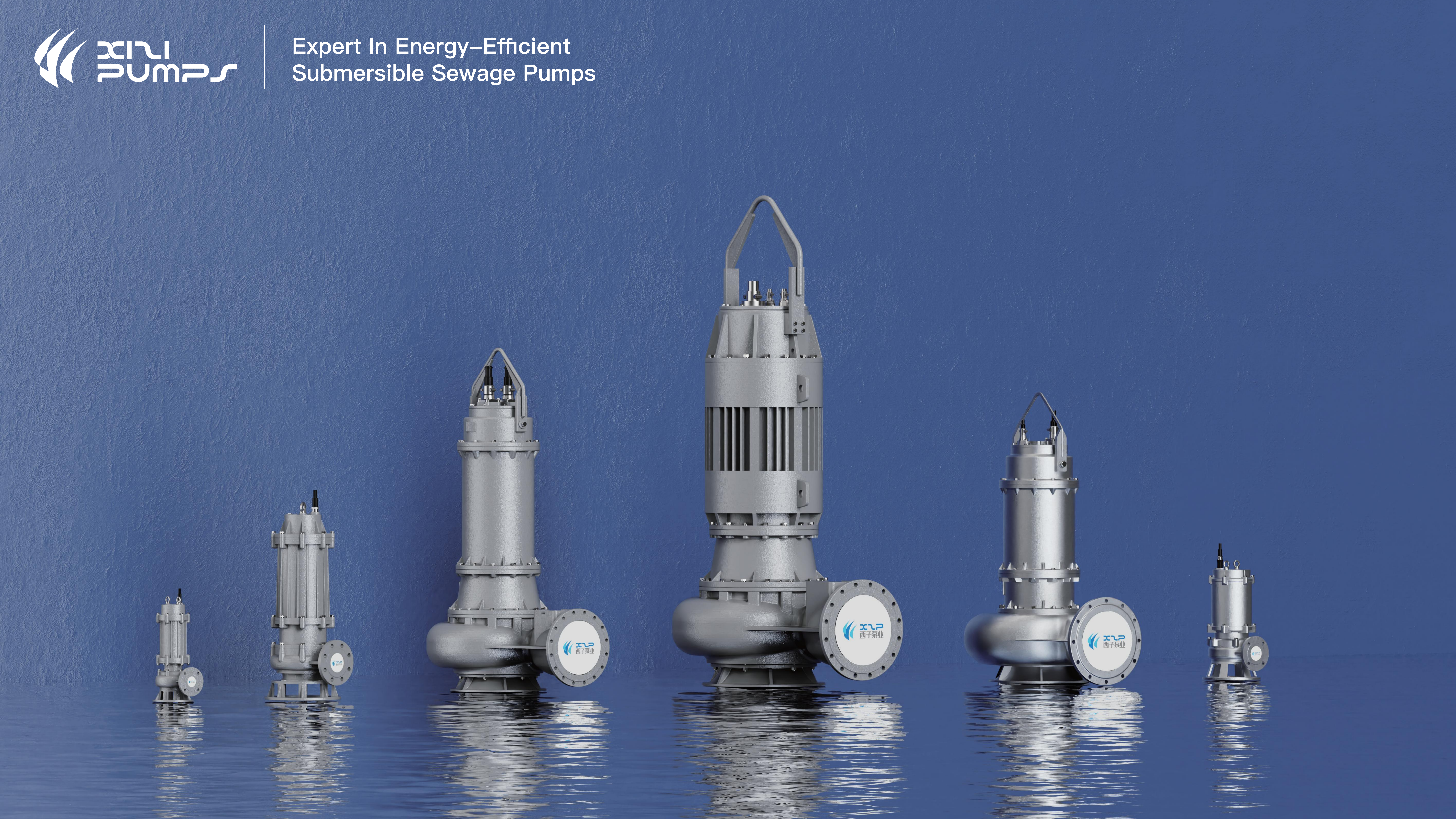
Efficient
Efficient products, efficient production, and efficient service. Products utilize hydraulic components based on internationally advanced hydraulic models and high-efficiency motors, resulting in wide high-efficiency ranges, high efficiency, and stable performance.

Innovative
Boasts a professional R&D and technical team with numerous national invention patents.

Customization
A complete production center and after-sales team enable the customization of personalized products and services for different customer needs, providing the best pump and system solutions.

Focusing on sewage pump and system R&D and manufacturing for 32 years

Catalogue


Expert In Energy-Efficient
Submersible Sewage Pumps

XIZI PUMPS

Submersible Sewage Pump Series Overview

For 30 years, Xizi Pumps has been dedicated to the research and development as well as manufacturing of submersible sewage pumps. Our pump products have been successfully applied in various industrial and municipal fields around the world. With the aim of providing better reliability and higher unit efficiency to our customers, we have, on the basis of summarizing over 30 years of experience in submersible pump production and manufacturing, drawn on and introduced advanced international hydraulic model design technology, and carried out comprehensive optimization design in mechanical structure, sealing, protection and other aspects. We meet the highest demands of customers in terms of efficiency, service life, maintenance and economic benefits. The high standards of Xizi submersible sewage pumps are based on decades of hydraulic design experience and extensive professional knowledge. Our goal is to provide customers with world-class pump products and services.

Xizi submersible sewage pumps are divided into four series:

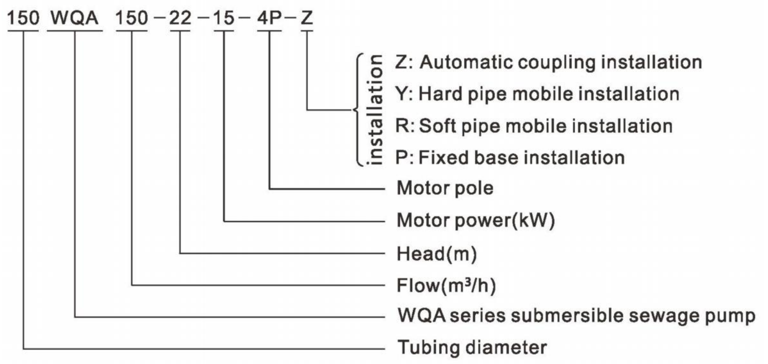
- WQA : High-efficiency submersible sewage pump
- WQE : Economic submersible sewage pump
- WQF : Stainless steel submersible sewage pump
- WQA-QG : Cutting submersible sewage pump

The products comply with the national standard GB/T24674 "Submersible sewage and sludge pumps".

Scope of Application:

Municipal engineering, sewage treatment, building construction, water conservancy projects, industrial wastewater discharge, environmental remediation, and other fields.

Model definition:



Product Features:

● The impellers of the WQA/WQE/WQF models all adopt a closed structure, with the main forms being: ① Double-channel type ② Blade type
The double-channel impeller has good performance in passing through debris; the blade-type impeller is energy-saving and has high hydraulic efficiency.
The impeller of the WQA-QG model is of a semi-open structure.

● The pump unit has a compact structure and short shaft extension to reduce vibration and noise. It has undergone strict dynamic and static balance testing to reduce vibration value to the lowest point,and extend the expectancy life of bearing and mechanical seal.

● The cable adopts rubber flexible cable with good oil resistance and excellent mechanical performance,which can work continuously for a long time under 40℃.The interception capacity of the cable has sufficient margin to make the service life longer.The oil-resistant rubber flexible cable meets national standards of JB/T8735.2 and GB/T5013.4.
The water-jet cutting cable prevents accidental water immersion in motor cavity .The pump end cable should be properly fixed to avoid dragging of the sealing device of lead-out wire of the motor upper cover cable .

● The motor adopts high-performance squirrel-cage induction motor, which is specially designed and manufactured for submersible pumps. It Implements GB755 standards. Protection class: IP68. Insulation class: F (standard), H (optional)

● The shaft seal adopts single-end mechanical seal to completely isolate motor and pump. It is installed in series up and down to improve sealing reliability,and ensure the long-term and stable operation of pump.The designed service life of mechanical seal is 10,000 hours.

● The bearing adopts lifelong maintenance-free rolling bearings.The upper bearing is deep groove ball bearing or cylindrical roller bearing, which is mainly used to bear radial force. The lower bearing is mainly used to bear radial force and axial force. Given that various pump models will generate different radial force and axial force,some are designed as a double-row angular contact ball bearing, and some are combinations of a diagonal contact ball bearing and a cylindrical roller bearing or deep groove ball bearing to ensure sufficient load margin, making the pump unit run smoothly. The designed service life of the bearing is 50,000 hours.

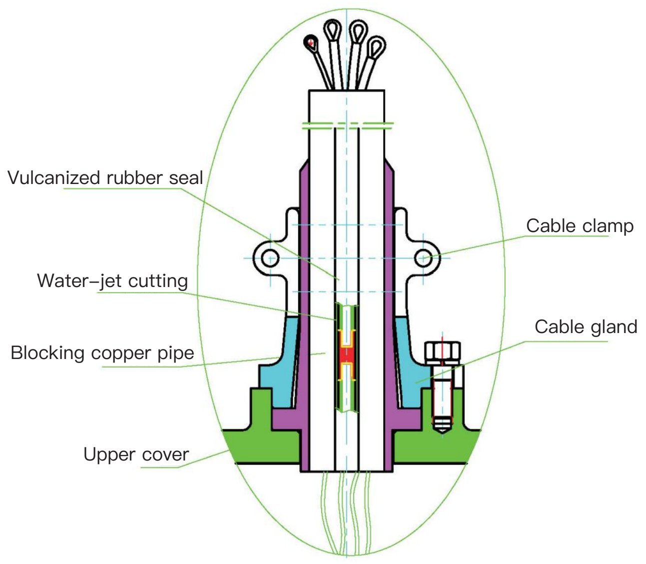
● Oil chamber: Adopt 32# engine oil which can lubricate mechanical seal and cool down the bearing.Executive standard is GB443-89. Add oil till it overflows from the filling hole to ensure that there is a certain gap in the oil chamber, so that the pressure will not rise significantly when oil temperature rises. This is to protect the mechanical seal in oil chamber, thereby protecting the motor.

● Temperature measuring element (thermal protector is equipped for 3kW and below): Each phase winding is pre-embedded with temperature measuring element PTC (115℃).When the pump is in abnormal working condition,and the motor winding temperature rise reaches the setting value, the temperature measuring element will work,the "overheating" indicator light on control cabinet will be turned on, the pump will stop working.This reminds the operator to check and find out overheating causes and solutions in time.
The pump can recover by itself after the winding temperature drops.

● Water leakage probe: The water leakage probe is used for leakage detection. When the mechanical seal or the sealing at impeller side is leaking, and the water amount in oil chamber is large to a certain extent, the two electrodes of the probe will be turned on and an alarm signal will be issued(the indicator lights on), to remind the operator to check mechanical seal in time and check whether the oil in oil chamber need to be replaced.

● Float switch: The float switch is installed in the empty cavity next to the bearing cavity on the lower side of motor. It is used to check whether the mechanical seal in oil chamber is damaged and whether the motor seal is leaking. When the water amount in the float switch cavity reaches a certain level, the float switch will float and send an alarm signal through control cabinet (the indicator lights on), to

Structural diagram of water-jet cutting cable



Technical descriptions:

● Xizi submersible sewage pumps can be equipped with flange coupling which implements GB/T17241.6-2008 standard.The standard configuration for coupling outlet flange DN250 and below is PN6 and for DN300 and above is PN10. Pump outlet flange pressure rating is detailed in related pump page.

For WQA/WQF submersible sewage pump 7.5kW and below,pump can be equipped with stirring (JY)
Some models of WQE can be used as stirring pumps. For details, please refer to the sample.

● The standard motor rated voltage of submersible sewage pump is 380V, and the rated frequency is 50Hz.
①11kW and above can be customized with frequency conversion electric pump unit.
②The 60Hz electric pump unit can be customized.
③Electric pump units with three-phase voltage below 690V can be manufactured.
④Single-phase 220V can be manufactured for the pumps 2.2kW and below. (except WQE)
⑤Frequency conversion can be used for the pumps 7.5kW and below.
⑥Three-phase 220V can be manufactured for the pumps 3kW and below.

● Wiring method and starting method of motor winding lead wire
①The Y-shaped wiring method is used for 3kW and below.The stator lead-out wire and a main cable have been connected in the wiring cavity with this method.It can use direct start.
②The internal delta (△) wiring method is used for 4-315kW. when the pump leaves the factory, the winding wires and 1, 2, 3, and 4 main cables have been connected in the wiring cavity according to power level. Based on the power grid capacity and load permit on customer's site, direct start, external electronic soft starter start or auto-coupling step-down start can be adopted accordingly. (Please refer to wiring diagram for the number of main cable)
③Single-phase 220V electric pump (0.55-2.2kW) adopts capacitor operation, double capacitor + centrifugal switch start.
④If there are other requirements (eg: Star Delta Start Y-), please specify separately when placing an order.

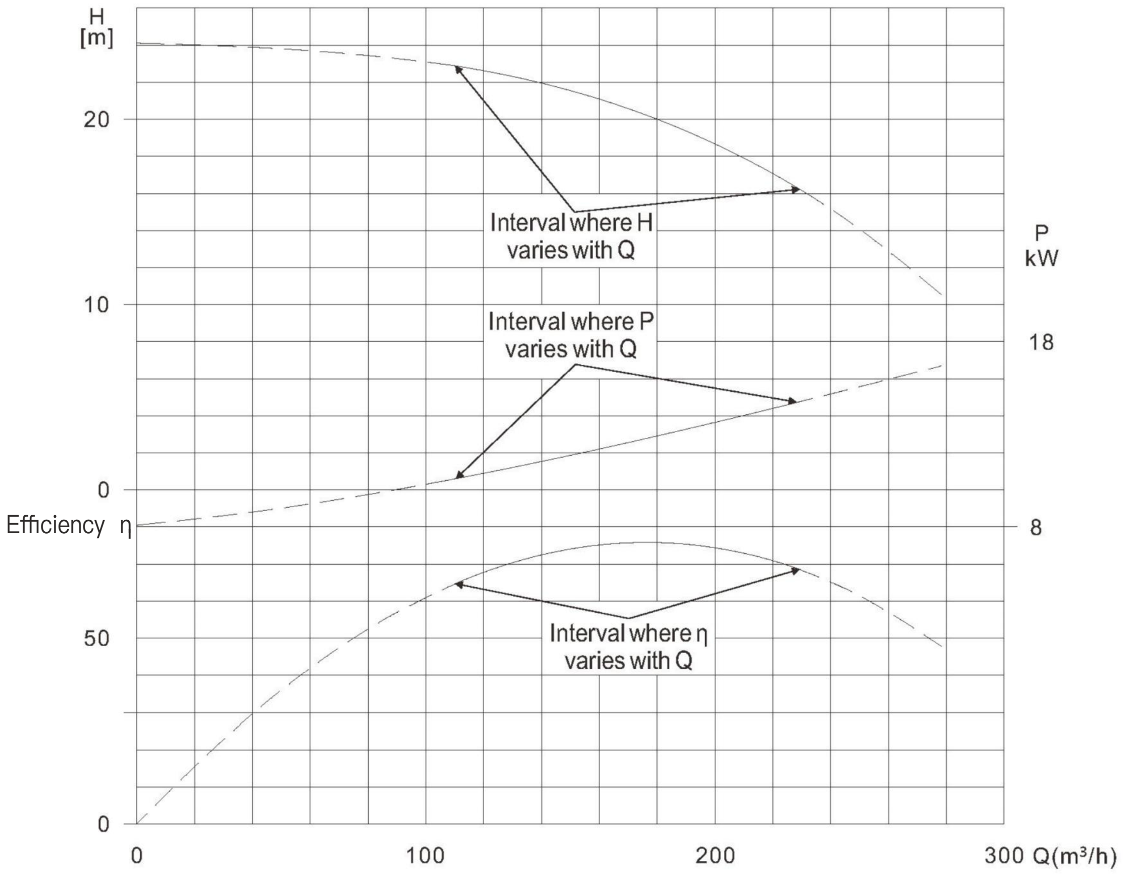
● Pump rotation direction
The impeller rotates counterclockwise looking from pump inlet.The impeller rotates clockwise looking from the motor.

● When the liquid level difference is large on customer's site , use two float switches to control the start/stop liquid level for main pump or large pump, while using one float switch for small pump or standby pump under super high liquid level.

● The terminal box can be installed on the spot when electric control cabinet is far away. When threading pipe (user owned) needs to be laid, confirm the inner diameter of the threading pipe according to the outer diameter of the cable. 7.When the liquid level difference is large on customer's site , use two float switches to control the start/stop liquid level for main pump or large pump, while using one float switch for small pump or standby pump under super high liquid level.
8. The terminal box can be installed on the spot when electric control cabinet is far away. When threading pipe (user owned) needs to be laid, confirm the inner diameter of the threading pipe according to the outer diameter of the cable.

Diagram and description of performance curve

Diagram of performance curve



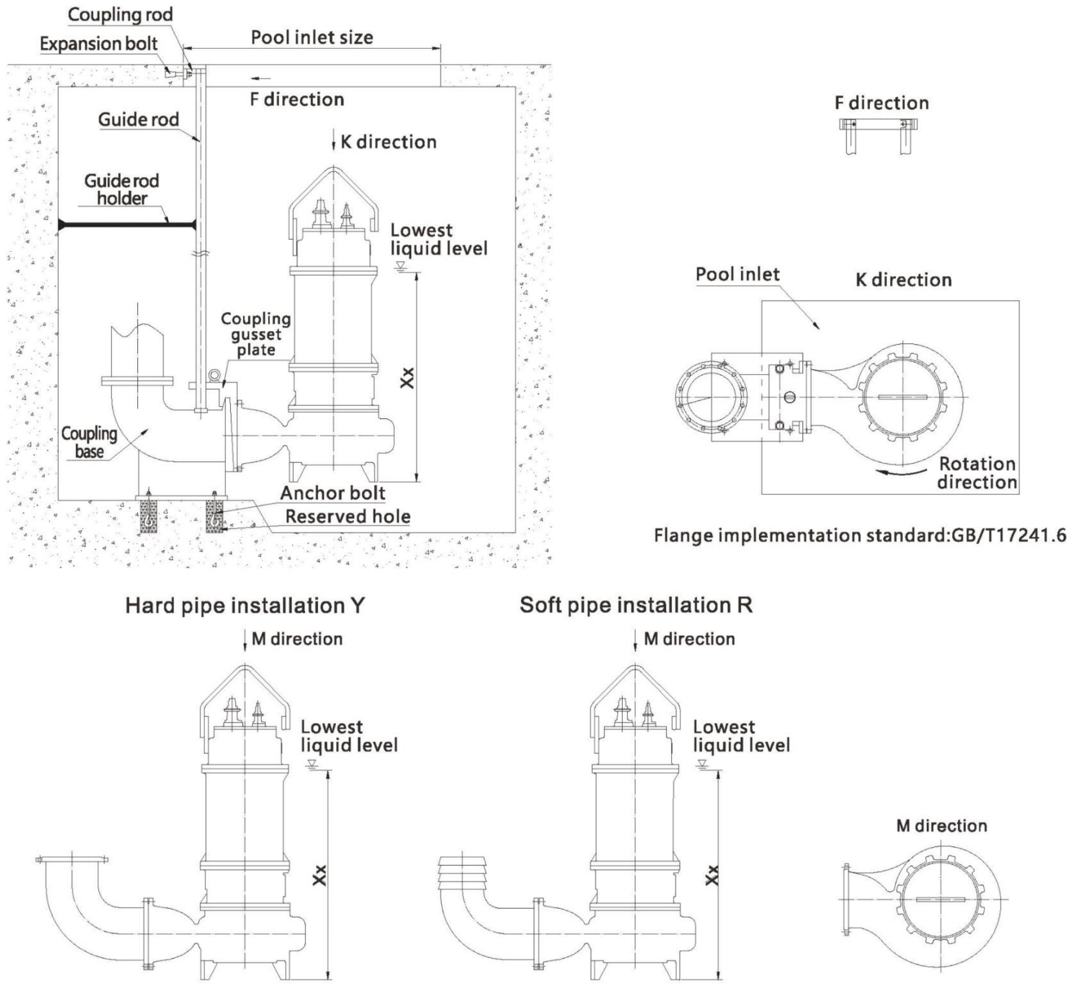
● The solid line part of the curve in the figure represents the reasonable pump operation range where pump efficiency is relatively high and economical.

When pump is running below the leftmost side of the solid line:
① Under long-term low-flow operation, the pump will be always in a state of holding pressure with high pump temperature, excessive vibration and shortened life.When the deviation from the reasonable working range is large, the pump will be damaged to a certain extent after long-term operation.
② Large radial force will be generated,bearing and mechanical seal damage is accelerated,rolling keys or even shaft breakage risk will be caused.
③ The efficiency will drop and electric energy will be wasted.

When pump is running higher than the rightmost side of the solid line:
The pump will have problems like vibration, noise, cavitation, even over-current and motor burning , etc.

Installation dimension drawing

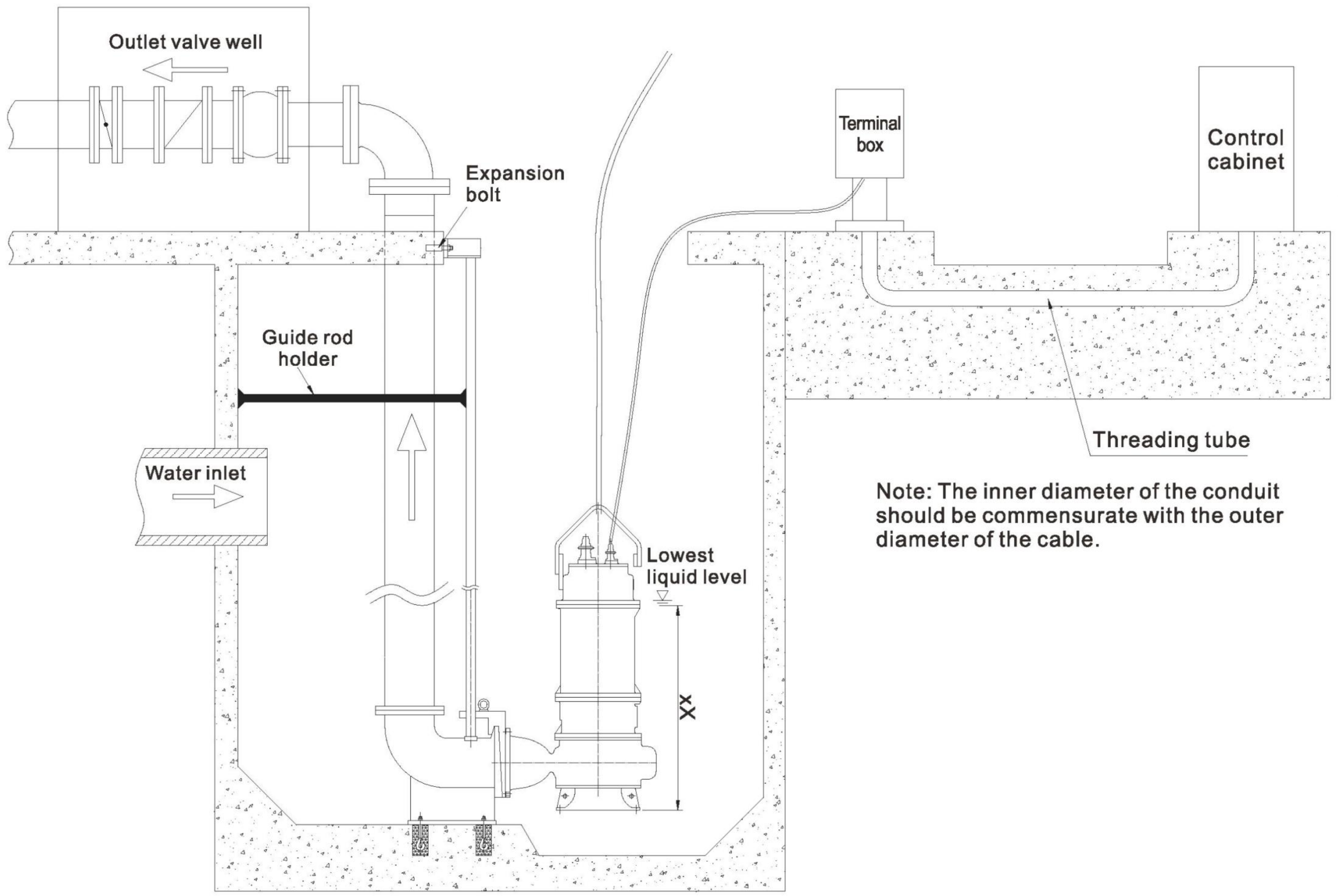
Dimension drawing of coupling installation



● The coupling railings are all fixed with expansion bolts which implement JB/ZQ4763-2006 standard.
When the length of guide rod is bigger than 5 meters, it is recommended that the user fix the guide rod every 3 meters.
● Calculate guide rod length according to the "pool depth" shown on the figure, and refer to the attached table for the length.
● ▽ indicates the minimum liquid level of pump operation. The pump operation liquid level should be higher than the minimum liquid level. It is best to submerge the pump completely under allowable site conditions, so that the motor can be fully cooled by the medium. The minimum liquid level can be controlled by float switch, ultrasonic level gauge, etc.
● The dimension of coupling base and pump outlet flange shall comply with GB/T17241.6 standard.

Long-distance layout plan and cable specifications

Long-distance layout plan: When the control cabinet is far away from pump room, a terminal box can be installed, which makes the operation more convenient.



Installation method

- **Fixed base installation (P):**

Fix the base on the foundation, fix the base with anchor bolts, and connect the outlet pipe. It is inconvenient to repair the single pump after running;

- **Mobile installation (Y, R):**

It is used for first aid or maintenance and construction, flexible and convenient to operate. It can be operated by connecting a hose or a hard pipe with the pump base support. It can also be suspended for use with a sufficiently rigid drainage pipeline.

- **Automatic coupling installation (Z):**

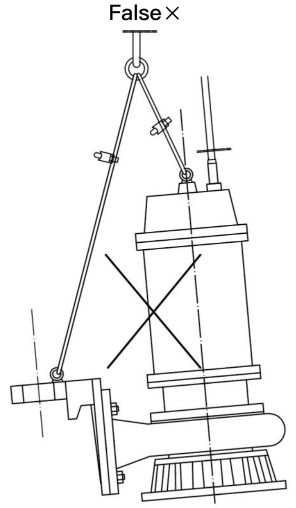
The coupling device includes: ① Coupling base, ② Coupling gusset, ③ Guide rod, ④ Coupling pull rod.

Automatic coupling installation: The coupling device links the pump and pipeline without fasteners, making the connection and disconnection of the pump and coupling base simple.

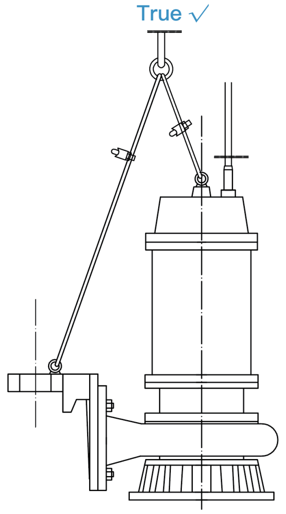
Operation essentials:

① Lay the cement foundation and reserve the anchor bolt holes according to the installation drawing.
② Fix the coupling base, coupling rod and guide rod, and the coupling base needs to be poured twice.
③ Connect the coupling gusset plate and pump body flange with bolts. Whether to install or lift the pump, you only need to lift the lifting chain and let the gusset plate semi-circular hole slide along the guide rod.
④ When installing with a coupling device, the pump must be adjusted to a vertical position using short chains. The lifting point must ensure the pump moves vertically up and down before hoisting can be performed. (As shown in the schematic below.)

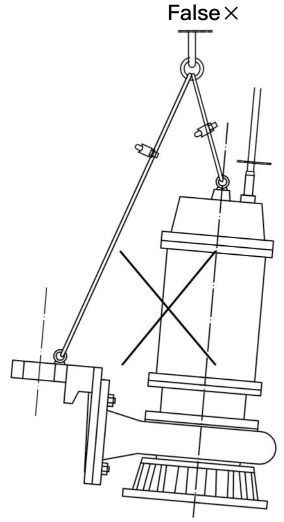
Installition structure diagram of automatic coupling



This situation can't lift pump

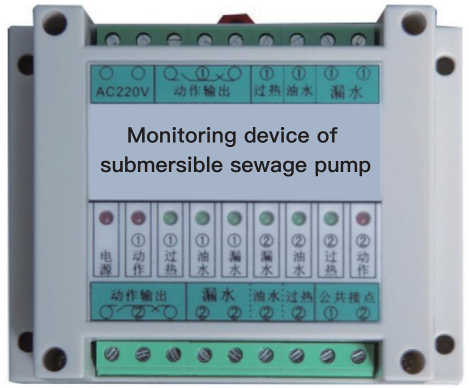


Adjust the pump to a **Vertical State** using short chains.



This situation can't lower pump

Special monitoring device of submersible sewage pump



● **Purpose and function**

XIZI submersible pump monitoring device (comprehensive protector) is an optional accessory specially designed for submersible pump protection device by our company, to realize the comprehensive protection function of the submersible pump.

It has monitoring and protection functions of motor stator temperature overheating (overheating), oil chamber water leakage (oil and water),wiring cavity and motor cavity water leakage(water leakage). It can monitor 2 pumps at the same time.

● **Specifications and technical parameters**

① Specifications

Length × width × height：115×90×40

Installation size：105×70,4– ϕ5; or install on the rail

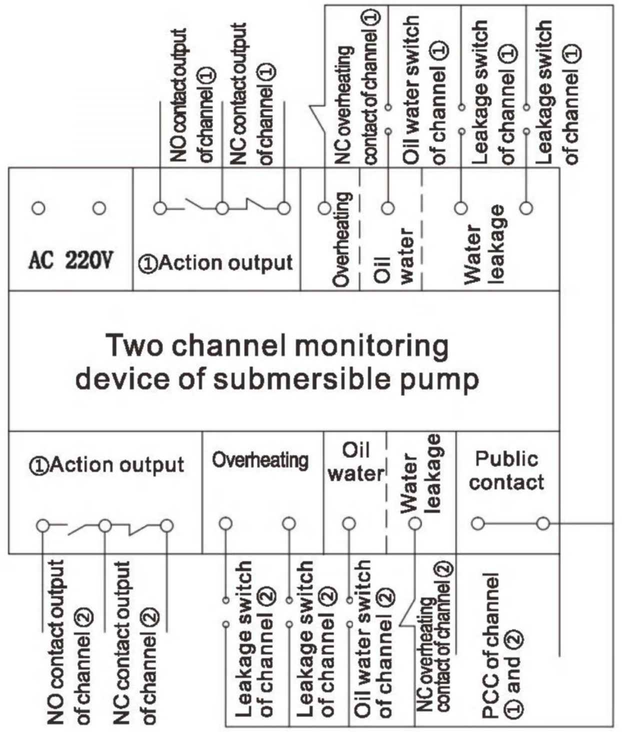
② Working environment

Ambient temperature：–20℃~50℃; Relative humidity≤ 85%RH; There is no corrosive gas in the surrounding environment

③ Power supply：AC220V / 50Hz

● **Electrical principle and terminal wiring diagram**

1. 220V access "AC 220V"
2. This product can monitor two water pumps at the same time. You can choose to connect to port ① or ② when there is only one pump.
3. The common wires of the "overheating", "oil water", "leaking" lead–out wires and the PCC lead–out wires, form the respective protection and monitoring circuits (Refer to schematic diagram).
4. "Action output" can be connected to "NO output" or "NC output" according to customer site.



● **Points for attention.**

The inside of the monitoring device is full of electronic circuits, and the insulation resistance value cannot be measured with a

Parameter table of submersible motor

Parameter table of submersible motor（380V / 50Hz）								
No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked–rotor current (A)	Rated torque (N•m)	Locked–rotor torque / Rated torque
1	0.55	2	1.8	0.82	73	8.1	1.87	2.2
2	0.75		2	0.83	75	10.9	2.53	2.2
3	1.1		2.5	0.84	77	17.4	3.71	2.2
4	1.5		3.6	0.84	79	23.4	5.03	2.2
5	2.2		4.9	0.85	81	32.8	7.38	2.2
6	3		6.4	0.87	83	46.1	9.96	2.2
7	4		8.2	0.88	85	58.7	13.24	2.2
8	5.5	2	11	0.88	86	80.3	18.09	2.2
		4	12	0.83	85	80.5	36.4	2.3
9	7.5	2	15	0.88	87	106.5	24.66	2.2
		4	16	0.84	87	107.1	49.63	2.3
10	11	2	22	0.89	88	156.8	35.81	2.2
		4	23	0.84	88	155.4	71.85	2.2
		6	25	0.78	87.5	154.7	108.2	2
		8	26	0.75	86.9	164.3	143.9	2
11	15	2	29	0.89	89	209.3	48.84	2.2
		4	30	0.85	89	223.5	97.97	2.2
		6	31	0.81	91.2	225.6	147.2	2
		8	34	0.76	88	219.8	196.2	2
12	18.5	2	35	0.9	90	254.3	60.24	2.2
		4	36	0.86	90.5	270.8	120.4	2.2
		6	37	0.83	90.7	275.9	180.7	2
		8	42	0.76	88.6	264.7	242	1.9
13	22	2	42	0.9	90	303.8	71.38	2
		4	43	0.86	91	319.5	143.2	2.2
		6	44	0.83	91.2	331.5	214.9	2
		8	48	0.78	89.1	308.9	287.8	1.9
14	30	2	58	0.93	90.3	364.2	97	2.2
		4	60	0.86	92	411.9	194.7	2.2
		6	61	0.84	91.5	408.8	292	2
		8	63	0.79	91	415.8	389.8	2
15	37	2	70	0.88	93.7	512.2	119.6	2
		4	71	0.87	92	501.2	239.6	2.2
		6	73	0.86	91.5	510	360.6	2.1
		8	78	0.79	91.5	502.9	480.7	2.0
16	45	2	82	0.91	92.3	622.2	145	2
		4	85	0.87	92.8	604.8	290.8	2.2
		6	86	0.86	92.5	597.8	438	2.1
		8	94	0.79	92	610.5	584.7	2.0
17	55	2	101	0.9	92.3	758.5	176.9	2
		4	104	0.87	92.6	741.6	355.8	2.2
		6	105	0.86	92.4	723.1	535.4	2.1
		8	114	0.81	92.3	728.7	714.6	1.8
18	75	4	140	0.87	93.1	993.6	483.6	2.2
		6	147	0.86	93.2	981.4	726.3	2
		8	152	0.81	92.8	977.5	974.5	1.8
19	90	4	167	0.89	93.5	1050	580	2
		6	172	0.87	93.5	1008	870	1.7
		8	180	0.82	93.1	1090	1161	1.9

No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
20	110	4	201	0.88	94.5	1200	710	1.8
		6	206	0.86	94.5	1230	1066	1.7
		8	217	0.82	94.1	1324	1420	1.9
21	132	4	241	0.88	94.8	1395	850	1.8
		6	244	0.87	94.5	1440	1280	1.8
		8	260	0.83	93.3	1927	1703	1.5
22	160	4	286	0.9	94.5	1800	1030	2.1
		6	294	0.88	94	1965	1550	1.8
		8	314	0.83	93.3	2204	2064	1.5
23	185	4	333	0.9	94.6	1947	1191	2.0
		6	340	0.88	94	2204	1792	1.6
		8	362	0.83	93.8	2625	2380	1.5
24	200	4	359	0.9	94.6	2112	1288	2
		6	368	0.88	94	2484	1937	1.6
		8	390	0.83	93.9	2835	2574	1.5
25	220	4	396	0.89	95	2758.7	1419.5	2.2
		6	404	0.88	94.2	2812	2132	1.8
		8	430	0.83	93.9	3118	2831	1.6
26	250	4	445	0.9	95	2664	1607	1.7
	250	6	460	0.88	94.4	3105	2424	1.8
	250	8	488	0.83	94	3548	3218	1.6
27	280	4	498	0.9	95	2988	1800	1.7
	280	6	510	0.89	94.4	3428	2715	1.8
	280	8	548	0.83	94	3975	3604	1.6
28	315	4	560	0.9	95	3360	2025	1.7
	315	6	570	0.89	94.5	3735	3054	1.8
	315	8	615	0.83	94	4298	4457.4	1.6

Parameter table of submersible motor（Single phase 220V / 50Hz)

No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
1	0.55	2	5.2	0.92	70	15	1.88	1.7
2	0.75		5.4	0.92	72	20	2.56	
3	1.1		7.8	0.95	75	30	3.73	
4	1.5		10.2	0.95	76	45	5.1	
5	2.2		14.7	0.95	77	65	7.42	

Parameter table of submersible motor（Three phase 220V / 50Hz)

No.	Motor power (kW)	Pole number (P)	Rated current (A)	Power factor (cosψ)	Motor efficiency (%)	Locked-rotor current (A)	Rated torque (N•m)	Locked-rotor torque / Rated torque
1	0.55	2	3.2	0.82	73	14.03	1.87	2.2
2	0.75		3.5	0.83	75	18.88	2.53	
3	1.1		4.4	0.84	77	30.14	3.71	
4	1.5		6.3	0.84	79	40.53	5.03	
5	2.2		8.5	0.85	81	56.81	7.38	
6	3		11	0.87	83	79.85	9.96	

WQA high-efficiency submersible sewage pump ►

WQA

High-efficiency Submersible sewage pump

WQA Submersible Sewage Pump is an ultra-high-efficiency submersible sewage pump specifically developed for the wastewater industry. Utilizing internationally advanced hydraulic models, most models achieve "National Grade One Energy Efficiency" making it a true full-head pump that thoroughly solves the problem of overcurrent for customers. At the same time, comprehensive optimization has been carried out in mechanical structure, sealing, and protection, and high-quality motor materials are used to ensure unit stability. The WQA full series includes 182 models.

Customization Available According to Different Working Conditions:
Stainless steel material (WQAF) / Special anti-corrosion coating for seawater
Internal circulation cooling submersible sewage pump / External cooling jacket sewage pump / High-pressure pump

Scope of Application:
Municipal engineering, sewage treatment, building construction, water conservancy projects, industrial wastewater discharge, environmental remediation, and other fields.

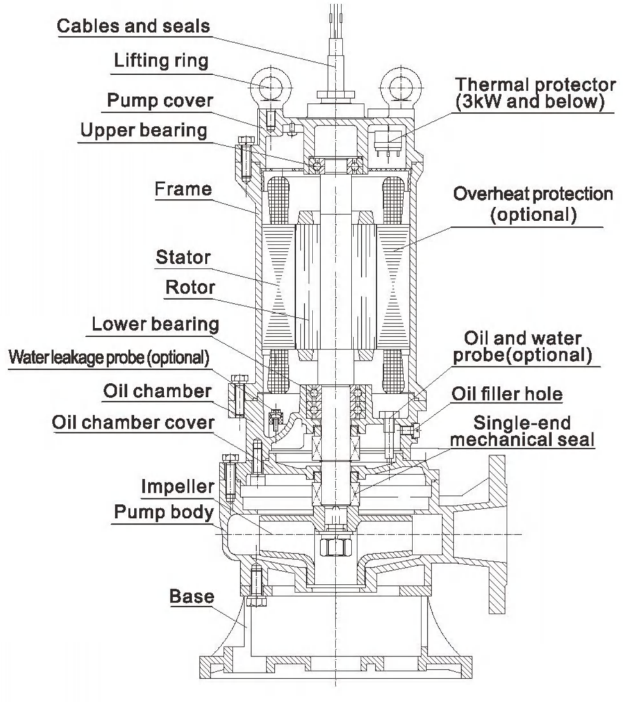
Operating Conditions:
Flow Rate: 7 ~ 4200 m^3/h
Head: 7 ~ 90 m
Medium Temperature: T ⩽ 40°C
Medium Density: ⩽ 1100 kg/m^3; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 10
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

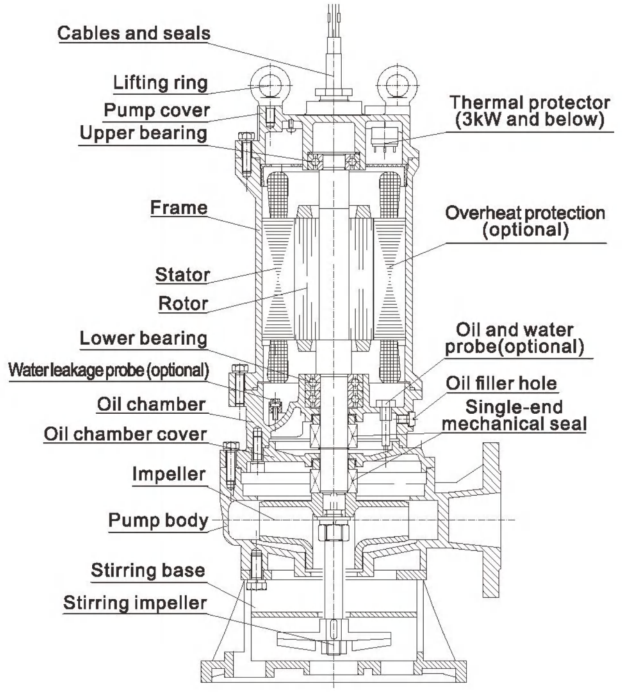
Product Advantages:
- "National Grade One Energy Efficiency": Pump efficiency exceeds the national standard by 2%–13%. Compared to most domestic products, it can achieve a lower power rating under the same flow and head conditions, resulting in more efficient and energy-saving operation.
- True Full-Head Pump: Wide high-efficiency range and stable performance.
- Water-Cut Process Cable: Effectively prevents liquid water and moisture from entering the pump interior.
- Multiple Detection and Protection Systems: Overheat protection switch, oil-water probe, float switch, all capable of real-time detection, and can achieve functions such as alarm, shutdown, and fault signal retention.
- Motor Inspection Port: Allows inspection of the motor cavity without disassembling the machine.
- Anti-Reverse Rotation Design: Impeller nut is designed with an anti-reverse rotation structure.
- Flexible Lifting Handle: Uses a tripod-style lifting method, with a handle that rotates flexibly for convenient, safe, and secure handling.
- Comprehensive Model Coverage: Easy and convenient selection. Can be flexibly customized according to different needs, suitable for various industries and working conditions.
- 100% Tested and Qualified Before Shipment: Complies with GB/T 12785.

External cooling jacket submersible sewage pump

Internal circulation cooling submersible sewage pump

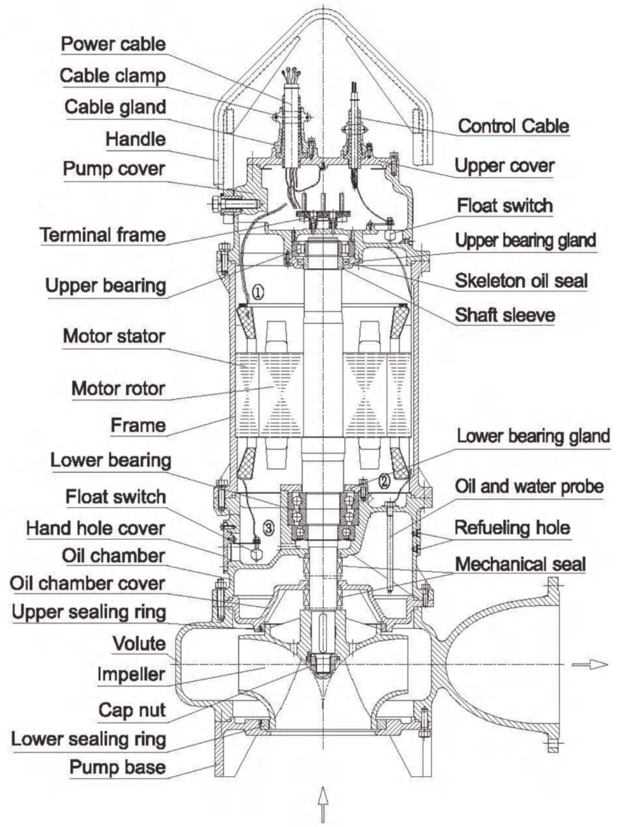
Structure diagram of WQA products





structure diagram for Stirring Pump （JY）

Structure diagram for 7.5kW and below



Structure diagram for 11kW–4P and above

Main materials and configuration table

7.5kW and below

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		HT200	/
2	Impeller		HT200	304
3	Oil chamber cover		HT200	/
4	Oil chamber，Motor frame Pump cover，Upper cover，Pump base		HT200	/
5	Motor insulation class		F（155℃)	H（180℃)
6	Shaft		2Cr13	304
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC （AQ1）	SiC/WC （Q1U1）
10		Pump side friction pair	SiC/SiC （Q1Q2）	SiC/WC （Q1U1）
11		Spring	Stainless steel	/
12		Rubber parts	(NBR)	(FKM)
13	O–ring		(NBR)	(FKM)
14	External leakage fastener		A2–70	A4–70
15	Protection device (optional)	0.55–7.5kW	Thermal protector for 3kW and below	Water leakage probe/Float switch /Overheating protection

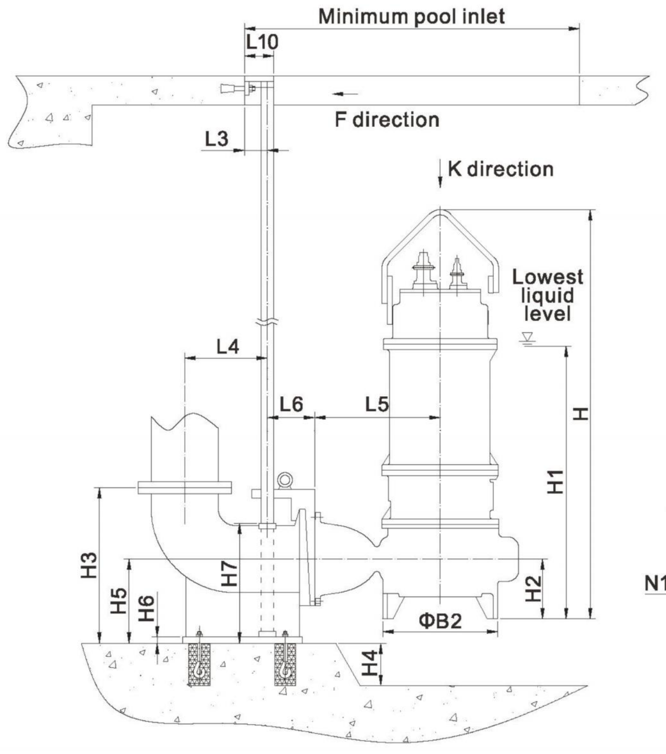
11kW and above

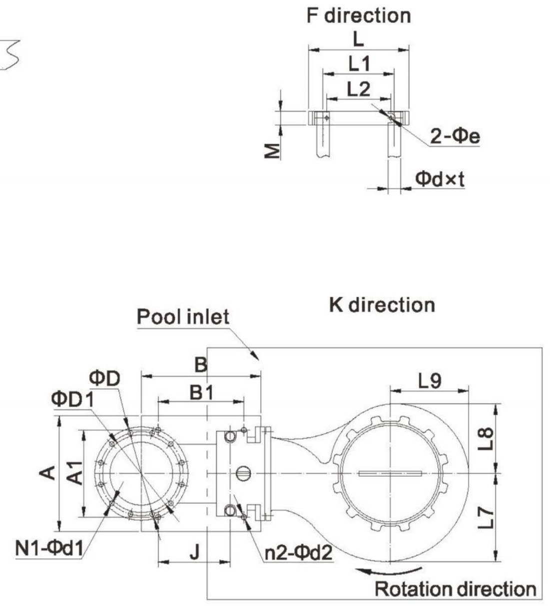
No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		HT250	QT
2	Impeller		HT250	304/ QT
3	Oil chamber cover		HT250	QT
4	Oil chamber，Motor frame Pump cover，Upper cover，Pump base		HT200	/
5	Motor insulation class		F（155℃)	H（180℃)
6	Shaft		2Cr13	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC （AQ1）	SiC/WC （Q1U1）
10		Pump side friction pair	SiC/WC （Q1U1）	SiC/SiC （Q1Q2）
11		Spring	Stainless steel	/
12		Rubber parts	(NBR)	(FKM)
13	O–ring		(NBR)	(FKM)
14	External leakage fastener		A2–70	A4–70
15	Protection device	11–15kW Optional 18.5–315kW Standard	Water leakage probe/Float switch /Overheating protection	Bearing temperature and vibration sensor

Specification/model of submersible motor cable

Motor power (kW)	Cable model	Standard main cable specification	Star delta	Model of controlling cable
0.55	YZW 300/500V	YZW 4×0.75	/	YZW 4×0.75 (Non-standard)
0.75				
1.1				
1.5		YZW 4×1		
2.2				
3				
4		YZW 3×1.5+1×1		
5.5				
7.5		YZW 3×2.5+1×1.5		
11/15–2		YZW 3×2.5+1×4	2–YZW 3×2.5+1×1.5	YZW 4×1 (Non-standard)
11/15–4/6			2–YZW 3×6+1×4	YZW 6×1 (Non-standard)
18.5/22/30–2	YCW 450/750V	YCW 3×10+1×6		YZW 6×1 (Standard)
18.5/22–4/6				
37–2	YZW 300/500V	2–YZW 3×6+1×4		
30–4/6/8	YCW 450/750V	YCW 3×16+1×16	2–YCW 3×10+1×6	
37–4/6/8				
45–4/6		YCW 3×25+1×10	2–YCW 3×16+1×16	
45–8. 55		YCW 3×35+1×10		
75–4/6/8		YCW 3×50+1×16	2–YCW 3×25+1×10	
90–4		YCW 3×70+1×25		
90–6/8			2–YCW 3×35+1×10	
110		2–YCW 3×35+1×10		
132		2–YCW 3×50+1×16		
160–4/6		2–YCW 3×50+1×16	/	
160–8		2–YCW 3×70+1×25		
200–4/6. 185				
200–8		3–YCW 3×50+1×16		
220–4/6/8				
250–4/6/8		3–YCW 3×70+1×25		
280–4				
280–6/8		4–YCW 3×50+1×16		
315–4/6/8				

Installation dimension drawing



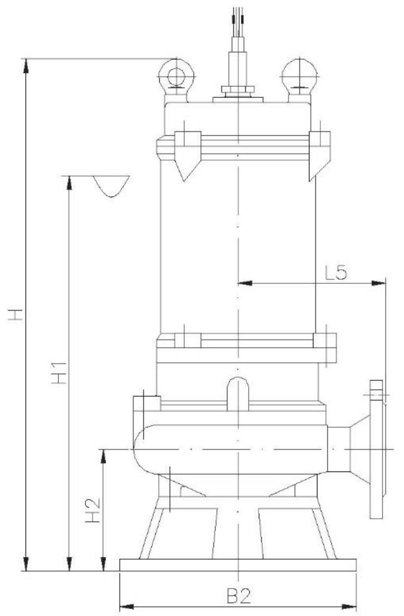


Flange implements GB/T17241.6 standard

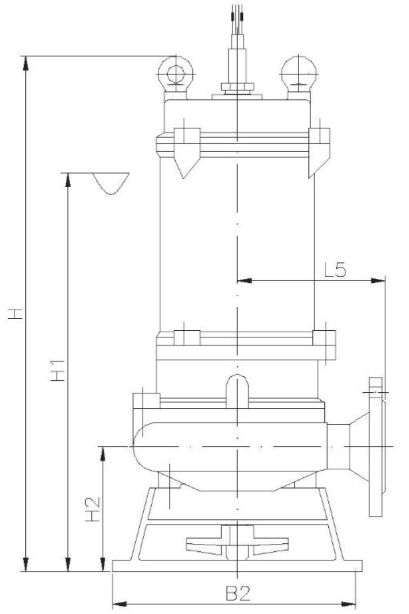
Flange coupling parts dimension table

Model	Flange Connection Size			Coupling Base Size						H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φd×t
DN	D	D1	N1–Φd1	A	A1	B	B1	J	n2–Φd2															
40	130	100	4–Φ14	108	65	138	68	68	4–Φ15	195	Adjust according to pump model	120	14	150	225	185	70	35	Φ14	65	125	85	13	Φ33×3
50	140	110		197	168	208	130	120	4–Φ19	250		165		/	265	215	105	42	Φ14	66	135	90	15	
65	160	130		238	190	252	155	135		265		170	16		280	235	125	50	Φ14	70	155	85	18	
80	190	150	4–Φ19	255	220	225	153	125		310		190			315	265	145	53	Φ14	80	155	100	18	Φ48×3
100	210	170		293	265	258	175	140		350		235	18		365	305	170	55	Φ15	100	200	110	20	
150	265	225	8–Φ19	400	285	410	300	210	4–Φ22	485		300	22	390	410	260	275	60			290	190	20	
200	320	280		400	300	445	350	250	4–Φ25	550		325		440							330	195	20	
250	375	335	12–Φ19	460	360	555	430	325		635		310	24	450							395	215	20	
300	445	400	12–Φ23	550	415	570	410	345		740		400	30	570	480	305	340	65		110	395	225	23	Φ60×3
350	505	460	16–Φ23	580	420	615	400	350		860		490	33	680	500	320	360			115	435	230	25	
400	565	515	16–Φ28	630	490	665	510	430		960		560	36	775							480	235	25	
500	670	620	20–Φ28	732	570	752	550	475	4–Φ28	1160		655	38	900	750	660	460	78	Φ20		565	250	26	Φ76×3.5
600	780	725	20–Φ31	830	720	850	700	590	4–Φ33	1320		710	40	1020	830	740	560				660	275	30	

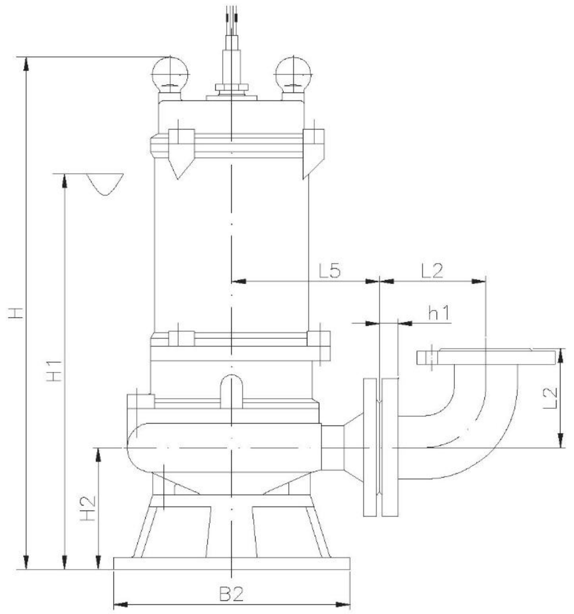
Pump appearance and mobile installation



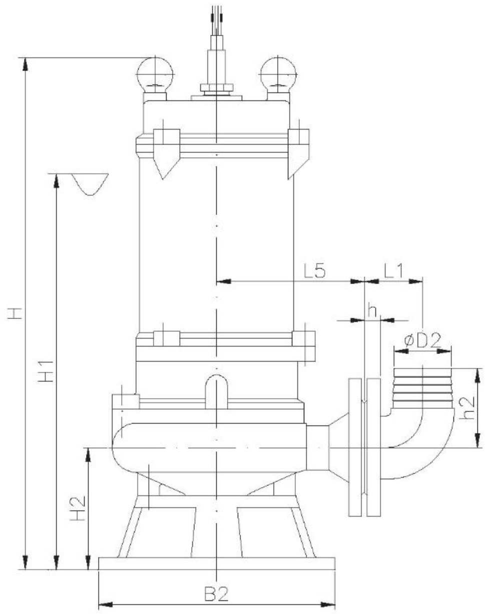
Dimension drawing of pump appearance



Dimension drawing of Stirring (JY) pump



Hard pipe(double flange joint/elbow) mobile installation



Soft pipe(joint) mobile installation

WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Stirring (JY) H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7		kg	PN
1	40WQA10–10–0.75	40	0.75	2950	220/380	20	135	200	460	380	110	90	95	100	35	23	6
2	40WQA8–15–1.1		1.1													24	
3	40WQA15–12–1.1															25	
4	40WQA12–15–1.5		1.5				150	240	490	400	125	100	110	115	40	29	
5	50WQA10–7–0.55	50	0.55				135	200	460	380	110	90	95	100	35	23	
6	50WQA10–10–0.75		0.75													24	
7	50WQA8–15–1.1		1.1													25	
8	50WQA15–12–1.1																
9	50WQA15–16–1.5		1.5				145	230	545	455	120	105	95	105	40	29	
10	65WQA25–8–1.5	65				25	160	230	570	480	125	105	110	115			
11	50WQA15–22–2.2	50	2.2				150	230	580	490	120	105	100	110		38	
12	50WQA25–15–2.2															37	
13	65WQA30–11–2.2	65					160	230	590	500	125	105	110	115		38	
14	80WQA40–10–2.2	80					170	230	605	515	130	110	110	125		41	
15	100WQA50–7–2.2	100				30	180	230	605	515	130	105	125	140		43	
16	50WQA15–30–3	50	3		380	25	150	230	590	500	120	105	110	115		43	
17	50WQA25–20–3																
18	50WQA15–35–4		4			20	170	260	635	535	140	120	110	120		61	
19	50WQA25–28–4																
20	50WQA20–40–5.5		5.5				180	260	665	565	140	115	110	125	45	68	
21	50WQA25–35–5.5																
22	50WQA25–40–7.5		7.5				190	320	730	620	155	135	125	135		95	
23	65WQA35–15–3	65	3			25	155	230	595	505	125	105	110	115	40	44	
24	65WQA35–20–4		4				170	260	625	525	140	115	115	120		62	
25	65WQA30–30–5.5		5.5						665	565					45	69	
26	65WQA30–35–7.5		7.5				195	320	735	620	160	135	130	135		97	
27	80WQA40–13–3	80	3				180	230	610	520	130	110	120	135	40	44	
28	80WQA50–15–4		4			30	175	260	650	550	145	115	125	130		63	
29	80WQA65–17–5.5		5.5						665	565					45	74	
30	80WQA45–25–7.5		7.5			35	205	320	750	640	165	135	135	145		100	
31	100WQA60–9–3	100	3				180	230	610	520	130	105	125	140	40	55	
32	100WQA60–11–4		4				180	260	655	555	150	120	130	150		64	
33	100WQA65–15–5.5		5.5						685	585					45	76	
34	100WQA80–17–7.5		7.5			30	215	320	765	650	175	135	155	170		103	
35	100WQA45–25–7.5																
36	150WQA100–10–5.5	150	5.5			35	200		720	610			145			125	
37	150WQA100–12–7.5		7.5						765	650						140	
38	150WQA130–8–5.5		5.5	1475		40	275	320	693	585	171	178	201	225		135	
39	150WQA150–10–7.5		7.5													145	

WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg	PN	
1	65WQA20-56-11	65	11	2950	380	20	220	320	800	600	160	145	155	160	153	6	no
	80WQA40-45-11	80	11				240								155		
	80WQA45-55-15		15						830	630					160		
2	80/100WQA80-45-18.5	80/100	18.5	2950		15	280	300	1170	880	155	155	160	175	246	6	
	80/100WQA50-65-22		22												255		
	80/100WQA80-53-22														255		
3	100WQA40-45-11	100	11	2950		20	240	320	800	600	160	145	155	160	162	6	
	100WQA70-40-15		15						830	630					172		
4	100WQA100-34-18.5	100	11	2950		40	240	280	1200	900	190	155	155	155	265	6	
	100WQA100-38-22		15												276		
5	100WQA80-25-11	100	11	1475		50	340	380	1120	830	205	210	220	230	263	6	
	100WQA80-30-15		15												286		
	100WQA100-30-18.5		18.5						1380	880					363		
	100WQA100-34-22		22												379		
6	100WQA80-75-37	100	37	2950		15	280	300	1170	880	155	155	160	175	295	10	
	100WQA80-65-30		30												290		
7	100WQA160-40-30	100	30	2950		40	300	280	1190	900	170	165	185	205	300	6	
	100WQA180-42-37		37												310		
	100WQA100-73-45	100	45	2950		30	320	300	1500	950	165	200	205	215	440	10	
	100WQA100-90-55		55												480		
8	150WQA150-15-11	150	11	1475		60	360	380	1140	850	210	210	230	250	275	10	yes
	150WQA150-21-15		15												300		
	150WQA220-20-18.5		18.5						1390	890					375		
9	150WQA130-30-22	150	22	1475		50	385	380	1380	880	205	220	235	255	390		no
10	150WQA150-44-30	150	30	1475		50	440	300	1550	970	220	245	255	265	630	6	
	150WQA200-32-30		30												630		
	150WQA180-30-30		30												630		
	150WQA150-48-37		37												655		
11	150WQA250-22-30		30	1475		50	440	300	1540	960	205	250	275	290	630	6	
	150WQA250-24-30		30												630		
	150WQA280-26-37		30												630		
	150WQA300-30-37		37												665		
12	150WQA240-42-45	150	45	1475		55	500	400	1630	1050	210	260	275	290	720	6	
	150WQA250-50-55		55												755		
13	150WQA280-56-75	150	75	1475		50	540	400	1880	1210	240	300	320	335	1009	10	
	150WQA300-60-90		90												1085		

WQA installation dimension table

No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg	PN	
14	200WQA340-7-11	200	11	1475	380	80	420	380	1180	890	245	240	280	310	295	10	yes
	200WQA400-8-15		15												315		
	200WQA400-10-18.5		18.5						1430	930					380		
	200WQA400-12-22		22												395		
15	200WQA240-12-11	200	11	1475		85	420	300	1130	830	205	230	270	300	293	6	
	200WQA300-13-15		15												315		
	200WQA300-16-18.5		18.5						1375	870					380		
	200WQA340-16-22		22												395		
16	200WQA340-22-30	200	30	1475		80	460	400	1560	980	215	250	280	310	655	6	no
	200WQA400-24-37		37												682		
	200WQA400-28-45		45						1640	1060					738		
	200WQA400-34-55		55												771		
17	200WQA550-32-75	200	75	1475		80	560	400	1900	1230	235	305	340	365	1070	6	
	200WQA600-36-90		90												1145		
18	200WQA480-50-110	200	110	1475		75	550	400	2150	1390	270	325	340	360	1585	10	
	200WQA500-55-132		132												1652		
	200WQA500-64-160		160						2270	1510					1757		
19	250WQA380-7-11	250	11	1475		75	450	460	1140	850	225	250	300	335	305	6	yes
	250WQA400-10-15		15												325		
	250WQA480-10-18.5		18.5						1390	890					388		
	250WQA480-12-22		22												405		
20	250WQA600-12-30	250	30	1475		80	530	400	1570	990	210	265	315	360	672	6	
	250WQA600-15-37		37												700		
	250WQA600-18-45		45						1650	1070					755		
	250WQA600-24-55		55												788		
21	250WQA500-13-30	250	30	980		70	560	475	1640	1060	270	290	320	360	690	10	no
	250WQA600-14-37		37						1720	1140					720		
22	250WQA650-16-45	250	45	980		90	570	550	1970	1300	285	300	335	370	975		
	250WQA700-20-55		55												1024		
23	250WQA700-27-75	250	75	1475		85	520	460	1935	1270	255	280	315	345	1095	6	yes
	250WQA750-30-90		90												1170		
24	250WQA600-40-110	250	110	980		40	630	650	2280	1530	270	370	420	445	1865	10	no
25	250WQA650-40-110	250	110	1475		100	660	460	2180	1430	280	325	360	390	1615	6	
	250WQA800-40-132		132												1685		
	250WQA800-50-160		160						2300	1550					1788		

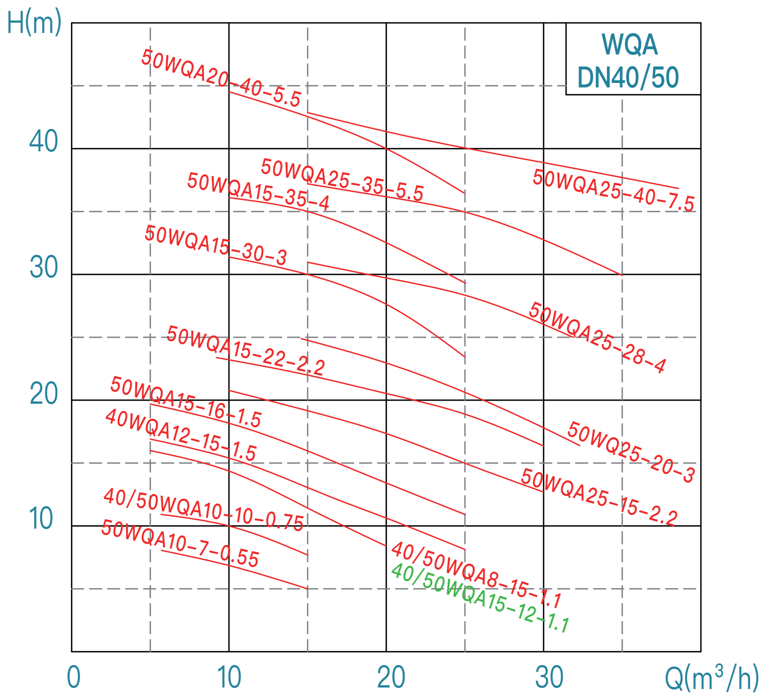
WQA installation dimension table

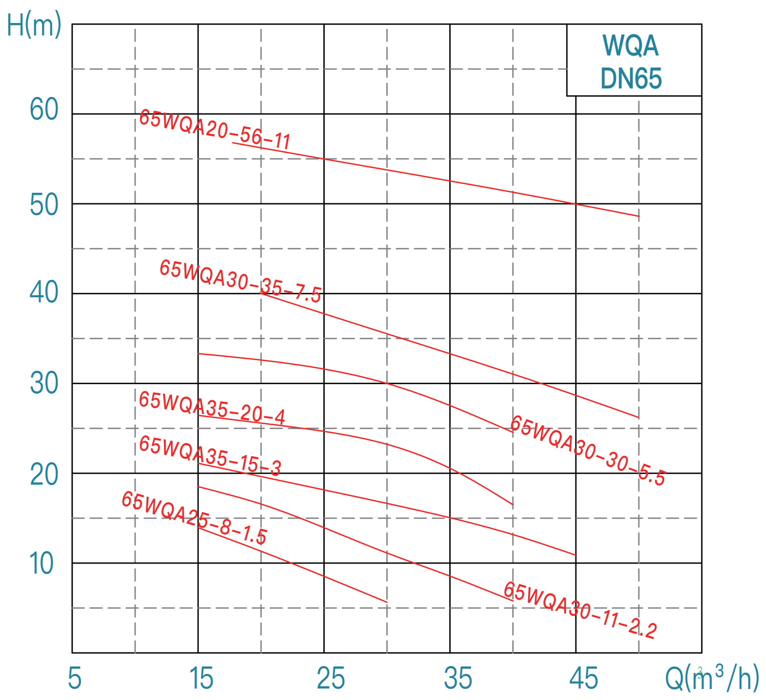
No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg	PN	
26	250WQA600-68-185	250	185	1475	380	70	600	460	2270	1510	250	350	380	405	1855	10	no
	250WQA600-74-200		200												1896		
	250WQA600-80-220		220						2350	1610					1963		
	250WQA600-85-250		250												2060		
	250WQA700-90-280		280						2650	1750					2350		
27	300WQA600-6-15	300	15	980		90	575	500	1450	950	255	325	380	440	510	10	yes
	300WQA600-8-18.5		18.5						1500	1000					542		
	300WQA700-8-22		22												558		
28	300WQA800-9-30	300	30	980		90	600	550	1705	1120	285	340	380	425	707	10	
	300WQA800-10-37		37						1785	1200					737		
	300WQA900-12-45		45						1970	1300					983		
	300WQA900-16-55		55												1035		
29	300WQA800-22-75	300	75	980		70	700	650	2180	1420	270	415	475	525	1635	10	
	300WQA900-24-90		90												1721		
	300WQA1000-27-110		110						2300	1540					1891		
30	300WQA900-35-132	300	132	980		45	630	650	2420	1660					1997		
31	300WQA900-20-75	300	75	1475		120	565	525	1985	1320	275	325	390	440	1100	10	
	300WQA900-24-90		90												1175		
32	300WQA1000-25-110	300	110	1475		100	670	525	2205	1450	285	350	400	445	1650	10	
	300WQA1000-30-132		132												1717		
	300WQA1100-36-160		160						2325	1570					1840		
33	300WQA1250-36-185	300	185	1475		70	650	525	2280	1520	265	375	425	470	1910	10	no
	300WQA1300-38-200		200												1950		
	300WQA1300-42-220		220												1975		
	300WQA1300-50-250		250						2410	1650					2236		
	300WQA1400-52-280		280												2340		
34	350WQA950-8-30	350	30	980		110	740	520	1670	1080	275	410	480	545	792	10	yes
	350WQA1100-8-37		37						1750	1160					871		
	350WQA1150-10-45		45						1970	1300					1055		
	350WQA1200-12-55		55												1105		
35	350WQA1200-18-90	350	90	980		100	700	650	2180	1420	270	415	475	525	1610	10	
	350WQA1200-20-110		110						2300	1540					1812		
36	350WQA1100-16-75	350	75	980		130	850	625	2210	1450	295	445	515	575	1668	10	
	350WQA1400-15-90		90												1755		
	350WQA1600-17-110		110						2330	1570					1927		
	350WQA1700-20-132		132												2032		

WQA installation dimension table

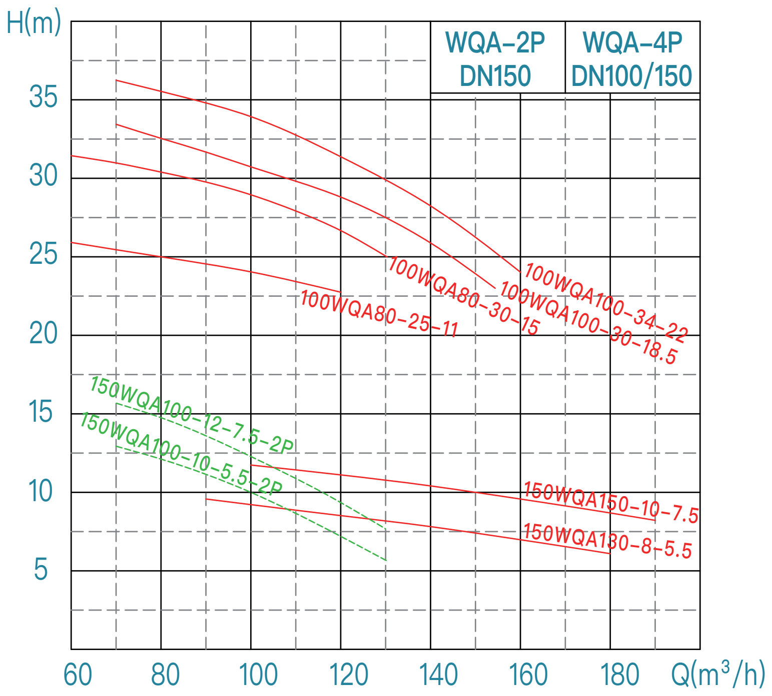
No.	WQA model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight	Pressure grade of pump outlet flange	Full-head?
		mm	kw	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg	PN	
37	350WQA1100-35-160	350	160	980	380	55	800	750	2730	1800	355	450	500	540	2570	10	yes
	350WQA1100-40-185		185												2620		
	350WQA1200-40-200		200												2675		
	350WQA1400-40-220		220												2800		
	350WQA1600-40-250		250												2990		
	350WQA1700-42-280		280												3070		
38	400WQA1600-11-75	400	75	980		120	850	650	2280	1520	350	495	565	635	1766	10	
	400WQA1800-12-90		90												1855		
	400WQA1900-15-110		110						2400	1640					2024		
	400WQA2300-14-132		132												2130		
39	400WQA1600-25-160	400	160	980		120	860	650	2750	1820	330	490	560	615	2620	10	
	400WQA1800-28-185		185												2670		
	400WQA1800-30-200		200												2730		
40	400WQA1900-26-220	400	220	980		150	850	670	2800	1870	335	470	555	635	2850	10	
	400WQA2200-28-250		250												2995		
	400WQA2200-32-280		280												3070		
41	400WQA1200-6-30	400	30	740		100	850	660	1960	1290	350	500	580	650	1700	10	
	400WQA1300-7-37		37												1737		
	400WQA1400-8-45		45						2060	1390					1816		
	400WQA1500-10-55		55												1910		
42	400WQA2000-9-75	400	75	740		100	800	800	2400	1640	435	475	575	655	2050	10	
	400WQA2200-10-90		90						2520	1760					2350		
	400WQA2200-12-110		110												2680		
	400WQA2200-15-132		132						2910	1990					2735		
	400WQA2400-17-160	400	160	740											3000		
	400WQA2500-18-185		185												3200		
	400WQA2500-19-200		200												3400		
43	500WQA2500-8-90	500	90	740		100	800	800	2520	1760	435	475	575	655	2330	10	
	500WQA2500-10-110		110												2390		
	500WQA3000-10-132		132						2910	1990					2780		
	500WQA3000-14-160		160												3130		
	500WQA3000-16-185		185												3210		
	500WQA3000-18-200		200												3290		
44	500WQA3000-20-220	500	220	740		95	1000	800	2770	1840	365	620	710	790	3470	10	
	500WQA3000-23-250		250												3550		
	500WQA3000-25-280		280						2950	2020					3740		
45	600WQA3500-12-185	600	185	740		150	960	870	2950	2020	460	570	690	785	3380	10	
	600WQA4000-12-200		200												3460		
	600WQA4000-14-220		220												3640		
	600WQA4000-16-250		250												3720		
	600WQA4200-17-280		280						3130	2200					3910		
	600WQA4200-20-315		315												4090		

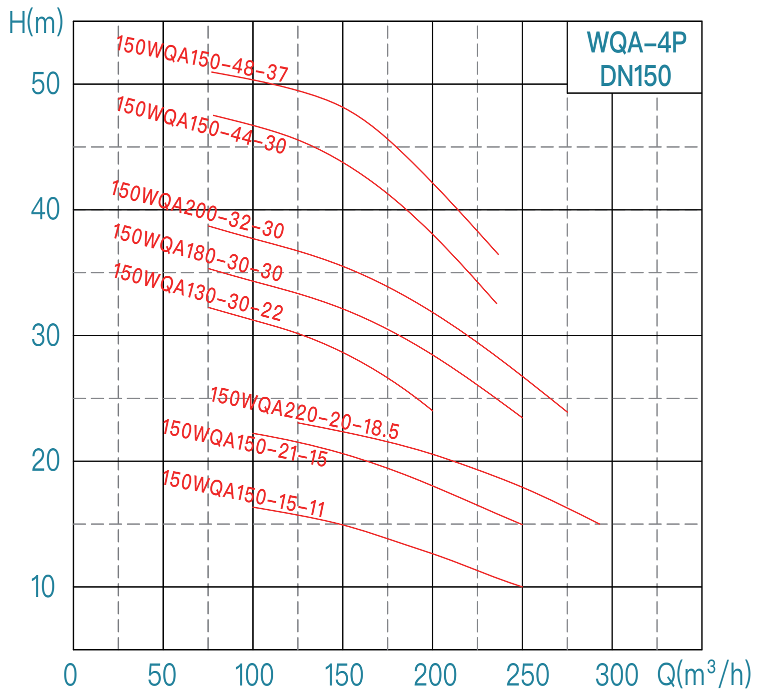
WQA performance curve chart（50Hz）

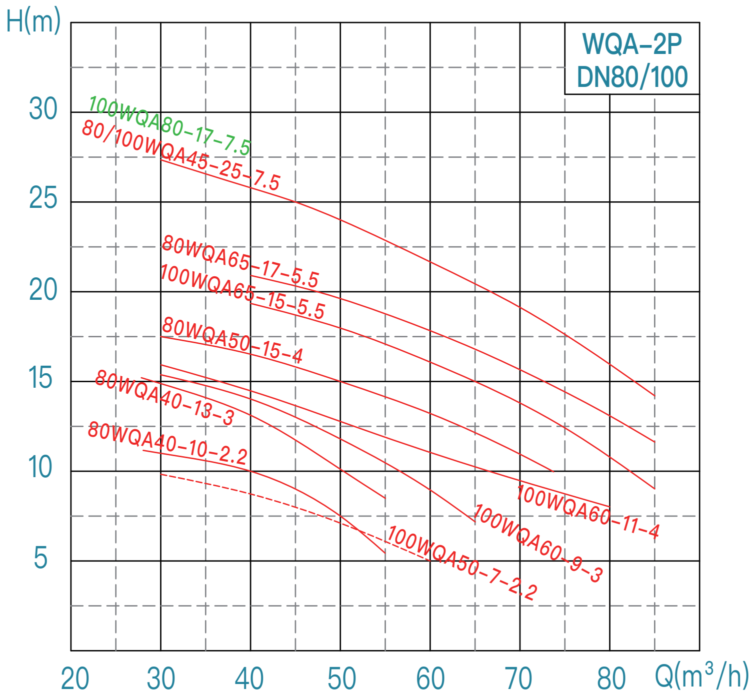


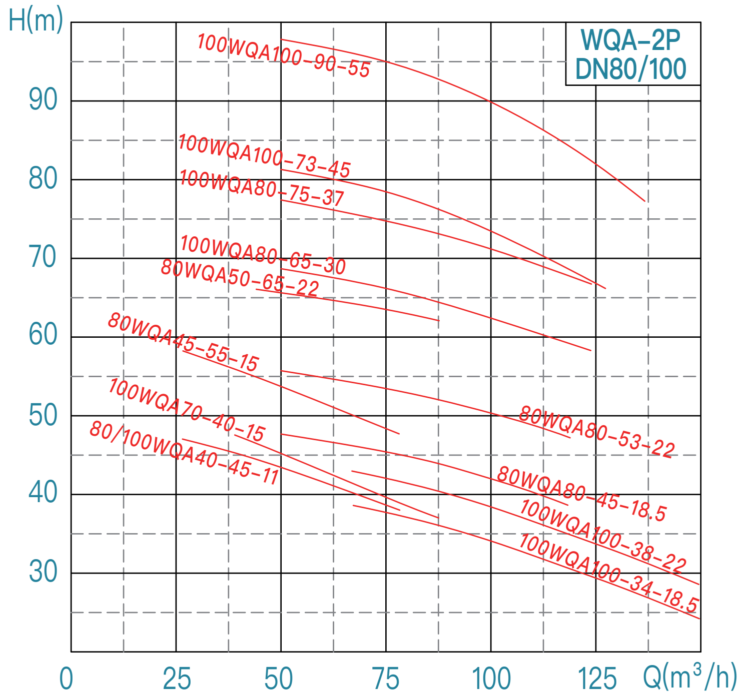


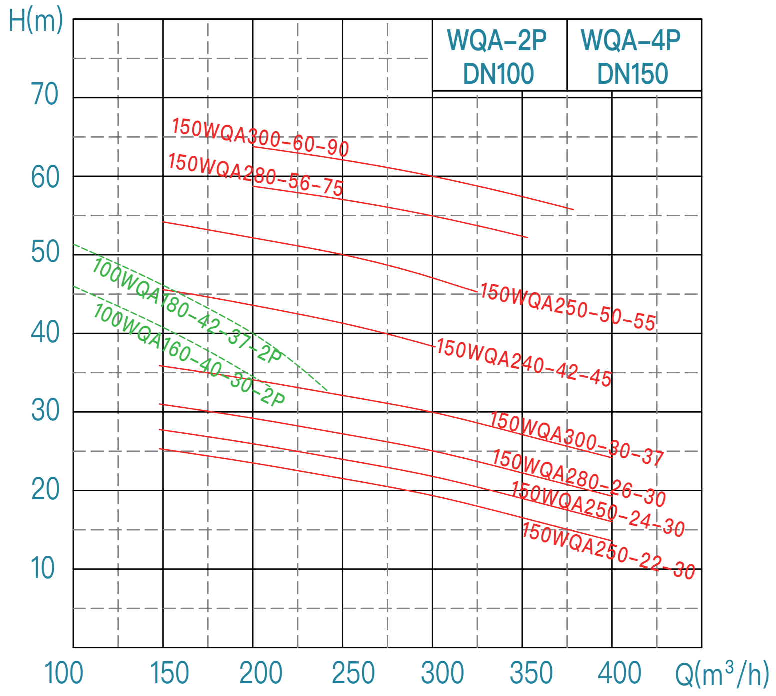
WQA performance curve chart（50Hz）

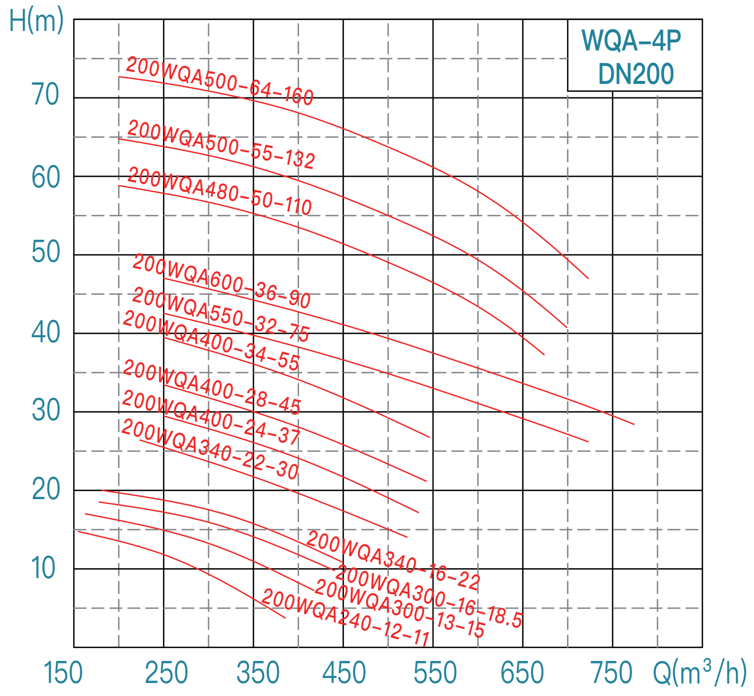




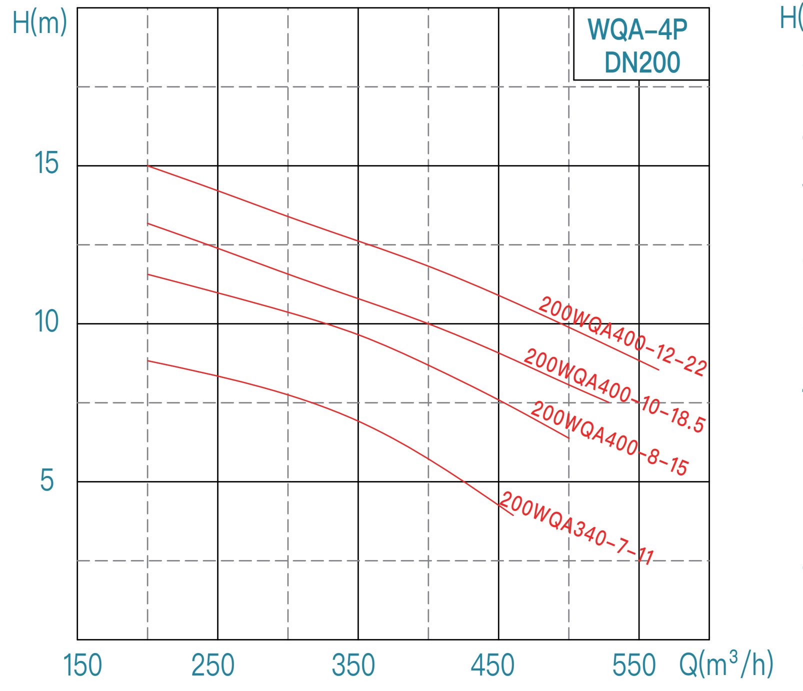


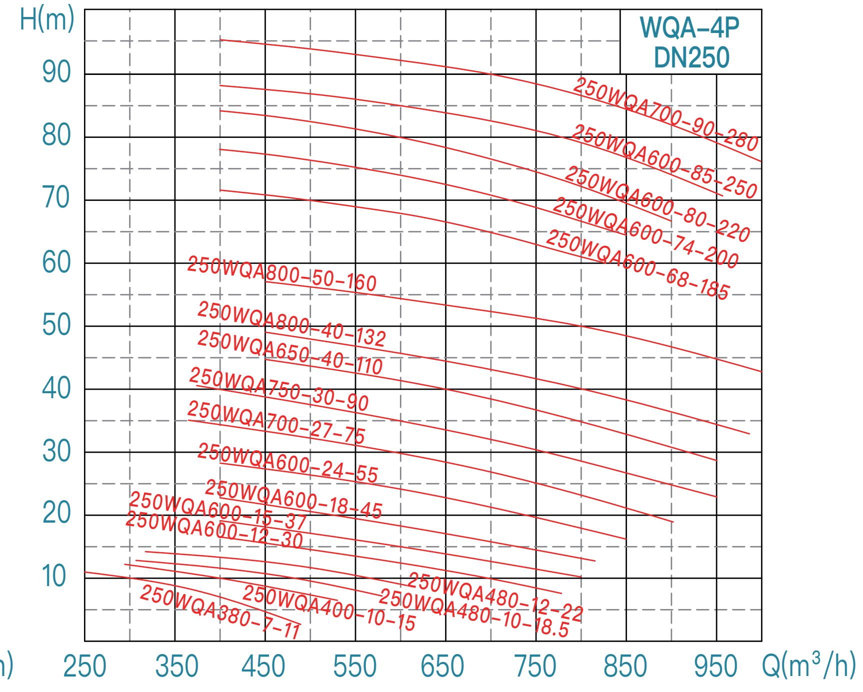


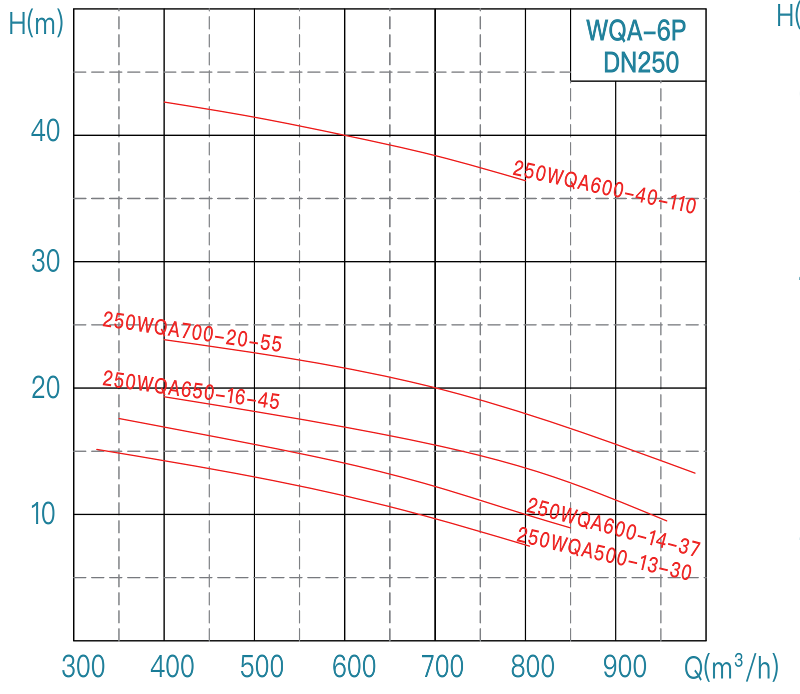


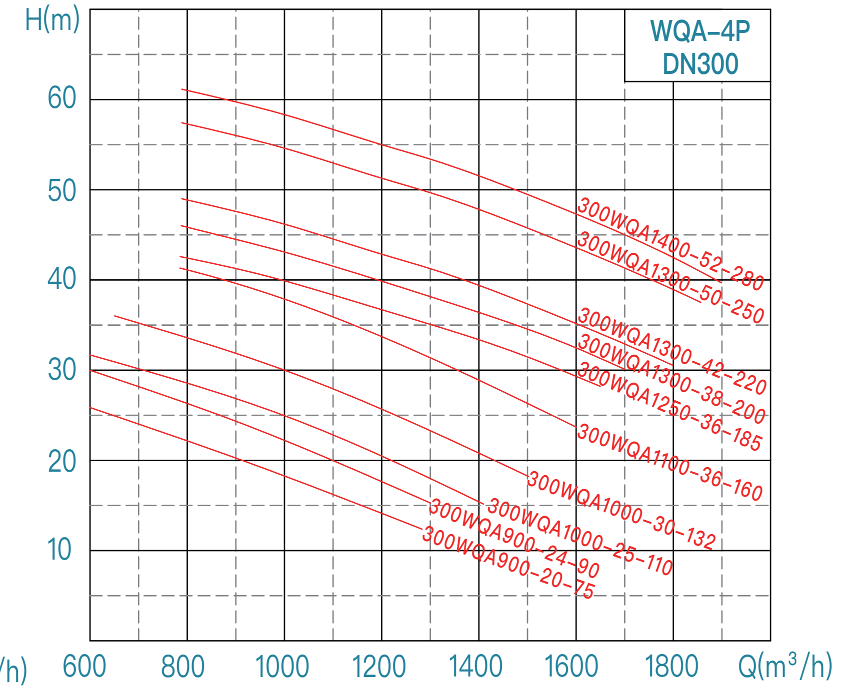


WQA performance curve chart（50Hz)

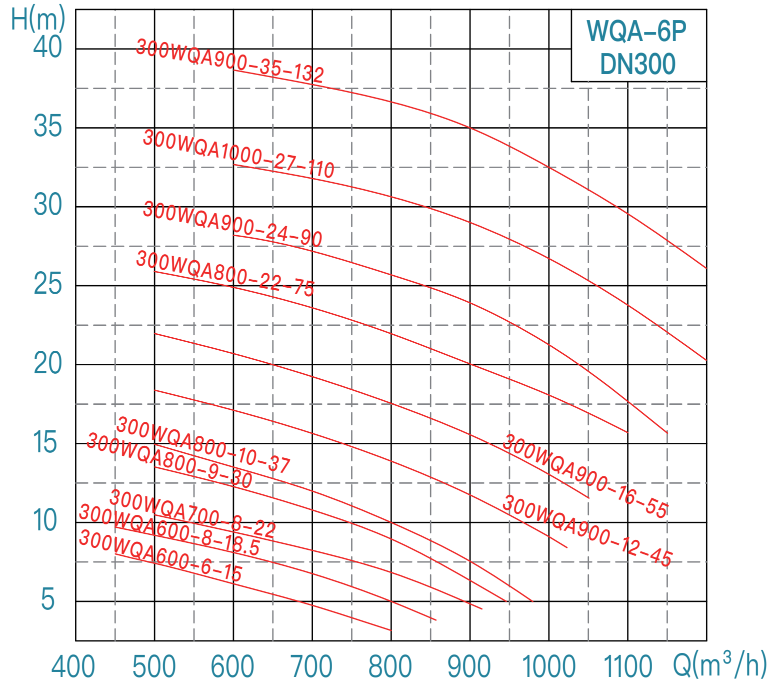


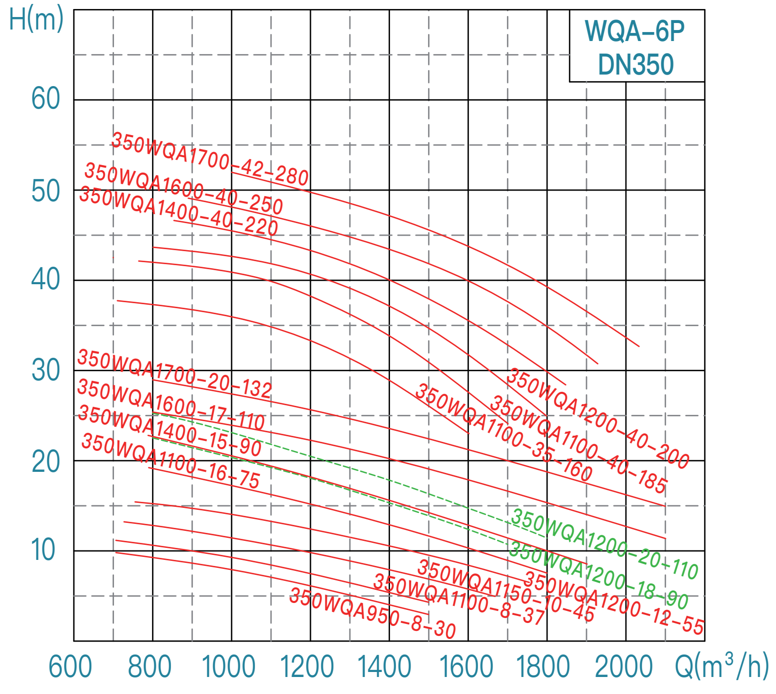


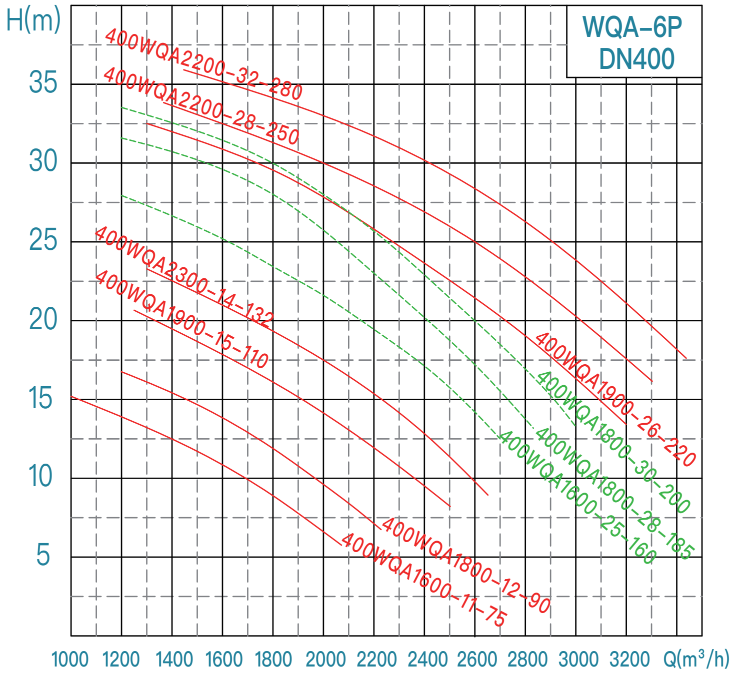


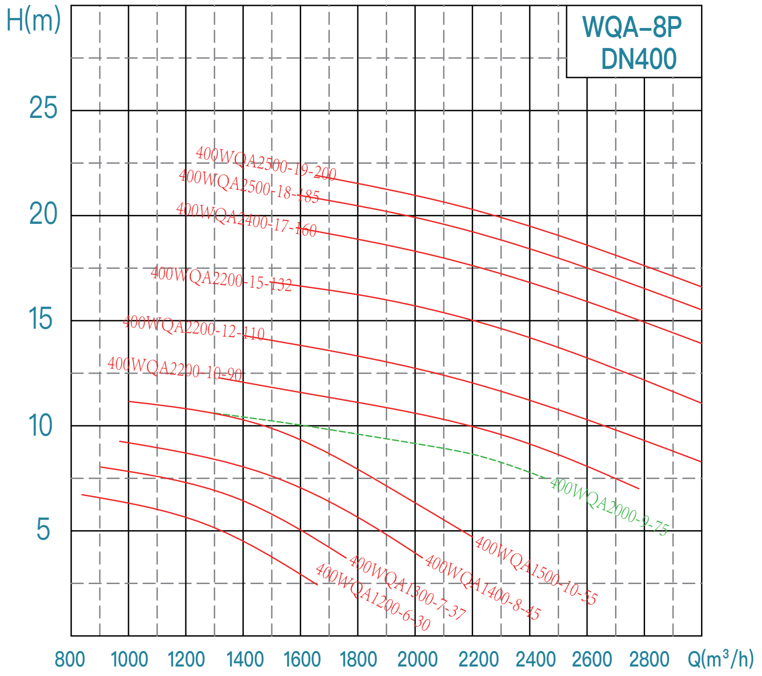


WQA performance curve chart（50Hz)

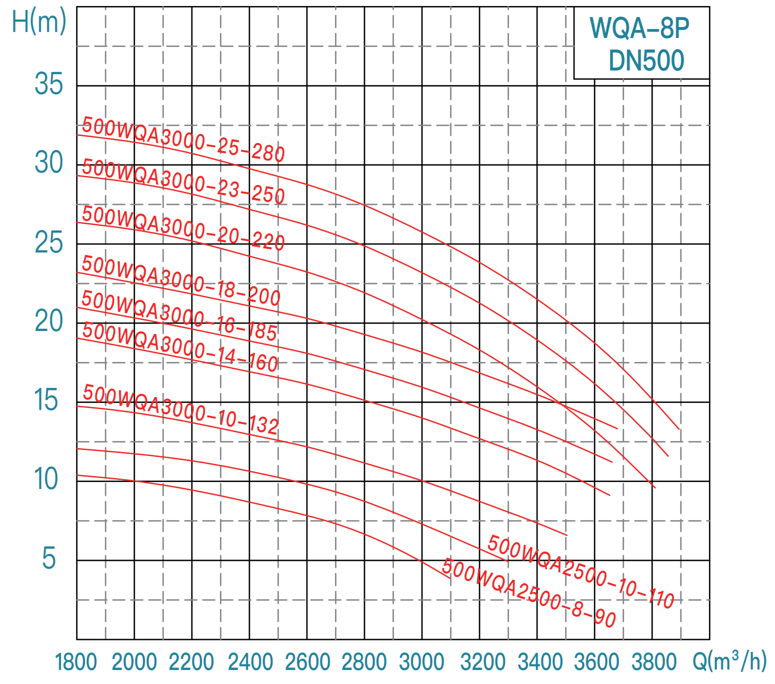


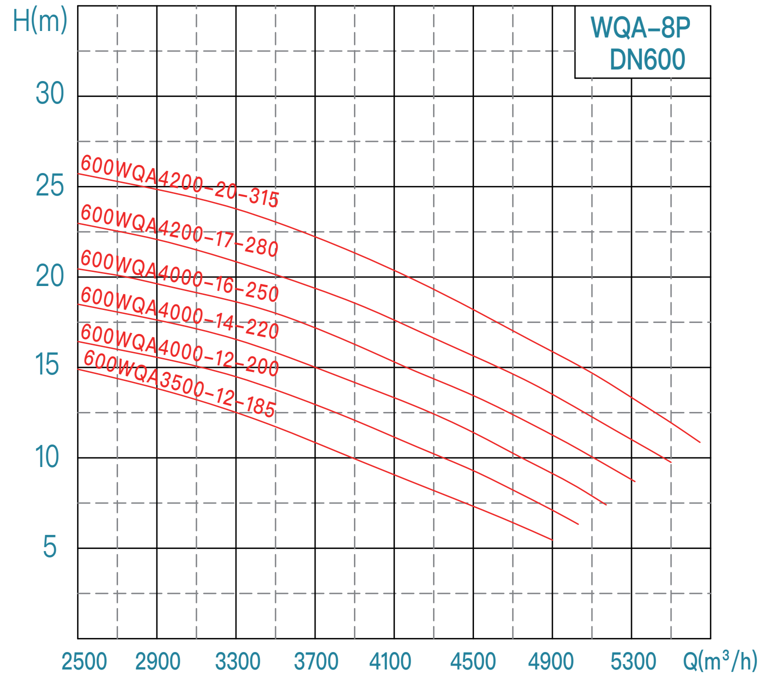






WQA performance curve chart（50Hz）



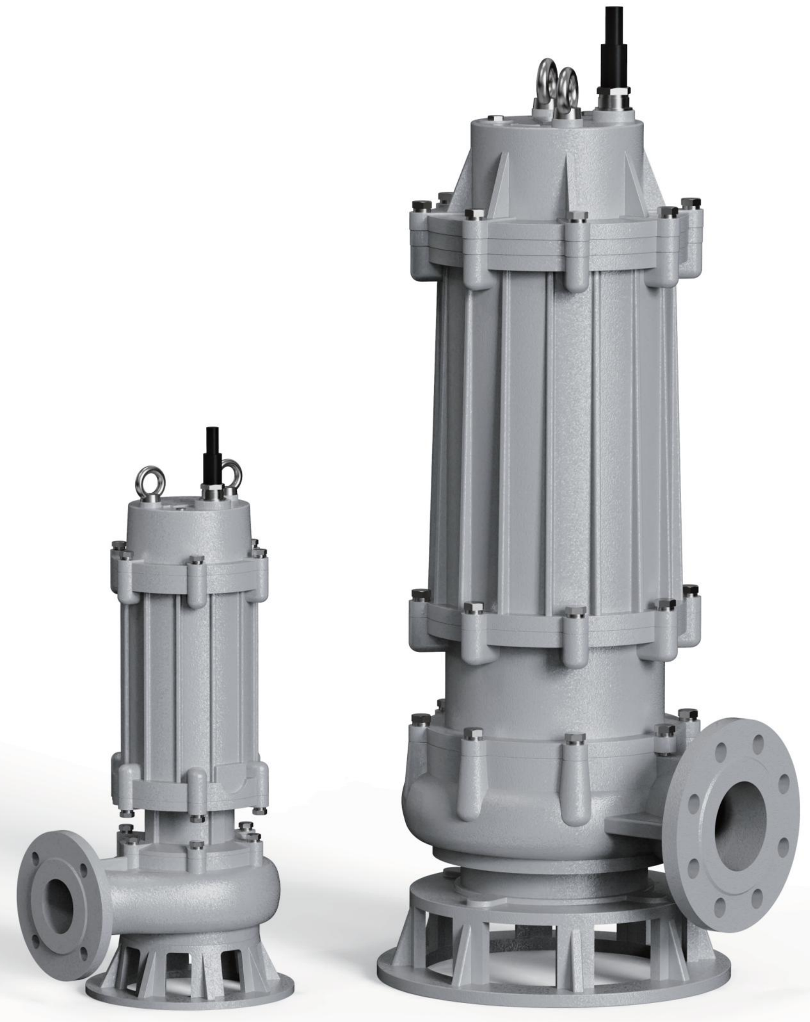


WQE economic
submersible sewage pump ►

WQE

Economic

Submersible sewage pump



Product Introduction:
The WQE series submersible sewage pumps are designed to provide better reliability and higher efficiency for our customers. Based on over 30 years of experience in submersible pump manufacturing and technological accumulation, we have optimized the hydraulic model. Additionally, we have comprehensively optimized the mechanical structure, sealing, and protection aspects, and adopted high-quality motor materials to ensure the stability of the unit. The WQE series consists of a total of 116 models.

Scope of Application:
municipal engineering, sewage treatment, flood control and drainage, hydraulic engineering, industry sewage, environmental protection ...

Operating Conditions:
Flow Rate: 7 ~ 2500 m^3/h
Head: 7 ~ 75 m
Medium Temperature: T ≤ 40°C
Medium Density: ≤ 1100 kg/m^3; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 10
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

Structure diagram of WQE products

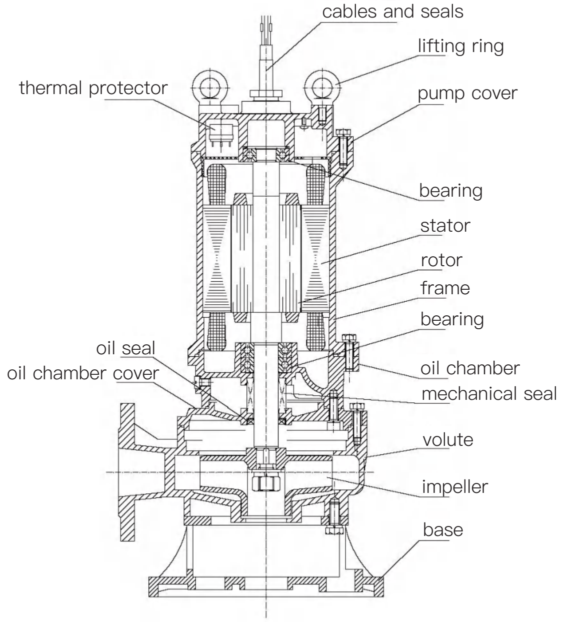
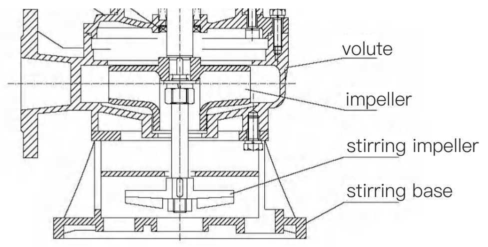


Figure 1



structure diagram for stirring base

Structure diagram for 7.5kW and below

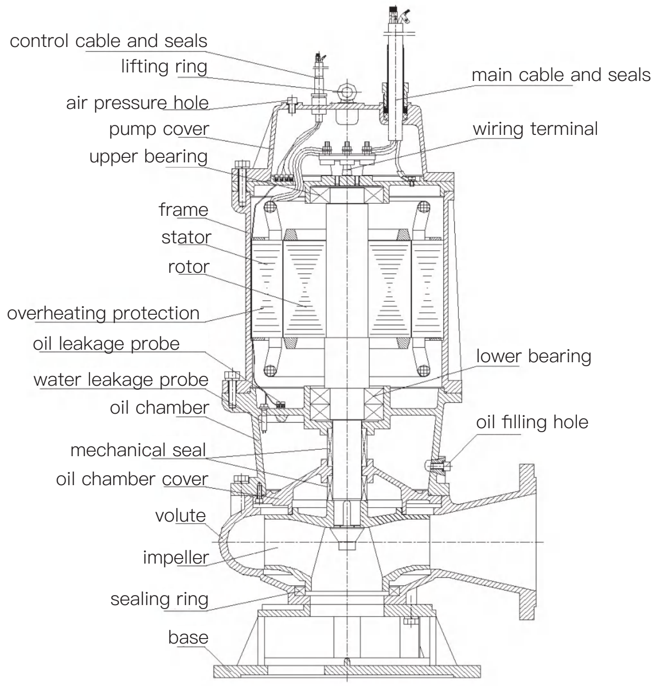


Figure 2

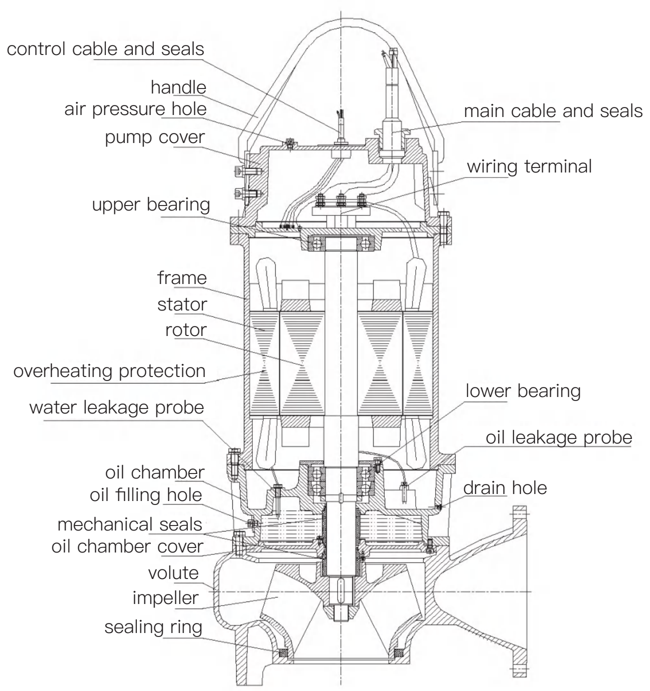


Figure 3

Structure diagram for 11kW and above

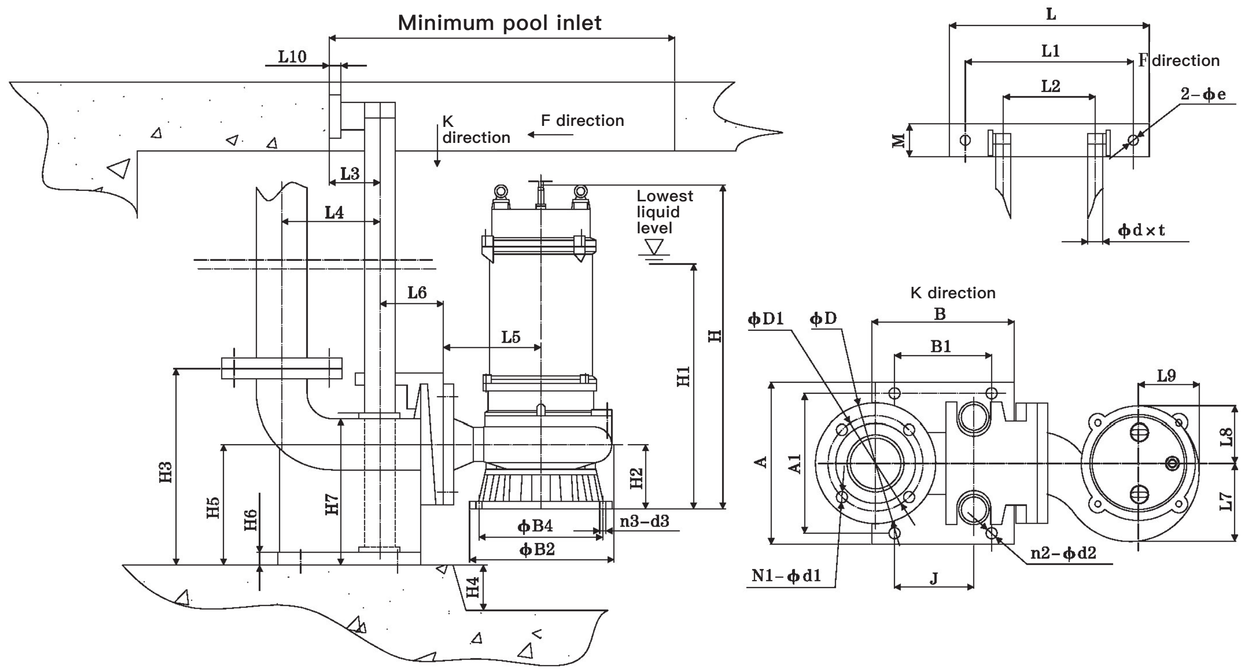
Main materials and configuration table

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		HT200	/
2	Impeller		HT200	/
3	Oil chamber cover		HT200	/
4	Oil chamber，Motor frame Pump cover，Upper cover，Pump base		HT200	/
5	Motor insulation class		F（155℃）	H（180℃）
6	Shaft		2Cr13	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	/
9		Motor side friction pair	Graphite/SiC （AQ1）	SiC/WC （Q1U1）
10		Pump side friction pair	SiC/SiC （Q1Q2）	SiC/WC （Q1U1）
11		Spring	Stainless steel	/
12		Rubber parts	(NBR)	/
13	O-ring		(NBR)	/
14	External leakage fastener		7.5kW and below：8.8 11kW and above：A2-70	7.5kW and below：A2-70
15	Protection device (standard)	75-6P，90-6P 110kW-132kW	Water leakage probe/Float switch /Overheating protection	Thermal protector for 3kW and below

Specification/model of submersible motor cable

Motor power (kW)	Cable model	Standard main cable specification	Star delta	Model of controlling cable
0.55	YZW 300/500V	YZW 4×0.75	/	YZW 4×0.75 (Non–standard)
0.75				
1.1				
1.5		YZW 4×1		
2.2				
3				
4		YZW 3×1.5+1×1		
5.5				
7.5		YZW 3×2.5+1×1.5		
11/15–2P		YZW 3×2.5+1×4	2–YZW 3×2.5+1×1.5	YZW 4×1 (Non–standard)
11/15–4P			2–YZW 3×6+1×4	
18.5	YCW 450/750V	YCW 3×10+1×6		
22				
30		YCW 3×16+1×16	2–YCW 3×10+1×6	
37				
45		YCW 3×25+1×10	2–YCW 3×16+1×16	
55				
75–4P		YCW 3×50+1×16	2–YCW 3×25+1×10	
90–4P			2–YCW 3×35+1×10	
75–6P			2–YCW 3×25+1×10	YZW 3×1.5+1×1 (Standard)
90–6P			2–YCW 3×35+1×10	
110		2–YCW 3×35+1×10		
132		2–YCW 3×50+1×16		

installation dimension drawing



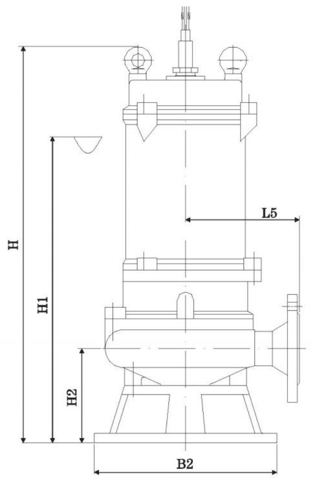
Flange coupling parts dimension table

Model DN	Flange Connection Size D	D1	N1–Φd1	Coupling Base Size A	A1	B	B1	J	n2–Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φdxt
40	130	100	4–Φ14	108	65	138	68	68	4–Φ15	195	Adjust according to pump model	120	14	150	225	185	70	35	Φ14	65	125	85	13	Φ33×3
50	140	110		197	168	208	130	120	4–Φ19	250		165		/	265	215	105	42	Φ14	66	135	90	15	
65	160	130		238	190	252	155	135		265		170	16		280	235	125	50	Φ14	70	155	85	18	
80	190	150	4–Φ19	255	220	225	153	125		310		190			315	265	145	53	Φ14	80	155	100	18	Φ48×3
100	210	170		293	265	258	175	140		350		235	18		365	305	170	55	Φ15	100	200	110	20	
150	265	225	8–Φ19	400	285	410	300	210	4–Φ22	485		300	22	390	410	260	275	60			290	190	20	
200	320	280		400	300	445	350	250	4–Φ25	550		325		440							330	195	20	
250	375	335	12–Φ19	460	360	555	430	325		635		310	24	450							395	215	20	
300	445	400	12–Φ23	550	415	570	410	345		740		400	30	570	480	305	340	65		110	395	225	23	Φ60×3
350	505	460	16–Φ23	580	420	615	400	350		860		490	33	680	500	320	360			115	435	230	25	
400	565	515	16–Φ28	630	490	665	510	430		960		560	36	775							480	235	25	
500	670	620	20–Φ28	732	570	752	550	475	4–Φ28	1160		655	38	900	750	660	460	78	Φ20		565	250	26	Φ76×3.5
600	780	725	20–Φ31	830	720	850	700	590	4–Φ33	1320		710	40	1020	830	740	560				660	275	30	

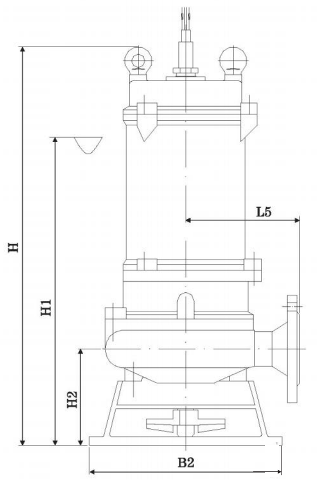
Lightweight flange coupling parts dimension table

Model DN	Flange Connection Size D	D1	N1–Φd1	Coupling Base Size A	A1	B	B1	J	n2–Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φdxt
50	140	110	4–Φ14	142	120	170	120	120	4–Φ14	235	100	150	11	180	225	185	72	36	Φ11	60	133	53	12	Φ33×3
65	160	130		150	128	195	142	142		245		138	12	177	265	215					142	55		
80	190	150	4–Φ18	170	148	210	158	158		272		158	13	220	280	235	77			77	153			Φ48×3
100	205	170		190	158	220	145	143	4–Φ18	340		198	14	257	315	265					178	82		
150	260	225	8–Φ18	262	220	306	230	240	4–Φ22	425	150	268	18	350	365	305	95	52	Φ15	83	230	100	18	

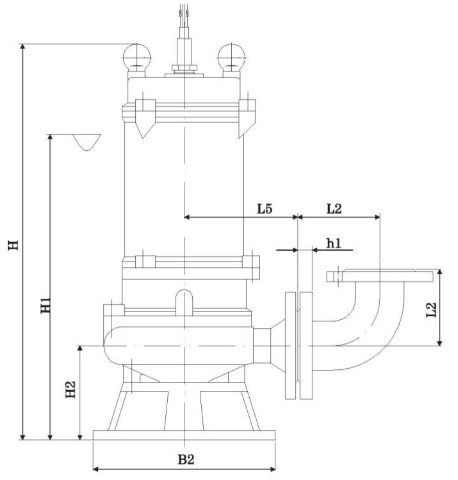
Pump appearance and moblie installition



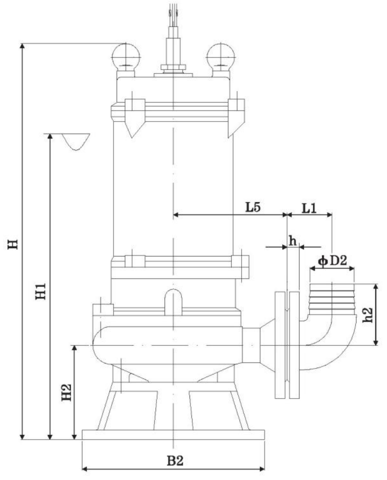
Dimension drawing of
pump appearance



Dimension drawing of
stirring pump （JY）



Hard pipe（double flange elbow/joint）
mobile installition



Soft pipe（elbow）
mobile installition

WQE installition dimension table

No.	WQE model	Diameter (mm)	Power (kW)	Rotating speed (r/min)	Voltage (V)	Passable particle diameter (mm)	Appearance dimension table (mm) L5	B2	B4	n3–d3	H	H1	H2	L8	L9	L7	304 impeller	Stirring（JY）H、H1、H2 needs to be increased	Weight (kg)	Pressure grade of pump outlet flange (kg)
1	40WQE10–10–0.75	40	0.75	2950	380	10	125	175	/	/	520	370	90	90	95	100	/	30	22	6
2	50WQE10–7–0.55	50	0.55																22	
3	50WQE10–10–0.75		0.75																22	
4	40WQE7–15–1.1	40	1.1				155	215			600	430	125	95	100	115			26	
5	50WQE7–15–1.1	50																	22	
6	50WQE15–15–1.5		1.5			15	150	235			570	440	125	105	110	125			22	
7	50WQE15–20–2.2		2.2								620	460							22	
8	65WQE25–10–1.5	65	1.5								580	450							22	
9	65WQE25–15–2.2		2.2								620	460							27	
10	80WQE40–10–2.2	80				30	165				640	480	135	105	110	135			27	
11	100WQE50–7–2.2	100				35	170				650	490	135	105	115	135			34	
12	50WQE15–27–3	50	3			25	150				640	480	130	115	125	135			34	
13	50WQE15–31–4		4				185	270			700	540	145	125	130	150			35	
14	50WQE15–37–5.5		5.5			30					720	560							37	
15	50WQE20–40–7.5		7.5			25	210	330			680	600	150	145	155	170		60	37	
16	65WQE30–15–3	65	3			30	165	235			650	500	125	120	135	145		30	40	
17	65WQE30–23–4		4				170	270			780	600	155	130	135	140			40	
18	65WQE30–29–5.5		5.5																39	
19	65WQE30–35–7.5		7.5			25	210	330			680	600	150	145	155	170		60	40	
20	80WQE40–12–3	80	3			30	180	235			660	500	135	110	135	140		30	43	
21	80WQE40–16–4		4				175	270			700	460	155	130	140	150			57	
22	80WQE40–22–5.5		5.5								720	580							57	
23	80WQE45–25–7.5		7.5				210	330			720	640	160	140	145	155		60	58	
24	100WQE60–9–3	100	3			35	180	235			670	500	130	110	120	135		30	58	
25	100WQE6–10–4		4				165	270			740	560	160		125	140			59	
26	100WQE65–15–5.5		5.5								760	580							64	
27	100WQE80–15–7.5		7.5				210	330			720	640	160	140	145	155		60	64	
28	150WQE100–9–5.5	150	5.5			45	210	270			780	600	175	125	155	160		30	64	
29	150WQE100–12–7.5		7.5				210	330			760	680	180	145	160	180		60	65	
30	150WQE135–10–7.5		7.5	1475			290	340			850	720	230	170	185	205			65	
31	65WQE20–56–11	65	11	2950		25	215	330			850	720	200	145	155	165	●		119	
32	80WQE40–45–11	80	11			30	215	330			850	720	220	145	155	165			120	
33	80WQE45–55–15		15				215	330			900	770	220	145	155	165			130	
34	100WQE80–25–11	100	11	1475		35	330	410	357	4–Φ14	920	730	210	220	235	255			220	
35	100WQE80–30–15		15								950	750							240	
36	100WQE100–25–15										950	750							240	
37	100WQE80–35–18.5		18.5								1050	850						80	280	
38	100WQE100–30–18.5										1050	850							280	
39	100WQE100–37–22		22								1150	950							300	

WQE installition dimension table

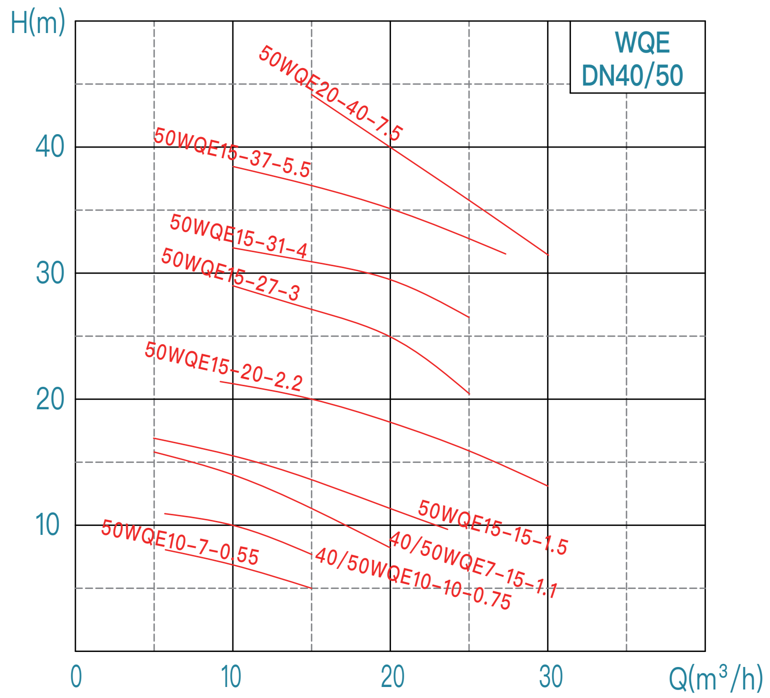
No.	WQE model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)										304 impeller	Stirring（JY）H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange
		mm	kW	r/min	V	mm	L5	B2	B4	n3–d3	H	H1	H2	L8	L9	L7			kg	kg
40	150WQE130–15–11	150	11	1475	380	45	355	430	380	4–Φ14	950	750	260	205	225	255	●	60	220	6
41	150WQE170–18–15		15				355	430	380	4–Φ14	1000	800	260	205	225	255		60	240	
42	150WQE180–20–18.5		18.5				355	430	380	4–Φ14	1100	900	260	205	225	255		80	290	
43	150WQE180–25–22		22				355	430	380	4–Φ14	1150	950	260	205	225	255		80	300	
44	200WQE300–7–11	200	11			55	365	430	380	4–Φ14	950	750	260	215	235	265		60	230	
45	200WQE250–13–15		15				365	430	380	4–Φ14	1000	800	260	215	235	265		60	250	
46	200WQE250–15–18.5		18.5				365	430	380	4–Φ14	1100	900	260	215	235	265		80	290	
47	200WQE250–17–22		22				365	430	380	4–Φ14	1150	950	260	215	235	265		80	310	
48	250WQE400–6–11 *	250	11			80	350	400	340	4–Φ22	1000	800	245	210	260	310	/	/	255	10
49	250WQE400–8–15 *		15				350	400	340	4–Φ22	1100	850	245	210	260	310			275	
50	250WQE400–9–18.5 *		18.5				370	400	340	4–Φ22	1120	910	235	235	285	325			315	
51	250WQE400–10–22 *		22				370	400	340	4–Φ22	1120	910	235	235	285	325			335	
52	100WQE120–42–30 *	100	30			35	360	515	435	4–Φ28	1380	970	285	260	265	270			435	6
53	100WQE130–45–37 *		37				360	515	435	4–Φ28	1500	1050	285	260	265	270			510	
54	100WQE120–55–45 *		45				360	515	435	4–Φ28	1600	1150	285	260	265	270			520	
55	100WQE150–60–55 *		55				360	515	435	4–Φ28	1650	1200	285	260	265	270			650	
56	150WQE180–30–30	150	30			45	380	520	467	4–Φ16	1250	1000	290	250	280	310	●	80	435	
	150WQE250–22–30						380	520	467	4–Φ16	1250	1000	290	250	280	310		80	435	
57	150WQE180–34–37		37				380	520	467	4–Φ16	1250	1000	290	250	280	310		0	500	
	150WQE250–27–37						380	520	467	4–Φ16	1250	1000	290	250	280	310		0	500	
58	150WQE250–34–45		45				380	520	467	4–Φ16	1300	1050	270	250	280	310	/	0	510	
59	150WQE200–50–55 *		55				410	515	435	4–Φ28	1600	1150	245	265	285	310		/	640	
60	200WQE360–16–30	200	30			55	390	520	467	4–Φ16	1250	1000	290	250	280	310	●	80	445	
61	200WQE350–22–37		37				390	520	467	4–Φ16	1250	1000	290	250	280	310		0	520	
62	200WQE400–25–45		45				390	520	467	4–Φ16	1350	1100	300	270	300	330		0	530	
63	200WQE300–38–55 *		55				400	515	435	4–Φ28	1650	1200	265	265	270	290	/	/	670	
64	250WQE600–9–30	250	30			70	390	520	467	4–Φ16	1300	1050	300	270	300	340	●	80	455	
65	250WQE600–13–37		37				390	520	467	4–Φ16	1350	1100	300	270	300	340		0	530	
66	250WQE600–17–45		45				390	520	467	4–Φ16	1400	1150	300	270	300	340	/	0	540	
67	250WQE600–20–55 *		55				440	515	435	4–Φ28	1700	1250	285	275	320	360		/	710	10
68	300WQE800–7–30 *	300	30				410	490	/		1400	1000	255	270	320	370			480	
69	300WQE900–7–37 *		37				410	490			1500	1050	255	270	320	370			550	
70	300WQE900–9–45 *		45				410	490			1550	1100	255	270	320	370			560	
71	300WQE800–15–55 *		55				410	490			1700	1250	255	270	320	370			730	
72	350WQE1000–6–30 *	350	30	980		90	520	590			1550	1100	280	345	410	460			505	
73	350WQE1000–7–37 *		37				520	590			1550	1100	280	345	410	460			656	
74	350WQE1000–10–45 *		45	1475			510	530			1600	1150	285	300	380	450			580	
75	350WQE1100–10–55 *		55				510	530			1700	1250	285	300	380	450			750	
76	150WQE200–55–75 *	150	75			40	440	515	435	4–Φ28	1700	1200	255	295	315	335			720	6
77	150WQE240–60–90 *		90				440	515	435	4–Φ28	1700	1200	255	295	315	335			880	

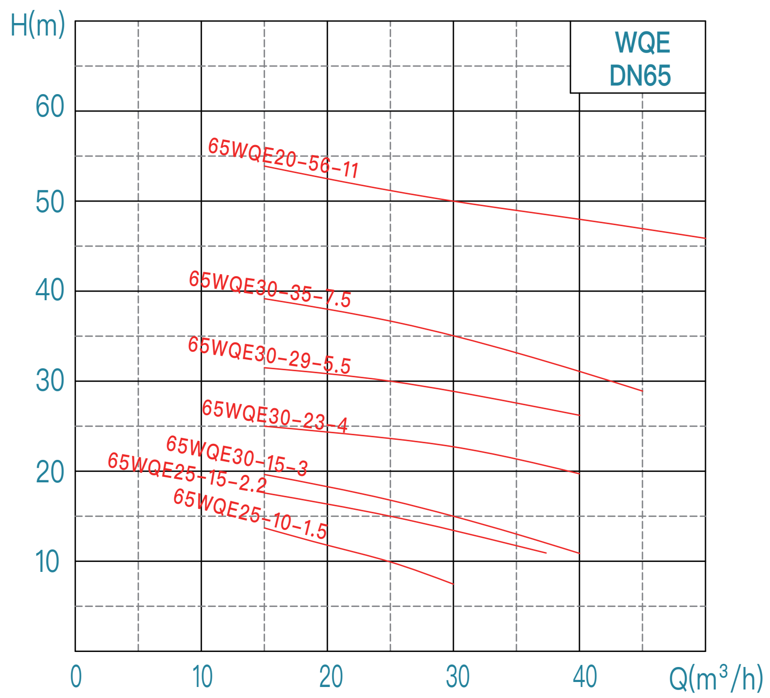
WQE installition dimension table

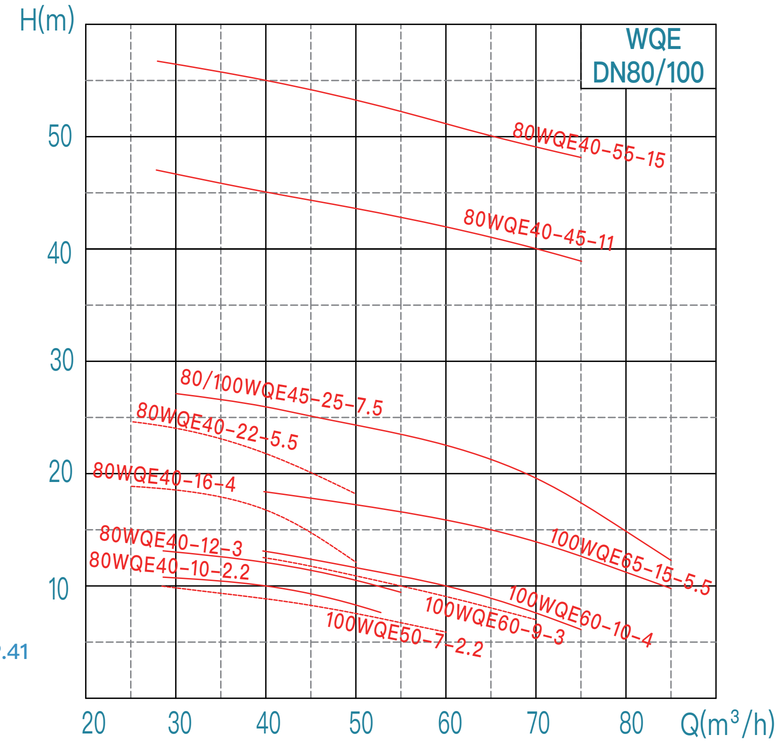
No.	WQE model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)										304 impeller	Stirring（JY）H、H1、H2 needs to be increased	Weight	Pressure grade of pump outlet flange
		mm	kW	r/min	V	mm	L5	B2	B4	n3–d3	H	H1	H2	L8	L9	L7			kg	kg
78	200WQE350–45–75 *	200	75	1475	380	50	460	515	435	4–Φ28	1750	1250	270	300	325	355	/	/	770	6
79	200WQE350–50–90 *		90				460	515	435	4–Φ28	1850	1350	270	300	325	355			930	
80	200WQE350–60–110 *		110				545	580	/		1900	1400	215	340	360	385			1300	
81	200WQE350–75–132 *		132				545	580			2000	1500	215	340	360	385			1390	
82	250WQE600–28–75 *	250	75			60	480	515	435	4–Φ28	1800	1300	305	310	350	385			760	10
83	250WQE700–30–90 *		90				480	515	435	4–Φ28	1900	1400	305	310	350	385			810	
84	250WQE600–38–110 *		110			75	540	730	660	4–Φ24	2000	1500	320	370	375	380			1300	
85	250WQE600–52–132 *		132				540	730	660	4–Φ24	2100	1600	320	370	375	380			1390	
86	300WQE800–18–75 *	300	75			75	460	515	/		1800	1300	255	280	320	360			830	
87	300WQE900–22–90 *		90				460	515			1900	1400	255	280	320	360			980	
88	300WQE900–27–110 *		110			90	485	580			2000	1500	255	385	400	420			1300	
89	300WQE800–35–132 *		132				485	580			2200	1600	255	385	400	420			1420	
90	350WQE1200–15–75 *	350	75	980		110	565	650			2000	1400	285	410	455	510			1165	
91	350WQE1100–18–90 *		90				565	650			2000	1400	285	410	455	510			1440	
92	350WQE1000–24–110 *		110	1475		90	565	650			2000	1500	285	410	455	510			1370	
93	350WQE1400–17–110 *		110	980		110	565	650			2100	1500	285	410	455	510			1640	
94	350WQE1000–30–132 *		132	1475		90	565	650			2000	1500	285	410	455	510			1450	
95	350WQE1500–18–132 *		132	980		120	710	730	660	4–Φ24	2350	1800	420	470	550	620			1650	
96	400WQE1300–8–55 *	400	55	980		110	590	640	/		1800	1300	310	380	455	510			926	
97	400WQE1400–12–75 *		75				590	650			2100	1500	315	410	455	510			1180	
98	400WQE1500–15–90 *		90				590	650			2100	1500	315	410	455	510			1470	
99	400WQE1500–17–110 *		110				590	650			2150	1550	315	410	455	510			1670	
100	400WQE2000–13–132 *		132			120	715	830			2250	1700	315	485	530	620			1680	
101	500WQE1800–7–55 *	500	55			110	690	800			1800	1300	365	425	520	600			985	
102	500WQE1800–9–75 *		75				690	800			2000	1450	365	425	520	600			1230	
103	500WQE2000–10–90 *		90				690	800			2000	1450	365	425	520	600			1500	
104	500WQE2500–10–110 *		110				690	800			2200	1650	365	425	520	600			1720	
105	500WQE2500–12–132 *		132			120	720	830			2300	1750	370	485	555	620			1740	

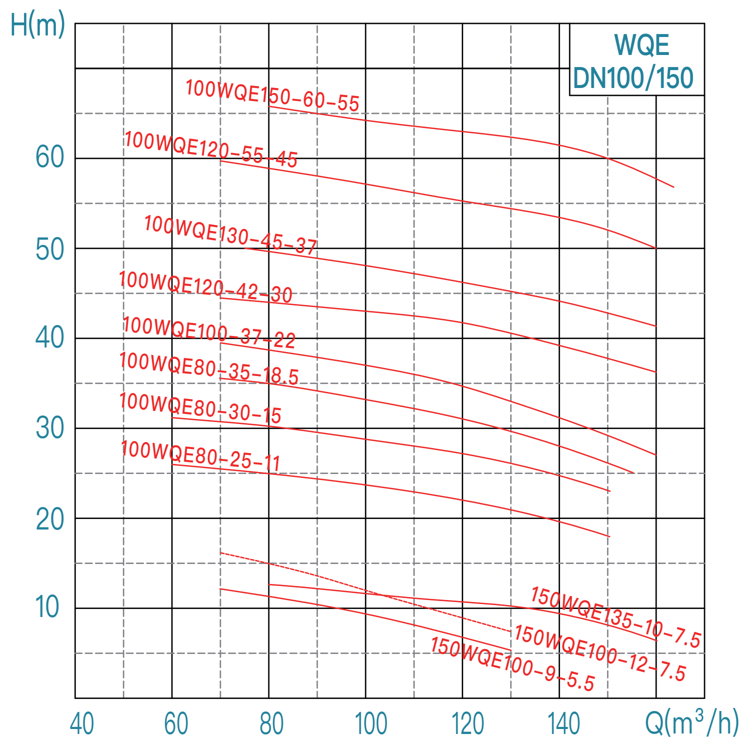
Note: The asterisked structure is shown in Figure 3. A slash represents "none".

WQE performance curve chart (50Hz)









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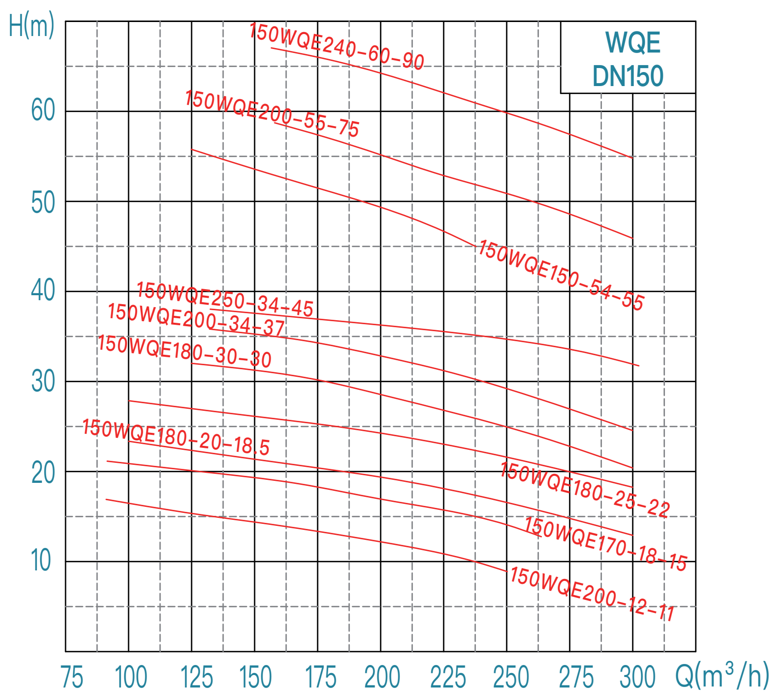
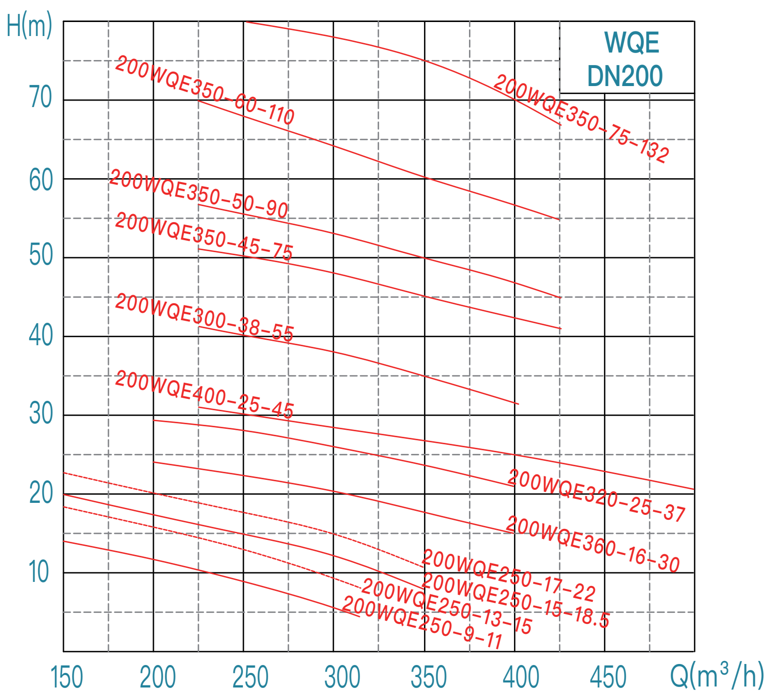
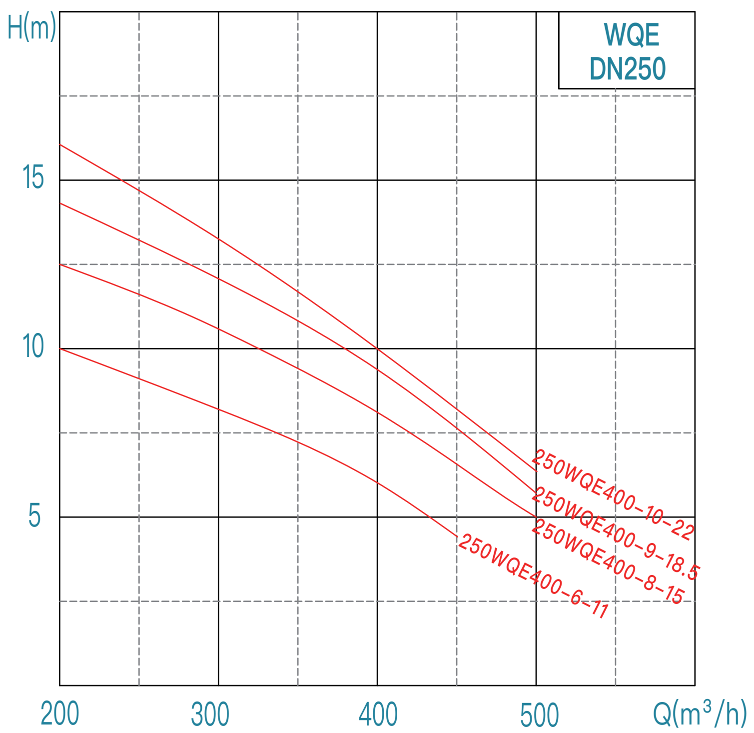
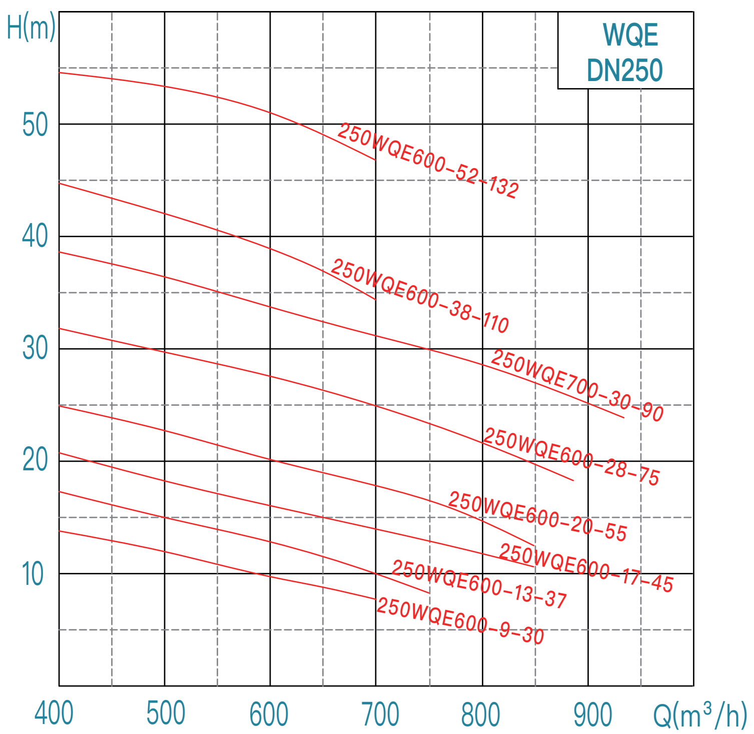
WQE performance curve chart (50Hz)

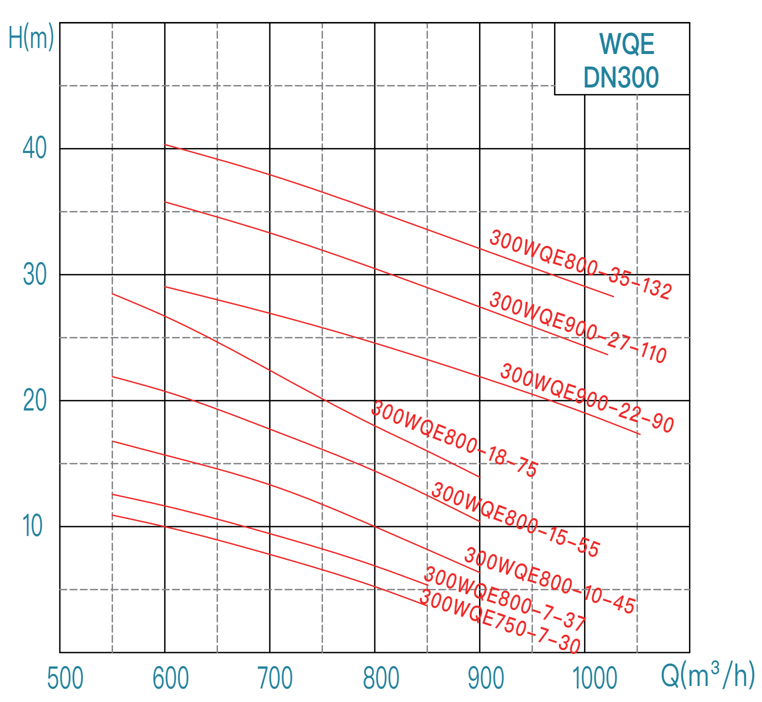


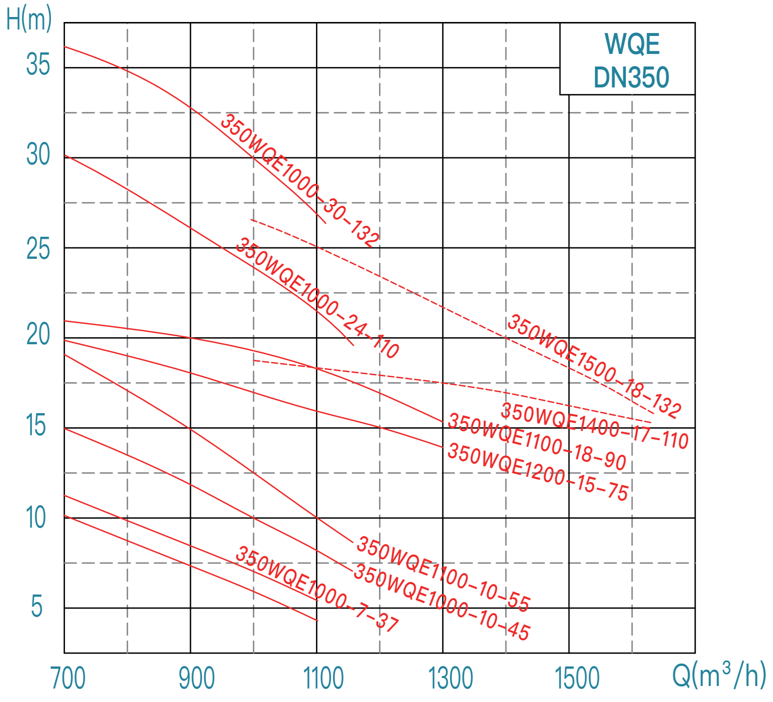


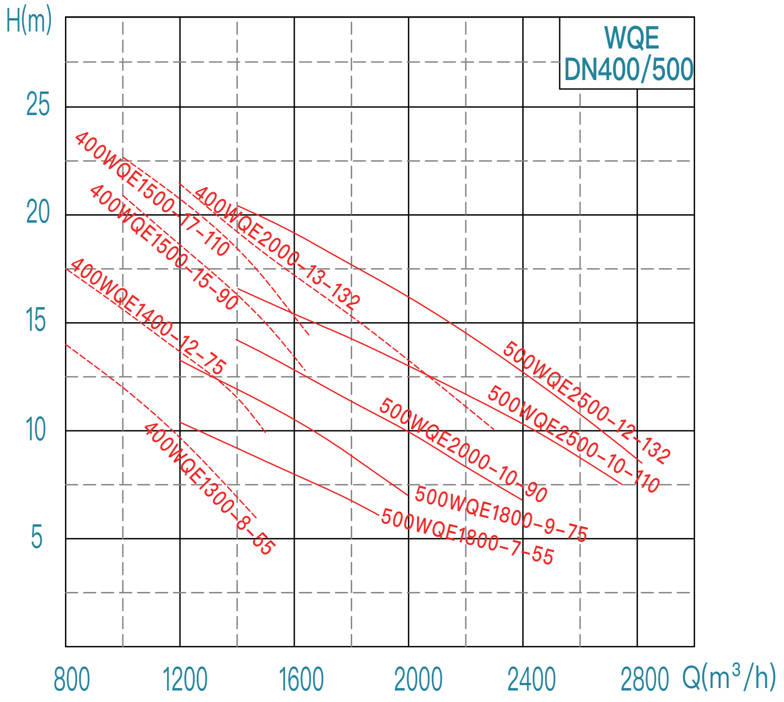




WQE performance curve chart （50Hz)







WQF stainless steel
submersible sewage pump ►

WQF

Stainless steel

Submersible sewage pump

Customizable Materials:

304/ 316/ 316L/

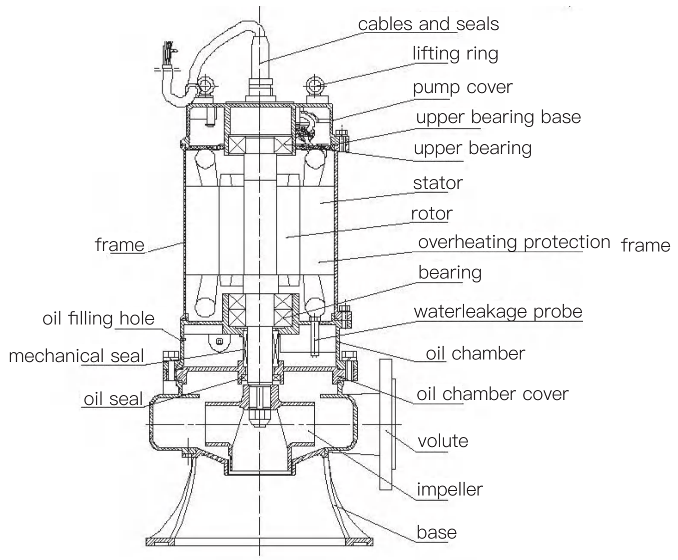
2205/ 2507

Product Introduction:
The WQF series stainless steel submersible sewage pumps from Hangzhou Xizi Pump Industry are specialized products developed with advanced technology. The pump's flow components are all made of stainless steel through precision casting, ensuring pure quality and an attractive appearance. They are self-pollution-free and corro-sion-resistant, expanding the application fields of water supply and drainage.

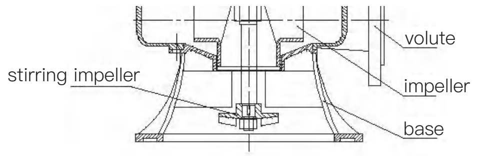
Scope of application:
municipal engineering、
sewage treatment、
chemical engineering、
hydraulic engineering、
seawater treatment、
environment treatment ...

Operating Conditions:
Flow Rate: 7 ~ 600 m^3/h
Head: 7 ~ 55 m
Medium Temperature: T ≤ 40°C
Medium Density: ≤ 1100 kg/m^3; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 11
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

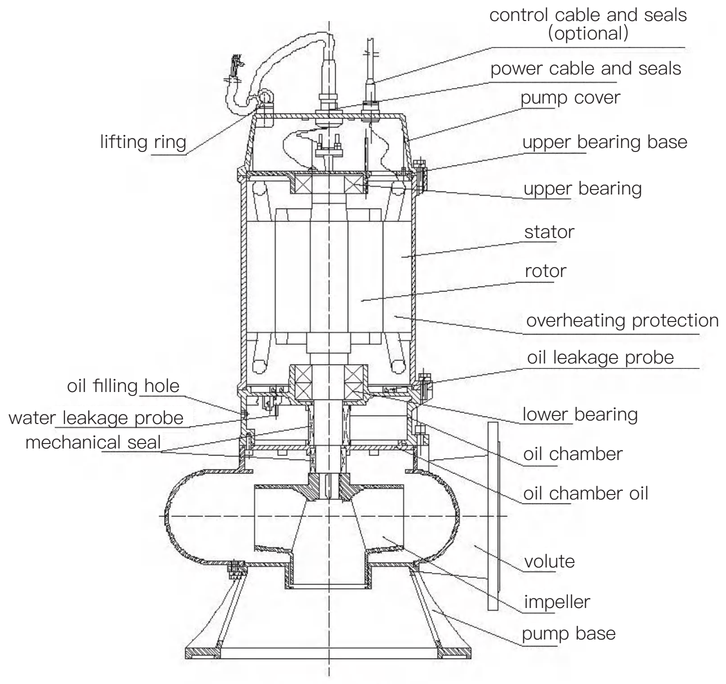
Structure diagram of WQF products



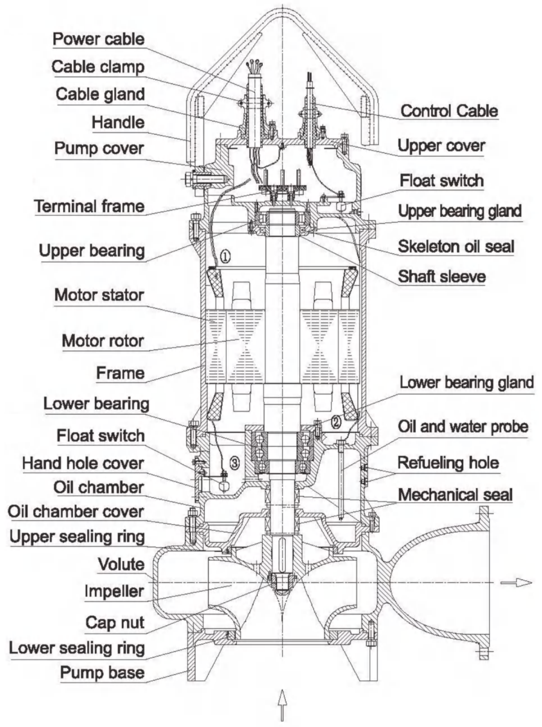
Structure diagram for 7.5kW and below



structure diagram for stirring base



Structure diagram for 11~45 kW



Structure diagram for 55kW and above

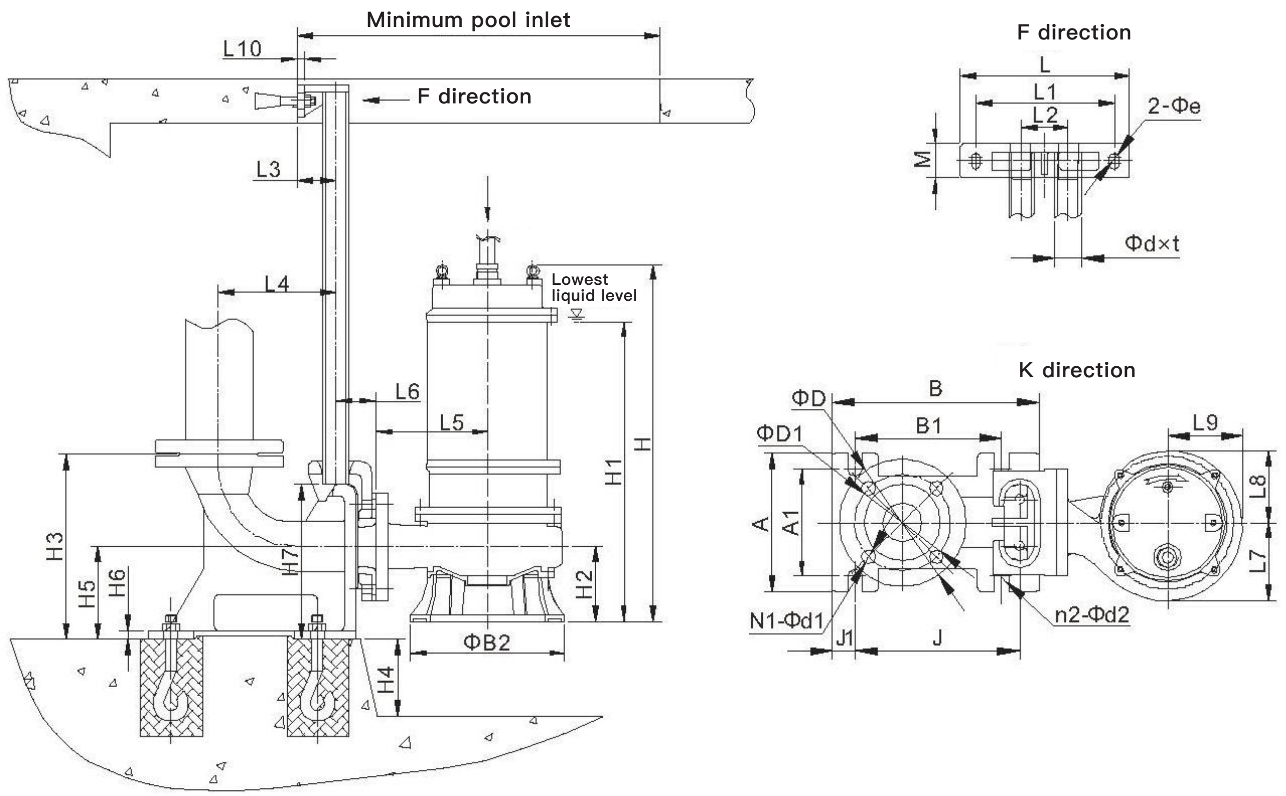
Main materials and configuration table

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		304	/
2	Impeller		304	/
3	Oil chamber cover		304	/
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		304	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		304	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/WC (Q1U1)	WC/WC (U1U2)
11		Spring	Stainless steel	/
12		Rubber parts	(P)	/
13	O-ring		(P)	/
14	External leakage fastener		A2-70	/
15	Protection device (optional)	45kW and below	Water leakage probe/Float switch /Overheating protection	Order remarks according to customer requirements.

Specification/model of submersible motor cable

Motor power (kW)	Cable model	Standard main cable specification	Star delta	Model of controlling cable
0.55	YZW 300/500V	YZW 4×0.75	/	Multicore cable
0.75				
1.1				
1.5		YZW 4×1		
2.2				
3				
4		YZW 3×1.5+1×1		YZW 4×0.75 (Non-standard)
5.5				
7.5		YZW 3×2.5+1×1.5		
11/15-2P		YZW 3×6+1×4	2-YZW 3×2.5+1×1.5	YZW 4×1 (Non-standard)
11/15-4P			2-YZW 3×6+1×4	
18.5	YCW 450/750V	YCW 3×10+1×6		
22				
30		YCW 3×16+1×16	2-YCW 3×10+1×6	
37		YCW 3×25+1×10	2-YCW 3×16+1×16	
45				
55		YCW 3×35+1×10		YZW 6×1 (Standard)

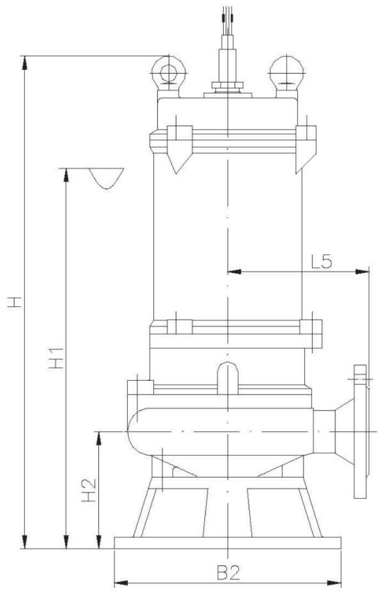
Installition dimension drawing



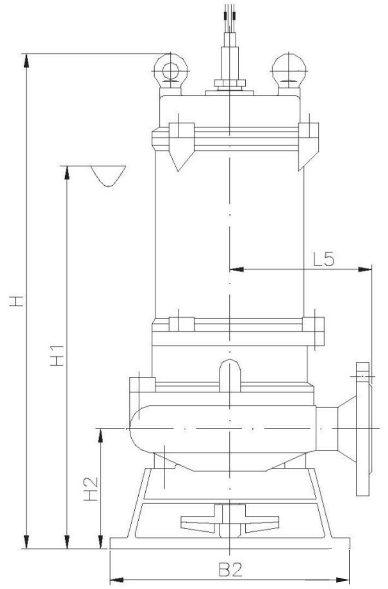
Fange coupling parts dimension table

Model	Flange Connection Size			Coupling Base Size																						
DN	D	D1	N1-Φd1	A	A1	B	B1	J	J1	n2-Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φd×t
50	165	125	4-Φ19	180	140	270	190	214	30	4-Φ19	240	100	120	12	200	220	180	60	44	Φ12	50	152	52	10	Φ33×3
65	180	145		190	150	280	190	221	30		240	100	120	12	200	220	180	60	44		50	149	55	10	
80	195	160	8-Φ19	210	170	300	210	237	30		270	150	130	12	225	250	210	80	50	Φ15	57	165	63	10	Φ48×3
100	215	180		230	190	300	210	237	30		300	150	150	14	260	250	210	80	50		57	185	63	10	
150	280	240	8-Φ23	300	250	460	330	367	50		380	150	180	18	320	330	280	120	60		62	305	76	12	
200	340	295		350	300	470	350	370	50		460	200	220	20	390	330	280	120	60		62	320	80	12	
250	395	350	12-Φ23	420	360	580	450	471	50	4-Φ25	500	200	240	20	450	340	280	160	70	Φ19	115	359	95	16	Φ76×3.5
300	445	400		460	400	630	500	521	50		560	200	260	20	490	380	320	200	70		115	399	96	16	
350	505	460	16-Φ23	520	460	680	550	570	50		640	300	300	20	550	380	320	200	70		115	439	88	16	

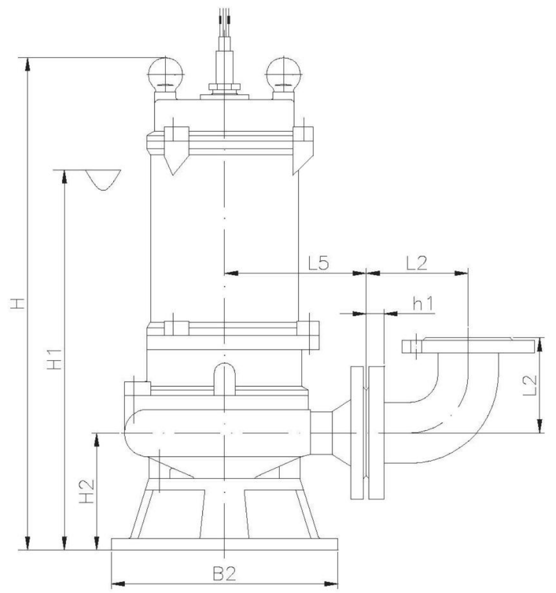
Pump appearance and moblie installition



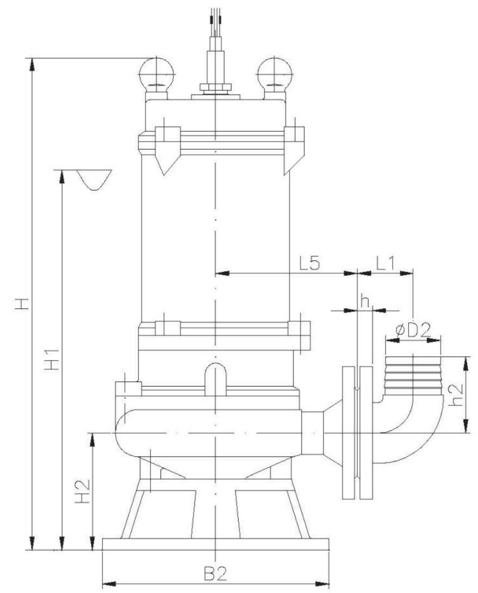
Dimension drawing of pump appearance



Dimension drawing of stirring pump （JY）



Hard pipe（double flange elbow/joint） mobile installition



Soft pipe（elbow） mobile installition

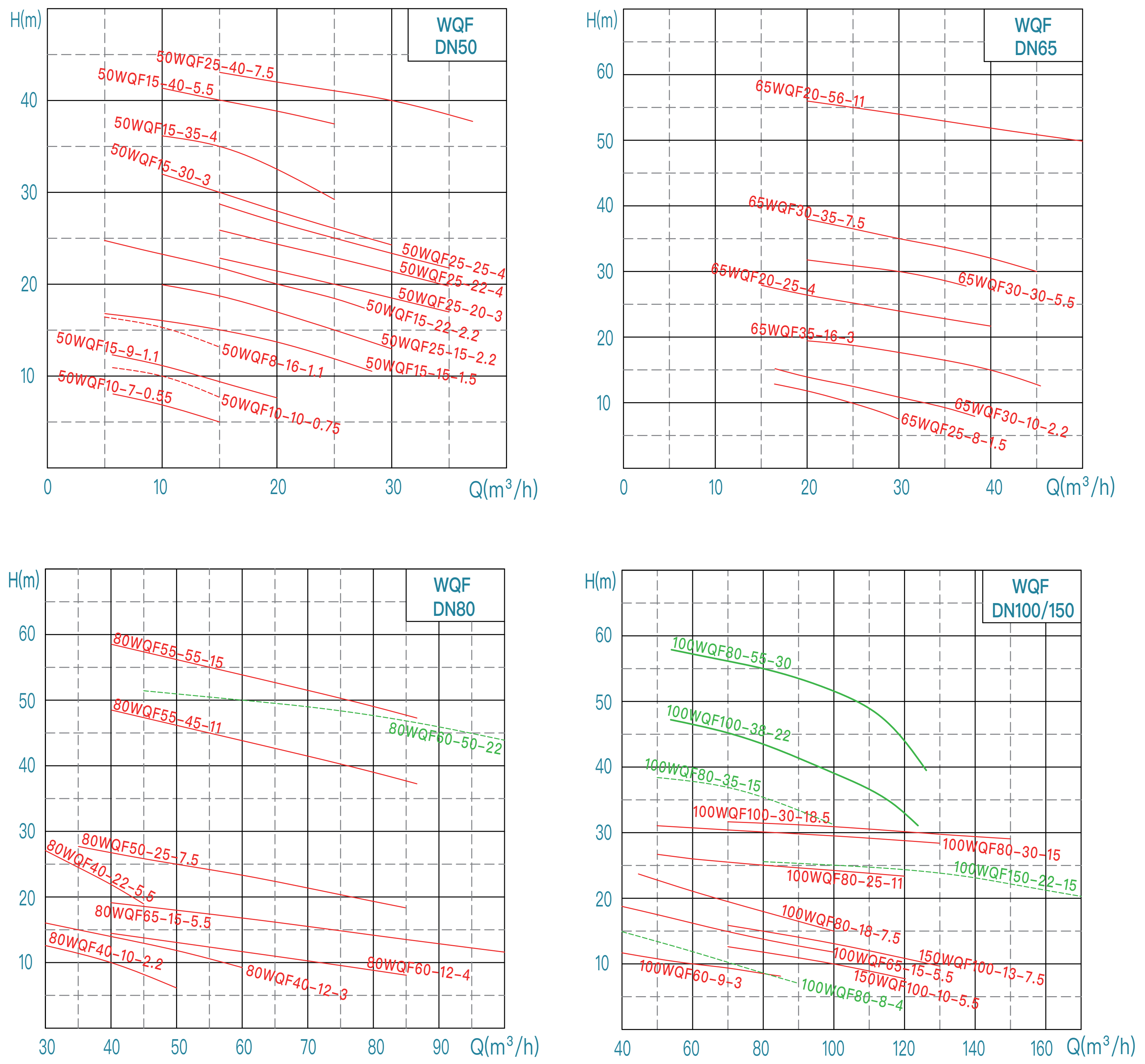
WQF installition dimension table

No.	WQF model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Stirring（JY） H、H1、H2 needs to be increased	Weight
		mm	kW	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7		kg
1	50WQF10–7–0.55	50	0.55	2950	220/380	15	135	190	490	420	100	90	95	105	40	21
2	50WQF10–10–0.75		0.75				135	190	490	420	100	90	95	105		21
3	50WQF8–16–1.1		1.1				135	190	490	420	100	90	95	105		23
4	50WQF15–9–1.1		1.1				135	190	490	420	100	90	95	105		23
5	50WQF15–15–1.5		1.5				150	200	530	460	100	95	100	105		28
6	50WQF15–22–2.2		2.2				150	200	530	460	100	95	100	105		31
7	50WQF25–15–2.2		2.2				150	200	530	460	100	95	100	105		31
8	50WQF15–30–3		3		380	20	150	200	550	480	100	105	110	115		36
9	50WQF25–20–3		3				150	225	560	490	110	95	100	105		36
10	50WQF15–35–4		4			15	170	240	600	520	134	120	125	130		49
11	50WQF25–25–4		4				170	240	600	520	134	120	125	130		49
12	50WQF15–40–5.5		5.5				170	240	600	520	134	120	125	130		54
13	50WQF25–40–7.5		7.5				185	320	680	560	140	125	132	140		85
14	65WQF25–8–1.5	65	1.5		220/380		160	200	530	460	95	90	100	115		28
15	65WQF30–10–2.2		2.2			25	160	200	530	460	95	90	100	115		32
16	65WQF35–16–3		3		380		160	225	560	490	110	90	100	115		36
17	65WQF30–22–4		4				165	240	600	520	120	120	130	136		50
18	65WQF20–30–4		4				165	240	600	520	120	120	130	136		50
19	65WQF30–30–5.5		5.5				165	240	600	520	120	120	130	136		55
20	65WQF30–35–7.5		7.5				200	320	680	560	140	125	132	140		85
21	80WQF40–10–2.2	80	2.2		220/38		165	250	550	480	130	105	115	125		33
22	80WQF40–12–3		3		380		165	250	570	500	130	105	115	125	0	38
23	80WQF60–12–4		4				165	240	640	550	150	120	130	140		51
24	80WQF40–22–5.5		5.5				165	240	640	550	150	120	130	140		60
25	80WQF65–15–5.5		5.5				165	240	640	550	150	120	130	140		60
26	80WQF50–25–7.5		7.5				190	320	710	590	175	130	140	150		85
27	100WQF60–9–3	100	3			30	180	250	580	510	140	105	120	135	40	39
28	100WQF80–8–4		4				175	240	650	560	155	125	137	150	0	51
29	100WQF65–15–5.5		5.5				175	240	650	560	155	125	137	150		62
30	100WQF80–18–7.5		7.5				190	320	730	610	180	130	150	170		85
31	150WQF100–10–5.5	150	5.5			40	200	240	650	560	150	115	130	150		66
32	150WQF130–8–5.5		5.5	1475			275	320	765	645	205	175	200	225		108
33	150WQF100–13–7.5		7.5	2950			200	320	760	640	200	160	180	195		89
34	150WQF135–10–7.5		7.5	1475			275	320	885	765	205	175	200	225		111
35	65WQF20–56–11	65	11	2950		15	240	320	820	680	165	137	145	152	/	106
36	80WQF50–45–11	80	11				240	320	820	680	165	137	145	152		106
37	80WQF50–55–15		15				240	320	850	700	165	137	145	152		115
38	80WQF60–50–22		22			20	250	440	970	720	177	175	180	185		184

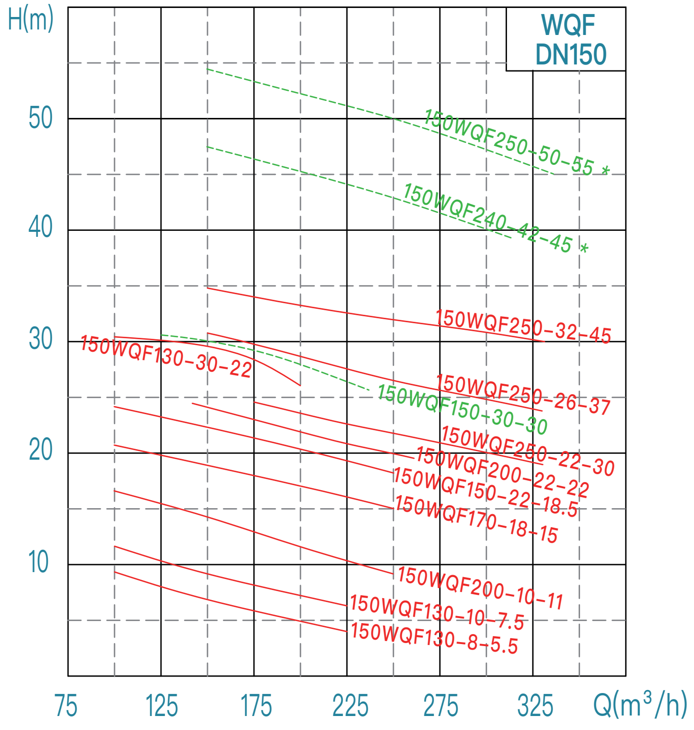
WQF installition dimension table

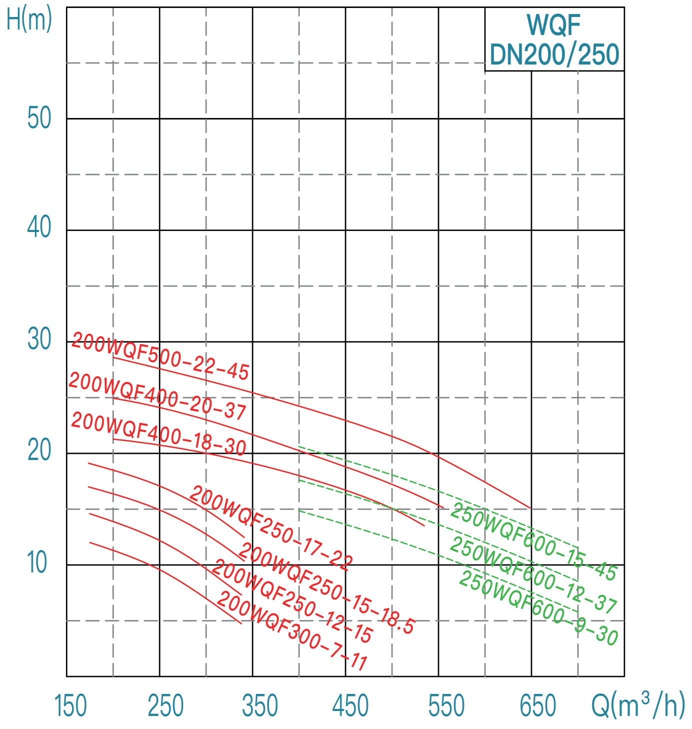
No.	WQF model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Stirring (JY) H、H1、H2 needs to be increased	Weight
		mm	kW	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7		kg
39	100WQF80–25–11	100	11	1475	380	50	280	400	970	780	221	195	212	227	/	195
40	100WQF150–22–15		15				280	400	970	780	221	195	212	227		212
41	100WQF80–30–15						280	400	970	780	221	195	212	227		212
42	100WQF80–35–15			2950		20	280	400	970	780	221	195	212	227		175
43	100WQF100–30–18.5		18.5	1475		50	280	400	1120	840	221	195	212	227		247
44	100WQF100–34–22		22	2950		20	280	400	1170	890	221	195	212	227		264
45	100WQF80–55–30		30				280	400	1170	890	221	195	212	227		244
46	150WQF200–10–11	150	11	1475		60	280	500	1100	830	243	223	240	258		195
47	150WQF170–18–15		15				280	500	1100	830	243	223	240	258		212
48	150WQF150–22–18.5		18.5				280	500	1150	860	243	223	240	258		255
49	150WQF200–22–22		22				280	500	1200	910	243	223	240	258		264
50	150WQF130–30–22						280	500	1200	910	243	223	240	258		264
51	150WQF150–30–30		30			50	350	500	1250	1000	230	242	260	278		383
52	150WQF250–22–30						350	500	1250	1000	230	242	260	278		383
53	150WQF250–26–37		37				350	500	1340	1100	230	242	260	278		440
54	150WQF250–32–45		45				350	500	1340	1100	230	242	260	278		449
55	150WQF 240–42–45 *					55	500	400	1630	1050	210	260	275	290		572
56	150WQF 250–50–55 *		55				500	400	1630	1050	210	260	275	290		605
57	200WQF300–7–11	200	11			70	280	500	1100	830	243	248	265	285		203
58	200WQF250–12–15		15				280	500	1100	830	243	248	265	285		221
59	200WQF250–15–18.5		18.5				280	500	1150	860	243	248	265	285		255
60	200WQF250–17–22		22				280	500	1200	910	243	248	265	285		273
61	200WQF400–18–30		30				320	500	1250	1000	245	250	270	285		391
62	200WQF400–20–37		37				320	500	1310	1050	245	250	270	285		458
63	200WQF500–22–45		45				320	500	1370	1150	245	250	270	285		467
64	250WQF600–9–30	250	30				400	530	1350	1130	290	218	258	300		400
65	250WQF600–12–37		37				400	530	1400	1180	290	218	258	300		467
66	250WQF600–15–45		45				400	530	1400	1180	290	218	258	300		475
67	200WQF400–8–15 *	200	15			80	420	500	1100	830	285	240	280	310		320
68	200WQFW400–10–18.5 *		18.5						1150	860						330
69	200WQF400–12–22 *		22						1200	910						340
70	200WQF340–22–30 *		30				460	400	1560	980	215	250	280	310		523
71	200WQF400–24–37 *		37													550
72	200WQF400–28–45 *		45						1640	1060						583
73	200WQF400–34–55 *		55													616
74	250WQF600–12–30 *	250	30				530	400	1570	990	210	265	315	360		550
75	250WQF600–15–37 *		37													572
76	250WQF600–18–45 *		45						1650	1070						605
77	250WQF600–24–55 *		55													638

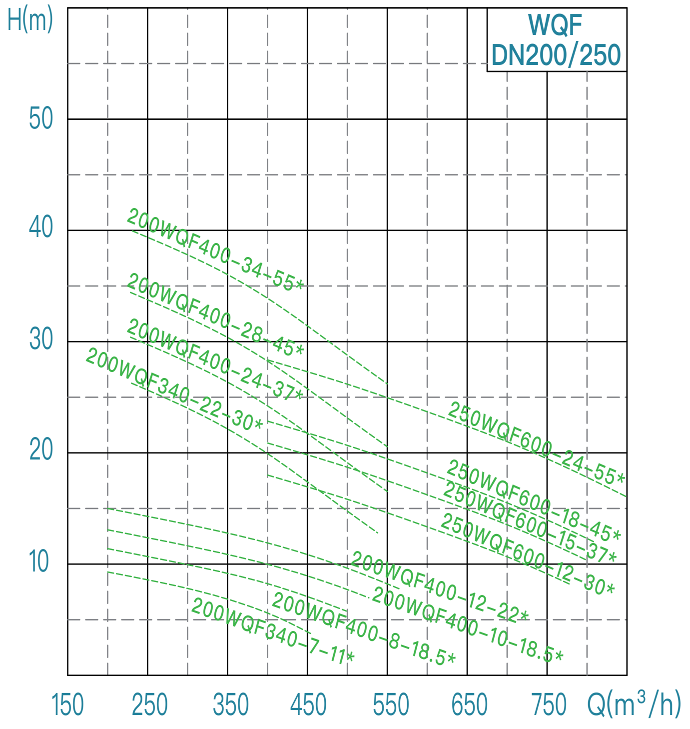
WQF performance curve chart (50Hz)



WQF performance curve chart （50Hz）







WQA-QG cutting
submersible sewage pump ►

WQA–QG

Cutting

Submersible sewage pump

Product Introduction:
The WQA–QG submersible sewage pump with a cutting impeller features a novel, aesthetically pleasing, and highly stable structure. Both the impeller and the cutting disc have sufficient hardness, providing excellent cutting performance for non–tough debris in the medium. Moreover, the cutting disc is equipped with a plum–shaped cutting edge to prevent entanglement at the inlet. (The length of debris that can be passed should not exceed twice the outlet diameter, and the particle diameter requirements are shown in the table.)

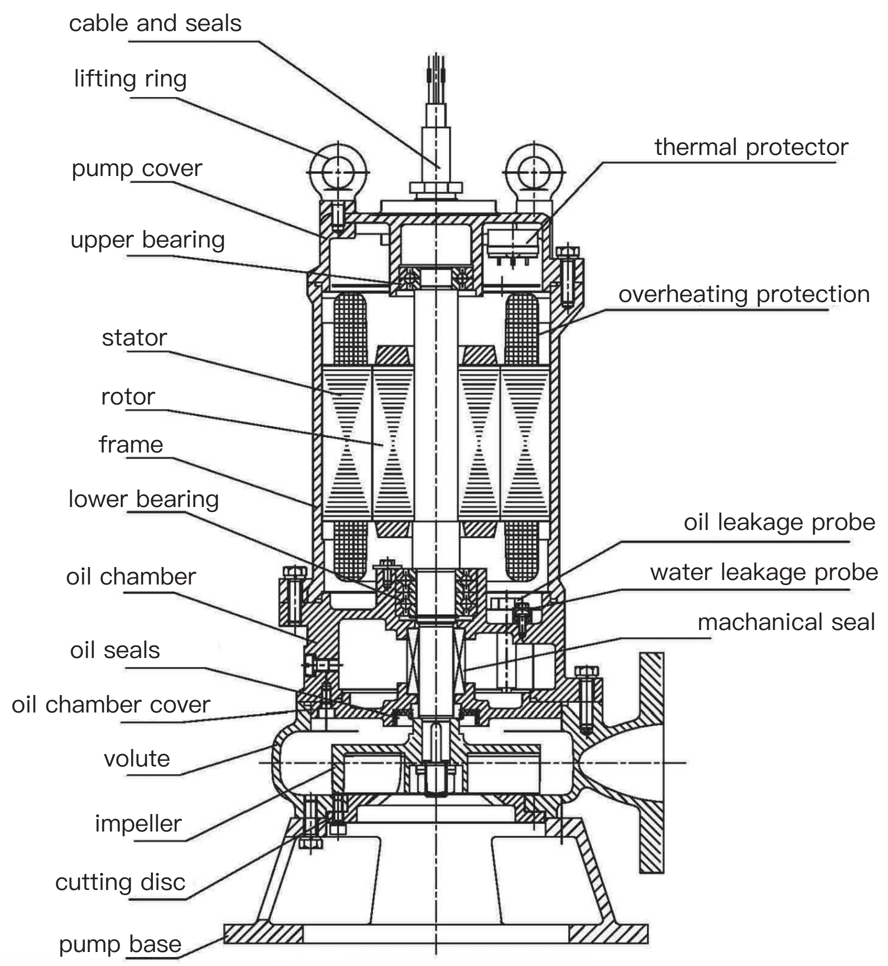
Limitations of Cutting Pumps
Although cutting pumps have strong cutting capabilities, they are not omnipotent. After prolonged use, the cutting head will wear out, leading to a decline in cutting performance. Additionally, the cutting head is prone to being entangled and blocked by long fibers and tough debris. Once this occurs, the cutting pump will fail to operate normally and may even result in damage to the cutting head. This indicates that cutting pumps may encounter difficulties in handling certain types of debris, such as long fibers and tough materials like hair and fabric scraps.

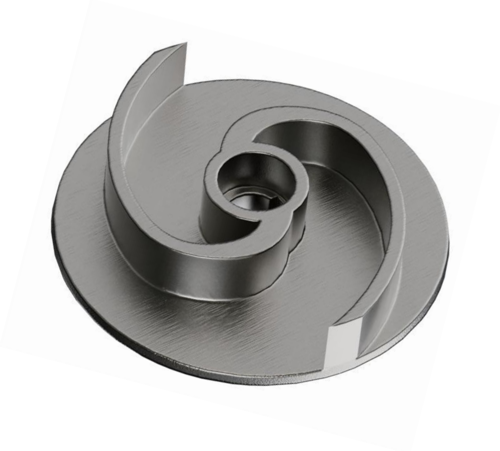
Scope of application:
municipal engineering、sanitary sewage、
building works、industrial sewage,
it can cut and convey ordinary plastic、fabrics、
paper、fibers and food residues, etc

Operating Conditions:
Flow Rate: 10 ~ 180 m^3/h
Head: 8 ~ 60 m
Medium Temperature: T ≤ 40℃
Medium Density: ≤ 1100 kg/m^3; Solid phase volume ratio in the medium is less than 3%
Medium pH Range: 4 ~ 10
Motor Insulation Class: Class F (standard), Class H (optional)
Protection Class: IP68
Minimum Operating Liquid Level: See "▽" in the installation dimension drawing (without motor cooling system); if the user's water level is lower than "▽", please contact the company's technical department.
The maximum diameter of solids in the medium should be less than the minimum size of the flow channel. The allowable solid passing diameter is detailed in the specific pump model size table.
Standard cable length is 9 meters. If a specific cable length is required, it can be specified in the order (Note: Voltage drop must be calculated).

Structure diagram of cutting pump

Structure diagram of WQA-QG products





cutting disc /cutting impeller

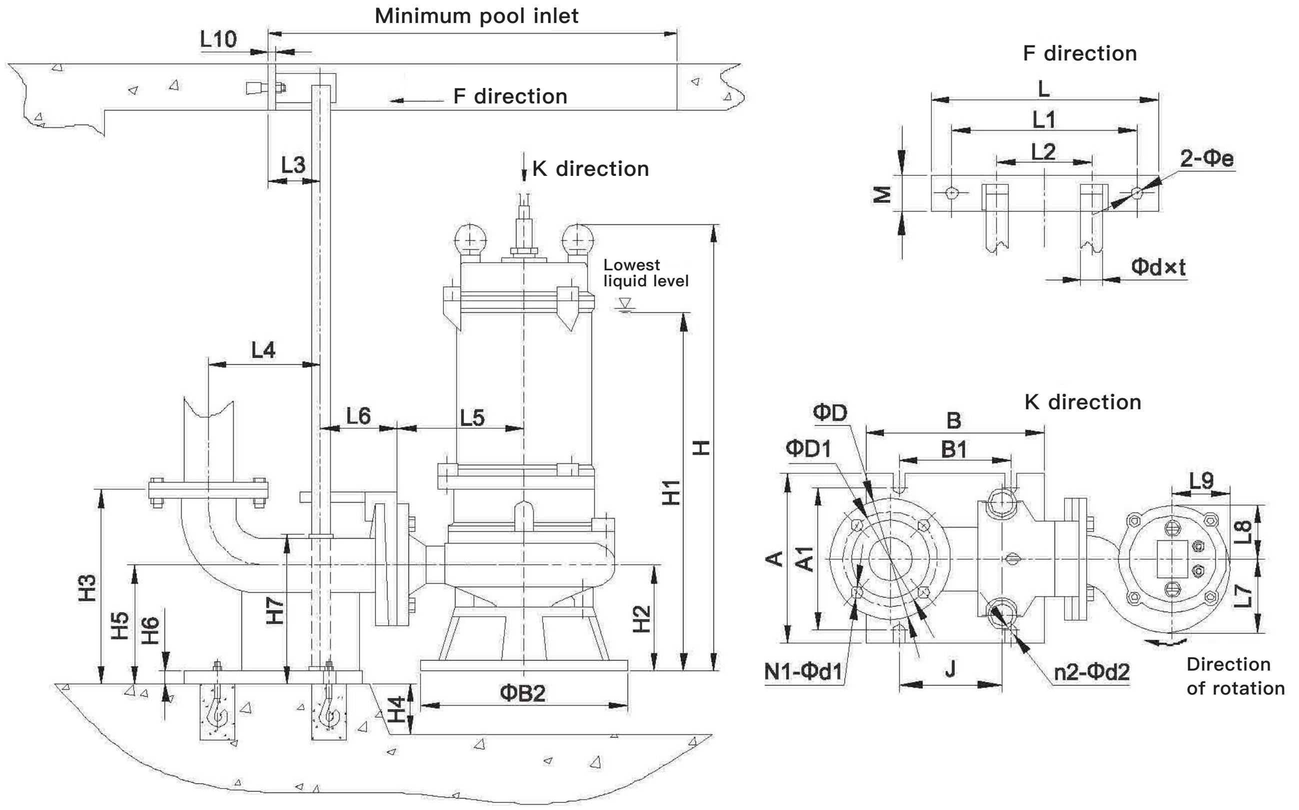
Main materials and configuration table

No.	Name		Standard	Optional
1	Volute (pump body, pressurized water chamber)		QT500	/
2	Impeller		QT500	/
3	Cutting disc		QT500	/
4	Oil chamber, Motor frame Pump cover, Upper cover, Pump base		HT200	/
5	Motor insulation class		F (155℃)	H (180℃)
6	Shaft		2Cr13	/
7	Bearing brand		Domestic	SKF/NSK
8	Mechanical seal	Brand	Domestic	Bergman
9		Motor side friction pair	Graphite/SiC (AQ1)	SiC/WC (Q1U1)
10		Pump side friction pair	SiC/SiC (Q1Q2)	SiC/WC (Q1U1)
11		Spring	Stainless steel	/
12		Rubber parts	(P)	/
13	O-ring		(P)	/
14	External leakage fastener		A2-70	/
15	Protection device (optional)	15kW and below	Order remarks according to customer requirements.	Water leakage probe/Float switch /Overheating protection

Specification/model of submersible motor cable

Motor power (kW)	Cable model	Standard main cable specification	Star delta	Model of controlling cable
0.55	YZW 300/500V	YZW 4×0.75	/	YZW 4×0.75 (Non-standard)
0.75				
1.1				
1.5		YZW 4×1		
2.2				
3				
4		YZW 3×1.5+1×1		
5.5				
7.5		YZW 3×2.5+1×1.5		
11/15-2P		YZW 3×6+1×4	2-YZW 3×2.5+1×1.5	YZW 4×1 (Non-standard)

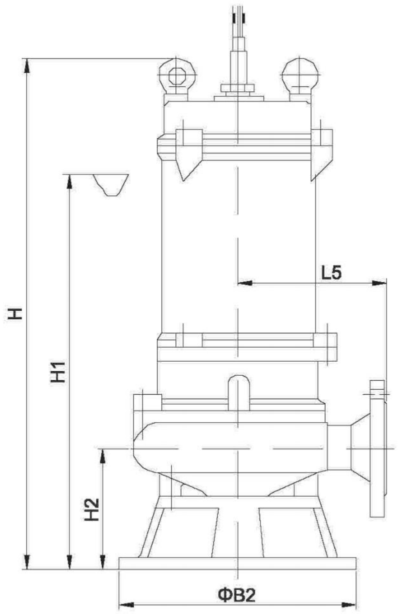
installition dimension drawing



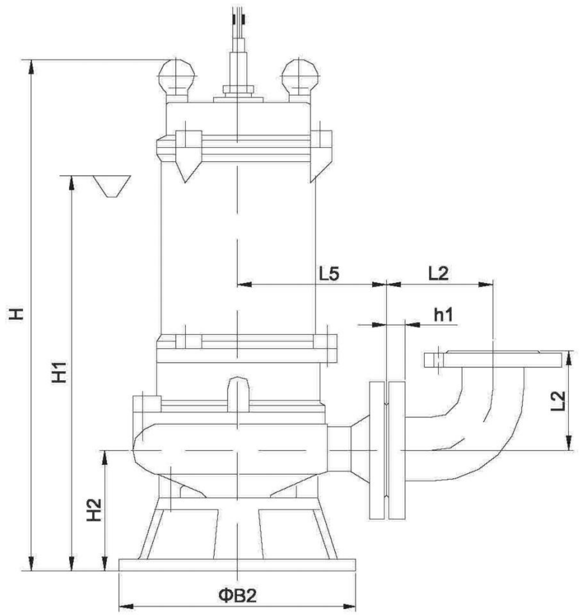
Flange coupling parts dimension table

Model DN	Flange Connection Size D	D1	N1-Φd1	Coupling Base Size A	A1	B	B1	J	n2-Φd2	H3	H4	H5	H6	H7	L	L1	L2	M	Φe	L3	L4	L6	L10	Φd×t
50	140	110	4-Φ14	197	168	208	130	120	4-Φ19	250	100	165	14	/	265	215	105	42	Φ14	66	135	90	15	Φ33×3
65	160	130		238	190	252	155	135		265		170	16		280	235	125	50		70	155	85	18	
80	190	150	4-Φ19	255	220	225	153	125		310		190			315	265	145	53		80	155	100	18	Φ48×3
100	210	170		293	265	258	175	140		350		235	18		365	305	170	55	Φ15	100	200	110	20	
150	265	225	8-Φ19	400	285	410	300	210	4-Φ12	485	150	300	22	390	410	260	275	60			290	190	20	

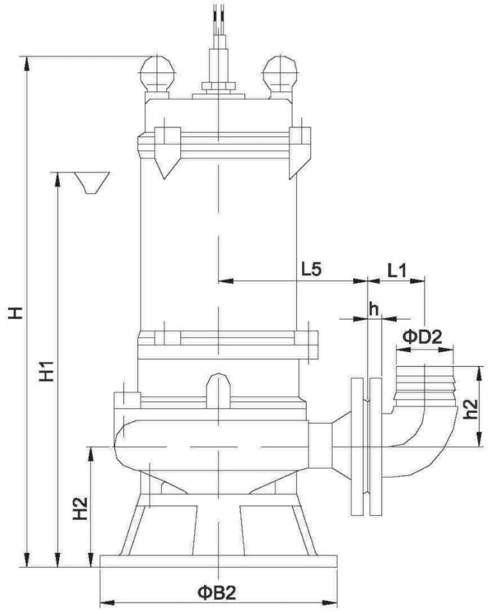
Pump appearance and mobile installition



Dimension drawing of pump appearance



Hard pipe（double flange elbow/joint）mobile installition

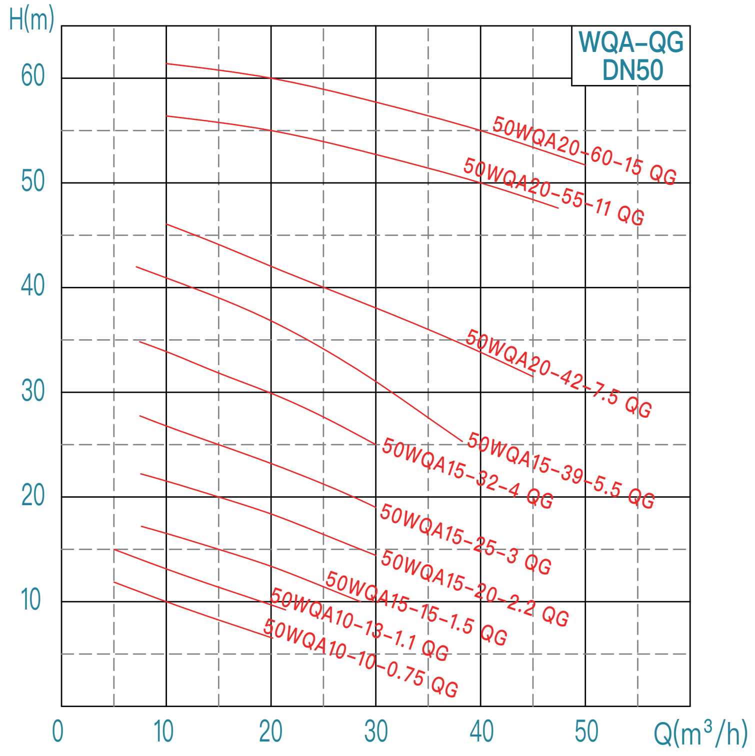


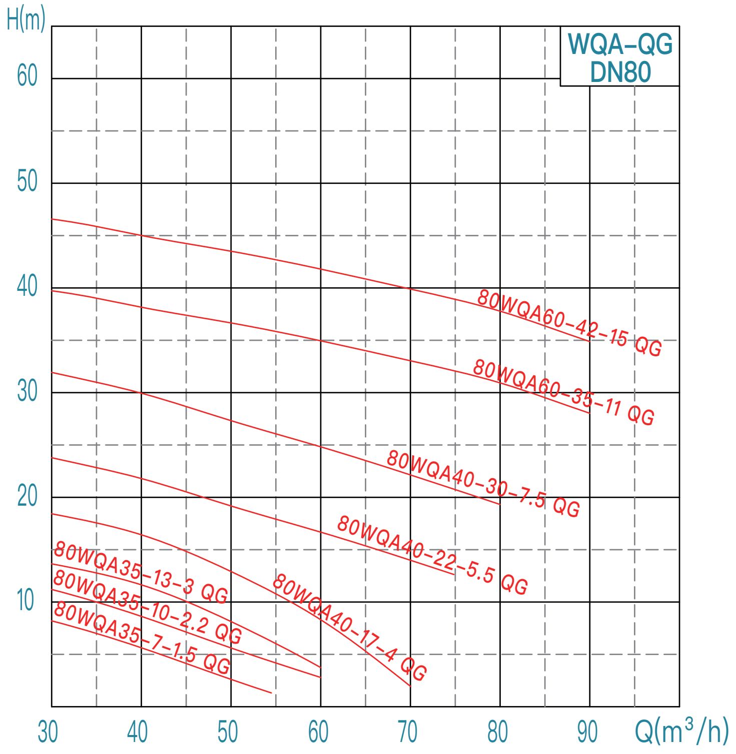
Soft pipe（elbow）mobile installition

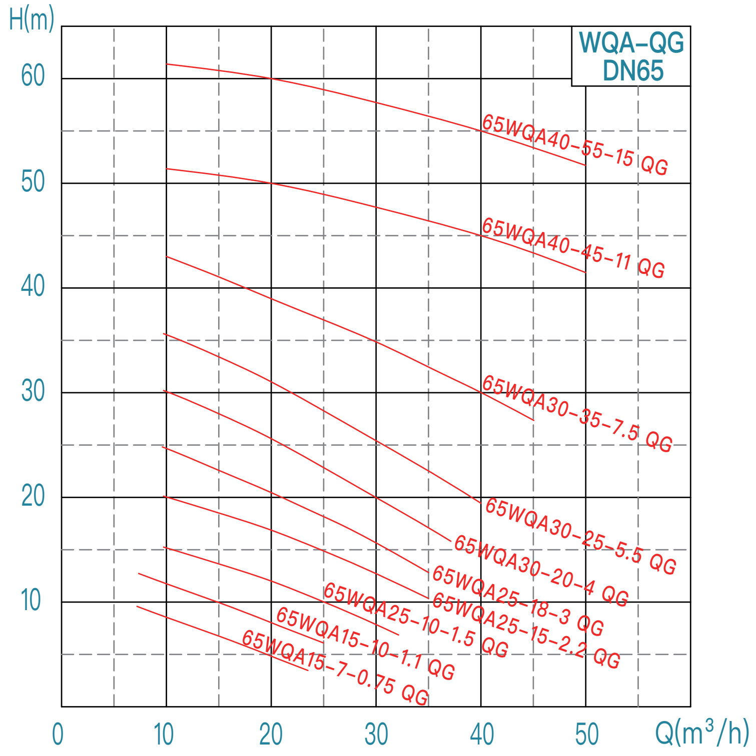
WQA-QG installition dimension table

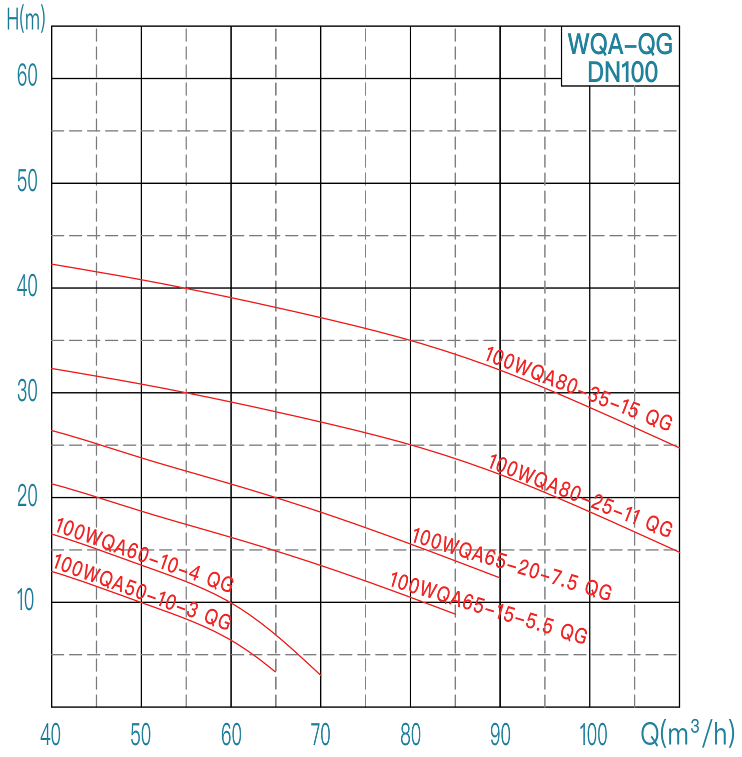
No.	WQA-QG model	Diameter	Power	Rotating speed	Voltage	Passable particle diameter	Appearance dimension table (mm)								Weight
		mm	kW	r/min	V	mm	L5	B2	H	H1	H2	L8	L9	L7	kg
1	50WQA10-10-0.75 QG	50	0.75	2950	220/380	15	135	200	460	380	100	82	90	95	26
2	50WQA10-13-1.1 QG		1.1												28
3	50WQA15-15-1.5 QG		1.5				145	270	485	400	115	100	105	110	32
4	50WQA15-20-2.2 QG		2.2						515	425					37
5	50WQA15-25-3 QG		3		380				535	445					42
6	50WQA15-32-4 QG		4				170	270	560	460	115	117	120	130	59
7	50WQA15-39-5.5 QG		5.5				180	320	650	540	135	140	135	145	70
8	50WQA20-42-7.5 QG		7.5				180	320	630	520	135	140	135	145	90
9	50WQA20-55-11 QG		11				190		710	600		140	145	150	130
10	50WQA20-60-15 QG		15						740	630					138
11	65WQA15-7-0.75 QG	65	0.75		220/380		135	200	460	380	100	82	90	95	26
12	65WQA15-10-1.1 QG		1.1												28
13	65WQA25-10-1.5 QG		1.5				145	270	485	400	115	100	105	110	32
14	65WQA25-15-2.2 QG		2.2						515	425					37
15	65WQA25-18-3 QG		3		380				535	445					42
16	65WQA30-20-4 QG		4				170	270	560	460	115	117	120	130	59
17	65WQA30-25-5.5 QG		5.5						585	485					67
18	65WQA30-35-7.5 QG		7.5				180	320	630	520	135	140	135	145	90
19	65WQA40-45-11 QG		11				190	320	710	600	135	140	145	150	130
20	65WQA40-55-15 QG		15						740	630					138
21	80WQA35-7-1.5 QG	80	1.5		220/380	40	170	270	495	405	125	110	115	120	40
22	80WQA35-10-2.2 QG		2.2						525	435					44
23	80WQA35-13-3 QG		3		380				555	455					49
24	80WQA40-17-4 QG		4				180	270	585	485	130	115	125	135	62
25	80WQA40-22-5.5 QG		5.5			25			610	510					73
26	80WQA40-30-7.5 QG		7.5				200	320	645	530	145	140	140	150	95
27	80WQA60-35-11 QG		11						720	610					137
28	80WQA60-42-15 QG		15						750	640					144
29	100WQA50-10-3 QG	100	3			40	170	270	555	455	125	110	115	120	49
30	100WQA60-10-4 QG		4				180	270	585	485	130	115	125	135	62
31	100WQA65-15-5.5 QG		5.5			25			610	510					73
32	100WQA65-20-7.5 QG		7.5				200	320	645	530	145	140	140	150	95
33	100WQA80-25-11 QG		11						720	610					137
34	100WQA80-35-15 QG		15						750	640					144
35	150WQA130-9-7.5 QG	150	7.5			50	240	320	670	560	165	145	160	175	128
36	150WQA140-14-11 QG		11						755	645					142
37	150WQA150-20-15 QG		15						785	675					150

WQA-QG performance curve chart (50Hz)

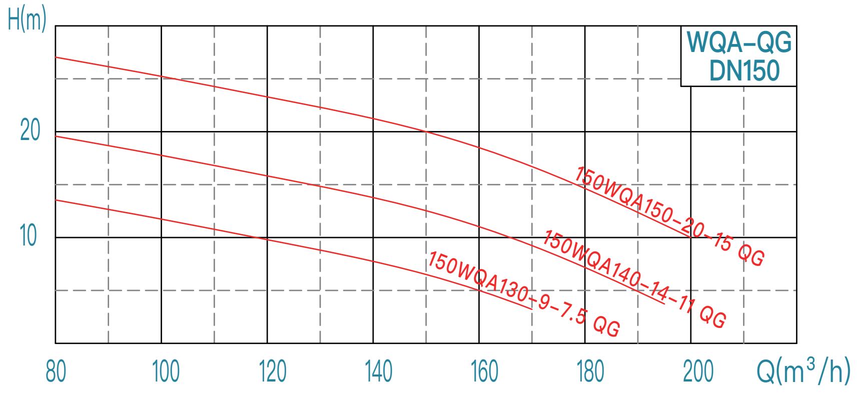








WQA-QG performance curve chart (50Hz)



Ordering, Installation and After-Sales Service Guidelines ►

Order description

To make the pump you choose more in line with working site requirements, users are warmly welcome to call our technical department.

Please specify when ordering:

1. Pump model, flow, head, speed, power, motor power, medium temperature, medium composition, medium density
2. Parts material: pump body, impeller
3. Installation method, outlet diameter
4. Please specify starting method in the order if there are some requirements.
5. The standard water pump cable is 9 meters, specify it if other lengths are required.
6. The complete set of supply parts is provided according to the determined installation method.
7. Necessary parts, optional parts and spare parts need to be ordered separately.
8. Follow the principle of "spread before turning" when connecting the spinal canal and the elbow joint on the discharge pipeline.
9. Fill in the selection form.

No.	Pump model:	Configuration: standard Optional (specify in the order)
1	Flow: Q= m³/h	Starting method:
2	Head: H= m	Wiring method:
3	Pump immersion depth:h= m	Medium temperature :Normal ℃; Max ℃; Min ℃
4	Outlet diameter: mm	Corrosion component:
5	Insulation class: F(standard) H(optional)	Oxide concentration(PPM):
6	Motor rated power: kW	Sediment content:
7	Voltage on site: V	Maximum particle diameter: mm
8	Pump speed: r/min	Fiber type and length:
9	Frequency: Hz	Ambient temperature:Normal ℃; Mak ℃; Min ℃

Supply list

	Supply list	Installation method				
		Automatic coupling installation(Z)	Fixed base installation (P)	Hard pipe mobile installation(Y)	Soft pipe mobile installation(R)	Single pump
Complete supply parts	Pump (9-meter cable)	√	√	√	√	√
	Coupling base	√				
	Gusset plate	√				
	Coupling rod	√				
	Pump base (1pc/set)	√	√	√	√	√
Essential parts	Guide rod	√				
	Anchor bolt	√	√			
	Expansion bolt	√				
	Hard pipe (1 pc/set)		√	√		
	Soft pipe (1 pc/set)				√	
Optional parts	Submersible pump monitoring device	√	√	√	√	√
	Stainless steel wire rope and rope clamp for lifting pump	√	√	√	√	√
	Spinal tube	√	√	√	√	√
	Mating flange	√	√	√		√
	Flange joint		√	√		√
	Gate valve/butterfly valve	√	√	√	√	√
	Check valve	√	√	√	√	√
	Electric control cabinet	√	√	√	√	√
	Terminal box	√	√	√	√	√
	Float switch (liquid level)	√	√	√	√	√
Spare parts	Impeller	√	√	√	√	√
	Bearing	√	√	√	√	√
	Mechanical seal	√	√	√	√	√
	Seal ring	√	√	√	√	√
	O-ring seal	√	√	√	√	√

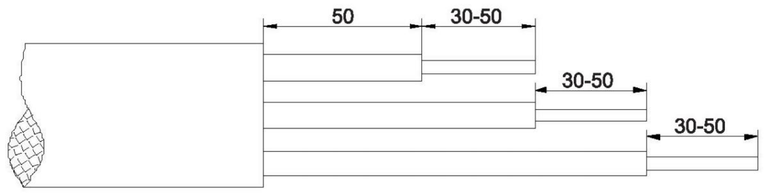
Installation instructions

● **installation**
1.Check the pump,motor, and fasteners before installation.
2. The weight of the pipeline should not be added to the pump to avoid deformation, vibration.etc.
3. The anchor bolts must be tightened during installation, and the guide rod and coupling gusset must be fixed.
● **Start up**
1. The liquid level shall not be lower than the minimum liquid level when starting, the minimum liquid level is shown in the installation diagram▽.
2. Start the motor, check rotation direction. The pump should rotate clockwise looking down from the motor side.
3. ⑴For centrifugal type impeller, close the valve and pressure gauge on the outlet pipeline.
⑵For mixed-flow and axial-flow impellers, open the valve according to the pump model.
4. Start the motor,turn on the pressure gauge.As the speed rises,gradually adjust the valve opening extent on the outlet pipe to achieve the required flow.
● **Stop**
1. First close the outlet valve, close the pressure gauge, and then cut off the motor power supply.
2. In case of long-term shutdown, it is recommended to lift the pump up, clean it up, put it in a dry place, and keep it properly.
● **Maintainance**
1. Check the motor current value regularly
①When the ambient temperature is motor rated temperature, the current must not exceed the rated current.
②When the ambient temperature is lower than motor rated temperature, the current value can be slightly increased, please refer to the motor data for details.
③When the current is abnormal, please find out the cause immediately.
2. After long-term operation, the pump should be shut down for inspection due to the wear of mechanical seals, bearings, impellers, seal rings and other parts. And wearing parts like mechanical seals, bearings, O-rings and seal rings should be replaced. The general inspection cycle is normally one year.The service life of the mechanical seal is determined by the on-site working conditions, and the designed service life is 8000 hours.
3. Check the oil chamber regularly. If oil emulsification occurs, replace 32# engine oil in time. Executive standard: GB443-89

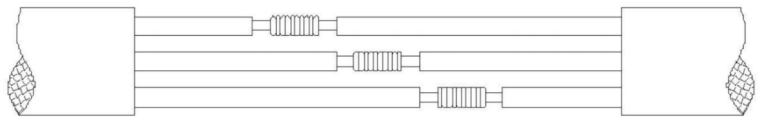
Diving cable connection

When the length of the cable cannot meet users' installation requirements, an external cable is required. The wiring should be operated by a full-time electrician. The cross-sectional area of the cable is determined by the installation length, motor power and starting method. The motor cable and the external cable joints are required to be reliably sealed, insulated, and have certain strength. The wiring process is briefly as follows.

1. Strip the cable on the sewage pump as shown in the figure below, and strip the copper core by 30~35mm long. Wipe it with gauze and copper wire to make it shine. Also strip the copper wire of the external three-core cable by 30~ 35mm,and wipe it clean with gauze.。

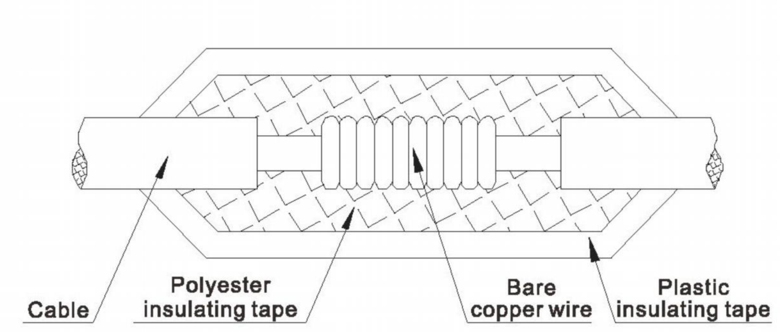


2. As shown in the figure below, insert the stripped copper wire parts of the three-core cable and external cable together, and then use a thin copper wire to bind tightly as shown in the figure, cut off the remaining part, flatten it with scissors, and do not tie your hands. Tie the three wires in the same way.

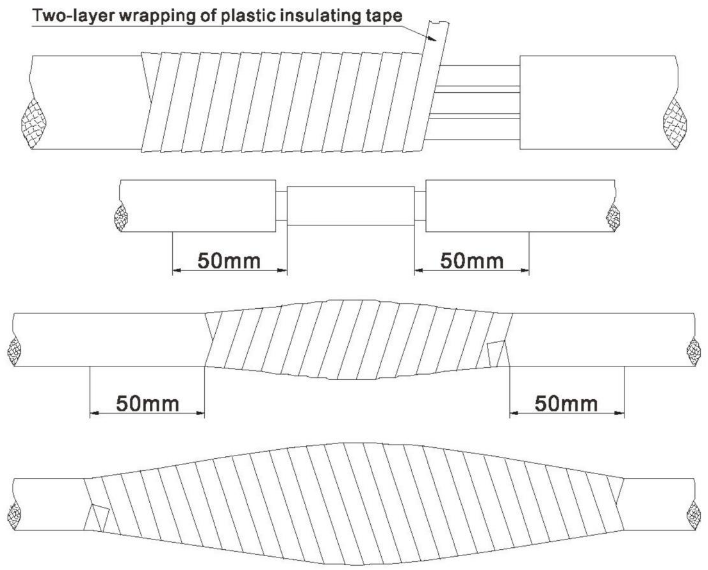


3. Prepare a small iron box or small iron pot that can stick the three wire ends, put solder into the pot and heat it on fire until it melts, and then apply proper amount of soldering oil to the three wire ends,quickly put into the pot and weld it firmly. The surface of the welding head is required to be smooth, free of burrs and false welding. If the welding is not strong or not smooth, it should be re-welded until it meets the requirements.

4. As shown in the figure below, when wrapping with polyester insulating tape, be sure to press half of the first circle (shift wrapping method) for 8~12 layers. After wrapping, wrap two more plastic tape layers for strength protection.



5. As shown in the figure below, wrap the three wires together with plastic tape by two layers. The first layer exceeds the end of each wire insulation layer by more than 50mm, and the second layer exceeds the first layer by more than 50mm. It needs to be tightly wound to exclude air space to the largest extent.



6. After wrapping, take a basin of cold water and completely soak the banded thread in the water. Measure the insulation with a 500V megohmmeter after 12 hours.The insulation should not be less than 50 megohms, otherwise it should be re-wrapped until it meets the requirements.

7. The grounding wire of the submersible electric pump also needs to be wrapped according to the wiring process requirements of the submersible cable.

Note: Customer needs to calculate voltage drop when connecting additional cable.Pay attention to voltage change during operation which should be generally controlled within±5% of the rated voltage.

Common malfunctions and solutions

Malfunctions	Cause analysis	Solutions
None or insufficient pump water flow	◎ Motor reverse rotation ◎ The impeller passage or pipeline is blocked ◎ Tubing resistance is too large ◎ The impeller is severely worn ◎ Gas mixed in the medium ◎ Check valve direction is reversed ◎ Head on customer's site is not enough ◎ Coupling leaks	◎ Adjust two phase power sequence ◎ Remove impurities, it is best to adda thin grille at the water inlet ◎ ①Reduce pipeline bends, increase diameter for long pipe ②Check whether the valve is open ③Check whether the impeller is scraped ◎ Replace the impeller ◎ Exhaust gas, increase liquid depth to reach the minimum water level requirement "Δ" ◎ Adjust the direction of the check valve ◎ Check whether customer's test method is correct after excluding the above reasons ◎ Check whether the installation is proper and ensure that the guide rod is vertical
The pump fails to start	◎ Lack of phase ◎ Impeller stucks ◎ Short circuit of winding joint or cable ◎ Stator winding burnt out ◎ Control cabinet error ◎ Low power supply voltage	◎ Check the circuit, eliminate problem of lack-phase ◎ Clear impurities ◎ Recover after checking with ohmmeter ◎ Replace the winding or stator, it is better to add a temperature measuring element ◎ Check the control cabinet and remove fault ◎ Solve power supply voltage problems
Large fluctuations in discharge pressure	◎ The liquid level of suction pool is too low ◎ The suction elbow is not tightly sealed, making air entering the pump ◎ Medium temperature is high	◎ Control the minimum liquid level of the suction pool ◎ Check the pipe connection and eliminate the cause ◎ Reduce medium temperature. If the temperature cannot be lowered, increase the pump minimum liquid level and replace high-temperature motor
Overloading current	◎ The head of the selected pump exceeds the actual head. Or the head of the device is greatly reduced, indicating that the pump is running at a large flow rate ◎ The density or viscosity of the medium is too high ◎ Bearing is damaged ◎ There are impurities at the neck ring ◎ Power supply voltage is too low	◎ Reduce outlet pipe valve opening or the outer diameter of small impeller, or replace a pump that matches the actual working conditions on site ◎ Dilute medium ◎ Replace bearing ◎ Clean up impurities ◎ Solve power supply voltage problems
Heating bearing	◎ Bearing wears ◎ Bearing grease is too little or too much ◎ Grease is degenerative	◎ Replace the bearing ◎ The amount of grease added is about 1/3~1/2 of the oil chamber cavity ◎ Change grease

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Version number：251201